

Guide to the Atlantium Hydro-Optic UV System

RZB300 System

Allen Bradley Control Module

For General Applications

**Supported Electrical Voltage:
400/440/480VAC**

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System and accessory specifications are subject to change without notice.

Control software version: **AB Ver. 11 - Display Ver. 10**

IMPORTANT NOTICE

It is strictly forbidden to alter/change the system hardware in any way.

Any change made without written permission from Atlantium shall void the warranty.



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1 Introduction

Congratulations on purchasing an Atlantium **RZB300 System** Hydro-Optic UV Water Treatment System. This user guide provides you with clear guidelines for preparing the site for installation, correct installation and software configuration procedures, and a good understanding of all system functions for maximum benefit from your Atlantium Unit.

The Atlantium **RZB300 System** systems guarantee water bio-security at optimized energy efficiency, and meet EPA criteria for providing bio-security.

Atlantium systems are designed to provide applications for the following Industries:

- | | |
|--------------------|-----------------|
| ■ Aquaculture | ■ Bio-Pharma |
| ■ Fish Hatcheries | ■ Bottled Water |
| ■ Dairy | ■ Beverages |
| ■ Municipal | ■ Breweries |
| ■ Power Generation | ■ Aquariums |
| ■ Ultra Pure Water | ■ ...and more |
| ■ Marine | |

Applications

- **Water Disinfection** - meets US Environmental Protection Agency (EPA) criteria for providing bio-security
- **Chlorine Replacement** - provides a chemical-free alternative to chlorine for treating source water. It easily inactivates microorganisms including those that are resistant to chlorine and heat.
- **Non-Thermal Pasteurization** - replaces energy-intensive heat pasteurization of water - meets US FDA criteria for pasteurized equivalent water.
- **Post-Active Carbon** - protects against breakthroughs and reduces bio-fouling downstream. Significantly reduces the frequency of aggressive cleaning, and lengthens the lifespan of RO membranes.
- **Dechlorination / Chlorine Destruction** - efficient, clean, non-chemical alternative to activated carbon filter or sodium bio-sulfite. Reduces risk, better protection, lower operating expenses.
- **Ozone Destruction** – efficient, cost-effective solution.
- **Ultra-pure water (UPW)** - efficient, cost-effective solution that complies with the uncommonly stringent specifications for *high-purity water*.
- And more....

Patented Innovative Technology

- Field proven worldwide, used by world-leading industries
- Based on Hydro-Optic engineering that optimizes UV efficiency and enables uniform dose distribution
- Medium Pressure High Intensity lamps provide a UV dose in the 200 to 454 nm spectrum which is much broader than low pressure UV (which yields a dose in the 254 nm range only). Hydro Optic Disinfection UV (HOD UV) is effective in cold & warm water and compromises microorganisms' repair mechanisms for long-term bio-security.
- Guaranteed required dose for complete microbial inactivation.
- Sustainable operation provides the security and stability that the industry demands.
- Uses "total internal reflection" of UV light within the outer quartz sleeve of the reaction chamber, enabling more effective treatment at a lower dose.

- Some enclosures/components as noted on the Atlantium main website under "About" <https://atlantium.com/about.html> include:
 - Nema 4x rating for indoor and outdoor use, protection against windblown dust and rain, splashing water and hose directed water.
 - IP56 rating protection from limited dust ingress. Protected from high pressure water jets from any direction

Guaranteed Dose Delivery

- Unique monitoring & control software ensures that required UV dose is being delivered at all times.
- Dedicated UV Intensity Sensor (UVIS) for each lamp for real-time monitoring.
- Automatically adjusts lamp power based on real-time analyses of critical parameters such as water flow rate, lamp intensity and water Transmissivity to UV (UVT).
- A user-friendly interface displays the monitored data in real-time including the actual delivered dose as well as the other critical parameters, so you can verify that you are actually getting the dose that you need.

Lower Electrical Consumption

- Optimized light distribution, through Hydraulic and Fiber optic principles applied to the UV system design, enable:
- Optimized UV power and uniform dose distribution.
- Efficient use of power, with lower electrical consumption than low-pressure based UV systems.

Minimum Down Time

- True in-line system.
- Quick lamp replacement (4 minutes) and better safety.

Integrates with your Process Line

- Field proven installation.
- With no additional software, you can integrate the Atlantium system into your current process using Atlantium's customizable settings.

Regulatory Compliance

- Meets the most stringent regulatory criteria including US Food and Drug Administration (FDA) and EPA.
- Automatically generated reports at the push of a button for proof of compliance and traceability.
- Provides validated minimum required dose and not average dose.

Efficiency and Easy Maintenance

- For total efficiency and dependable performance, the Atlantium system monitors the lamps for performance, ensuring that a lamp is replaced only when absolutely needed.
- By following the easy preventive maintenance programs, you keep the system working at peak performance.
- The **Viewport** allows you to safely peek at what's going on inside.

For further information about Atlantium products and technologies, visit our website at <https://atlantium.com/>

1.1 Before You Begin

Before you begin using this product, or any installation or service operation, please read the following safety information:

- Attention to these warnings helps prevent personal injuries and damage.
- It is your responsibility to use the product in an appropriate manner.
- This product is designed for use solely indoors.
- You are responsible if the product is used for other than its designated purpose or in disregard of Atlantium instructions. Atlantium shall assume no responsibility in any way for such use of the product.
- The product must be used for its design purpose, based on its product documentation and within its performance limits.
- Using the product requires technical skills. It is therefore essential that only skilled and specialized staff or thoroughly trained personnel be allowed to use the product.
- Keep the basic safety instructions and the product documentation in a safe place and pass them on to the users.
- Applicable local or national safety regulations and rules for the prevention of accidents must be observed at all times.

1.2 Tags and Their Meaning

The following indicators are used in the product documentation to warn the reader about risks and dangers

DANGER!

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.



WARNING!

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION!

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

ATTENTION!

Indicates the possibility of incorrect operation which can result in damage to the product.



Indicates a hazardous situation involving electricity which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving Electrostatic Discharge (ESD), which, if not avoided, can result in damage to the product.



Indicates a hazardous situation involving UV light which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving a hot surface which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving working in an enclosure with moving parts, which if not avoided, can result in serious injury to the hands.



Indicates a hazardous situation involving mercury, a poisonous material which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving caustic chemicals which, if not avoided, can result in death or serious damage.



Indicates that components or equipment are heavy and care is to be taken to avoid lifting incorrectly. Incorrect lifting can be dangerous to the personnel lifting and may result in dropping and damaging the components or equipment.



Indicates that components or materials to be discarded are classified as hazardous waste and must be disposed of appropriately.



Indicates information related to installation, safety or use of system.



2 About the RZB300 System Systems

The Atlantium **RZB300 System** includes a disinfection unit with a specially designed lamp chamber, a Ballast Module/Plug-style power supply configuration, and **Control Module**. The quartz disinfection chamber takes advantage of **Total Internal Reflection** and reflects UV light back into the water to uniformly distribute the UV dose. Water flows unimpeded, resulting in low head-loss. The specially designed short (Arc length 250mm/9.84 inch) MPHI lamps deliver a broad spectrum disinfection that irrevocably damages DNA and proteins in microorganisms. Atlantium's Quick Connect MPHI lamps, each housed within a thick quartz sleeve, significantly reduce risk of breakage and its consequences. A UV Intensity Sensor (UVIS) provides continuous measurement of each lamp's output.

In addition, a UVT Analyzer reports and tracks the water quality and a UV-proof Viewport allows you to view the inner workings. An optional cleaning system cleans the surface of the quartz sleeves automatically.

2.1 The RZB300 System Units

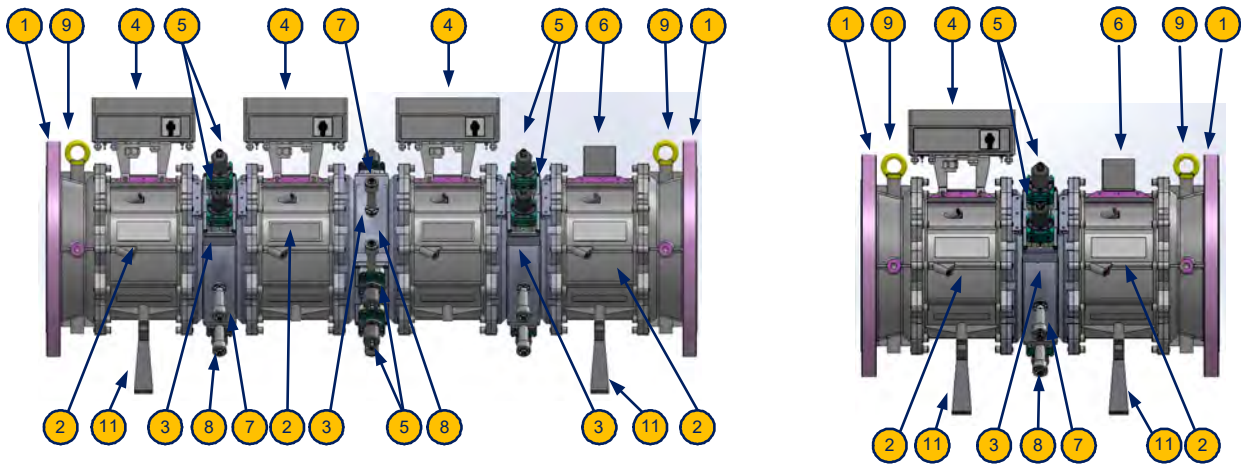


Figure 2-1: RZB300-13, RZB300-11

Table 2-1: Atlantium Unit Key

#	Item	Description
1	Flange or Ferrule interface	The unit connects to the facility piping via a flange interface as per customer order
2	Quartz Chamber	The quartz pipe in which the disinfection occurs
3	Lamp Chamber	The chamber that holds the lamp and optional Cleaning system
4	Connection Box	The box that sits on top of each lamp unit and holds the electrical circuitry
5	Lamp	Lamp encased in quartz sleeve assembly Medium Pressure High Intensity (MPHI)
	Arc length (mm / inch):	250 / 9.84
	Lamp total length (mm / inch):	417.8 / 16.5

Table 2-1: Atlantium Unit Key

#	Item	Description
6	Viewport	The UV-proof window to view the Unit working.
7	UVT Analyzer	The sensor that measures the UV transmittance through the water. Together with the UV Intensity sensor and the Control Module , it analyzes, tracks and reports the water quality.
8	UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds it to the real time dose calculation.
10	Cleaning system	A sophisticated cleaning system for the quartz sleeve
11	Triangle Support Brackets	Brackets that are used to support and balance a unit anchored to customer supplied supports or surface that hold up the unit at least 750mm / 29.5inch above the floor (see RZB300 System Interface Control Drawings on page 13).

2.2 RZB300 System Interface Control Drawings

For the ICDs of the systems in the **RZB300 System**, consult the accompanying ICD files.

2.3 Atlantium System Footprint

The Atlantium system has a small, compact footprint that facilitates in-line installation in any process train.

Table 2-2: **RZB300 Series** Physical Specifications

Item	Detail	
Number of lamps	Dependent on model supplied: RZB300-1X where X= No. of Lamps e.g. RZB300-12 refers to 2-lamp model Refer to ICD	
Atlantium Unit Construction Materials: Casing: Quartz Chamber:	Electro-polished stainless steel 316 High grade fused silica (Quartz)	
Standard Pipe Flange Interface Options:	DIN2576 DN350 PIN16	14" 150-lbs ANSI
Unit Length (mm / inch) RZB300-11 RZB300-14	837 / 32.9 1956 / 77	851.2 / 33.5 1977 / 77.8
Flange Inner Ø Diameter (mm / inch)	350 / 13.8	355.6 / 14
Flange Outer Ø Diameter (mm / inch)	505 / 19.9	533.4 / 21
Width at Widest Point (mm / inch): Without DPM:	505 / 19.9	533.4 / 21

Table 2-2: **RZB300 Series** Physical Specifications

Item	Detail	
Bolt Circle Diameter (mm / inch)	460 / 18.11	476.3 / 18.75
Bolt Hole Diameter (mm / inch)	22 / 0.87	28.4 / 1.12
Number of Bolts	16	12
Unit Volume (Liter / Gallon)	Refer to ICD	Refer to ICD
Weight (without water) (Kg / lb)	Refer to ICD	Refer to ICD
Service Clearance on each side (mm / inch)	750 / 29.5	750 / 29.5
Minimum Height above Floor Level (mm / inch)	750 / 29.5	750 / 29.5
Maximum Working Pressure	7 Bars / 102 psi	7 Bars / 102 psi
Maximum Flow Rate	Application Dependent	Application Dependent
Distance between Bolt Holes on Triangle Support Brackets (mm / inch)	Refer to ICD	
Bracket Nut / Bolt Size (mm / inch)	13 / 0.5	13 / 0.5

2.4 The Ballast Module

Ballast Module/Plug-style is a water-proof easy plug-n-play box that contains the ballast power supply and related circuitry. Its convenient plugs make it easy to service. Its modular design makes it serviceable and flexible and easier to fit into an existing plant. A Ballast Module is supplied for each lamp in the Atlantium system. Its heat sink frame complies with industry standards and contains one cooling fan on the top. It can be mounted on the wall or other suitable anchoring point. The location of the Ballast Module/Plug-style must be designated according to the maximum distance needed for positioning the **Cable Harness** of the Atlantium Unit. The **Cable Harness** is **5m / 16.4'** (also available in **10m / 32.8'** length). For systems with more than one lamp, the Ballast Modules can be positioned one above the other or next to one another. The Ballast Module must have at least 300 mm (11.8 inches) of clearance on its right and left sides for proper ventilation and 300mm (11.8 inches) of clearance on the bottom for cable routing. For more details on mounting, see **Installing the Ballast Module(s)*** on page 49.

Table 2-3: Ballast Module Physical Details

Physical	Details
Dimensions (W x D x H) (mm/inch):	313 / 12.3 x 228 / 9.0 x 548 / 21.6 39 / 15.7 x 228 / 9.0 x 548 / 21.6
Weight (Kg/Lb):	22/ 48.5
Heat Load per Ballast Module: (For calculating space cooling requirements)	210W
Protection Category:	IP 56.

2.5 Electrical Usage

Utilize a Residual Current Device (RCD) that is Type A with 3000A surge resistance.

Atlantium recommends ABB P/N F204 A-xx/0.03 AP-R (xx represents the current rating according to the system load. Select the appropriate RCD accordingly. Calculating the value, takes into account the power consumption of the system (see the table in the section below) and the mains voltage of your locale.

The tables below detail the Atlantium Unit's electrical usage and power consumption.

Table 2-4: **RZB300 System** Unit Electrical Usage

Electrical	Details
Phase:	3 phase
Voltage:	400/440/480 VAC per Ballast Module
Maximum allowed voltage tolerance:	±10% from the nominal voltage

2.5.1 Power Consumption and Local Voltage Requirements

The table below details the power consumption for the **RZB300** per segment according to local voltage.

Table 2-5: **RZB300-11/12** Power Consumption

# of lamps :			2			4		
Nominal lamp power (kW) :			8.4			16.8		
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required System Circuit Breaker	Required Branch Circuit Breaker	Current consumption each phase (A)	Required System Circuit Breaker	Required Branch Circuit Breaker
No cleaner	Direct Feeding	400	13.4	3x16A	-	26.8	3x32A	3x16A
		440	12.1	3x16A	-	24.2	3x32A	3x16A
		480	11.1	3x16A	-	22.2	3x32A	3x16A

For the power consumption of the system supplied, refer to the ICD

2.6 Surge Protection

Every Atlantium system **must** be protected by a device that prevents electrical surges and fluctuations from causing damage. There are three types of surge protection:

- UPS (Uninterrupted Power Supply) has an internal battery that allows the equipment plugged into it to continue to run in the event of a power outage until power is restored. It also has the benefit of performing the same functionality of surge protectors, protecting against spikes and surges. If needed, the UPS allows the system to be shut down cleanly if power is out for a lengthy period.

- Surge protection device protects your equipment from variations in electrical current, such as surges and spikes.
- OVPD module (Over Voltage Protection Device) - available from Atlantium. see below.

2.6.1 Surge Protector Devices

Surge protection devices are designed to protect industrial communication networks. This device recommended by Atlantium uses a combination of 3-electrode gas discharge tubes and fast clamping diodes. Typical applications include industrial processing equipment, transmission systems, I/O cards, probes, actuators, and displays. Surge protection devices are available via Atlantium.

Table 2-7: Surge Protector Details

Atlantium PN	DATA SHEETS	Module PN	Description
EP0002300	DOC002300	8859660000	SURGE PROTECTOR 400V 3P 40KA TNC CONFIG
EP0002000	DOC002000	4983-DS277-404	SURGE PROTECTOR 440-480 3P 40KA
EP0002600	DOC002600	4983-DS120-402	SURGE PROTECTOR L + N, 115V
EP0002800	DOC002800	1352740000	SURGE PROTECTOR L + N, 230V

2.6.2 The OVPD Module

The Over Voltage Protection Device (OVPD Module) may be required in locations where fluctuations in the voltage may occur. If supplied, see the **OVPD Module ICD** accompanying the documentation. Consult your Atlantium representative.

2.7 The Control Module

The Atlantium system is monitored and controlled from a **Control Module**. It can be mounted on a wall or suitable anchoring point up to 50 meters (164.04 ft.) from the Atlantium Unit. However, it should be located in a close proximity to the Atlantium Unit in a place that it is easy to reach the touch screen and read the monitoring information. See the **Control Module ICD** accompanying the documentation.

2.8 Environmental Requirements

The Atlantium system should be situated indoors in an area that is adequately cooled and with sufficient air flow. However, there have been many successful Atlantium installations in less friendly environments. Consult your Atlantium representative to choose the most suitable location.

Table 2-8: **RZB300 System** System Environmental Details

Environmental	Details
Maximum ambient air temperature:	45°C (113°F)
Maximum ambient air temperature in the Ballast Modules environment:	45°C (113°F)
Maximum atmospheric relative humidity:	95%
Water Temperature: Maximum – high temperature: For water temperature above 70°C (160°F) consult with Atlantium service engineers Minimum – cold temperature:	90°C(194°F) No limit

2.9 Regulatory Compliance

Atlantium Systems are designed to comply with the following regulatory standards:

- EMC directive 2004/108/EC
- European Low Voltage Directive (LVD), 2006 / 95 / EC for electrical safety
- Council Directive 98 / 83 / EC of 3 November 1998 for the quality of water intended for human consumption
- ISO 9001:2008 Quality Management Standard – Development, Design, Production and Sales of Water Disinfection Systems
- US Federal Performance Standards 40 CFR 141.720
- Long Term 2 Enhanced Surface Water Treatment Rule of January 5th, 2006
- All components have CE or UL 508 NRTL (MET) certification
- 3rd party validated by HDR/HydroQual, Inc.
- Resolucion exenta N° 2.327 del 31 de diciembre de 2010, en el marco de la Ley General de Pesca y Acuicultura
- GOST Standards Institute, Russian Federation
- National Institute of Public Health, Poland



3 Getting Ready for Installation

Using the guidelines and information in this section, you can prepare the piping and electrical setup for installing the Atlantium Hydro Optic Water Treatment System. If you have any questions, consult with an Atlantium application engineer.

3.1 Planning

Consider where to locate the unit from the perspective of your process. Prepare your installation plan according to the Mechanical and Electrical ICD drawings provided with the system and to planning points below, as well as the other guidelines in this manual.

3.1.1 Access

■ Installation Access

- Take into account that sufficient door space is required to transport the Atlantium Unit into the facility where it is to be placed.
- The Atlantium units are packed and transported in large wooden crates. Consider the need for a place near the installation area for unpacking of the units from the crates, prior to installation.

■ Service Access

- Ensure that sufficient space is required for maintenance personal to access the Atlantium Unit for periodic component replacement and maintenance.
- The installed units must also be accessible: for example if a unit is installed high above the ground, permanent stairs or walkways may be needed for convenient and safe service access
- Ensure sufficient space to access the back of the unit, to allow removal and replacement of lamps as well as to do maintenance on cleaning systems (where these are installed), connection boxes, etc.

■ Installation: vertical or horizontal

- Vertical installation of the Atlantium unit and piping, with an upwards direction of flow, is the preferred configuration for installation, where this is possible.
- If mounting the unit in a horizontal configuration, a slope of at least 5 % is recommended to minimize the risk of air pockets developing or remaining trapped in the Atlantium unit. The Customer is to supply the mechanical means for ensuring this slope.

3.1.2 Hydraulics

■ Piping

- Align the upstream and downstream piping with the designated location for the Atlantium unit.
- Prepare the flanges/ferrules and piping as specified according to **The Piping Infrastructure** on page 27. To avoid a step in the connection, make sure that the internal diameters where the Atlantium Unit and your piping match.
- **Do not** use sealing material that contains graphite when sealing the flanges/ferrules and gaskets as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- To promote laminar flow of water through the Atlantium unit:
 - ◆ The piping connected to either end of the unit should be of the same diameter as the unit itself. Connection of narrower piping may reduce flow and increase water temperature and turbulence.

- ◆ Allow a straight length of 10 pipe diameters (10D) before the Atlantium unit (upstream) and a straight length of 5 pipe diameters (5D) after the unit (downstream). In situations where less space is available, review with your Atlantium representative.
- ◆ In order to minimize turbulence and air pocket formation in a vertical installation, install a concentric reducer/expander adapter to streamline the transition between piping dimension and the internal dimension of the unit.
- ◆ For a horizontal installation, install an eccentric reducer/expander adapters, with downstream outlet higher than inlet, as shown in Figure 3-1 .

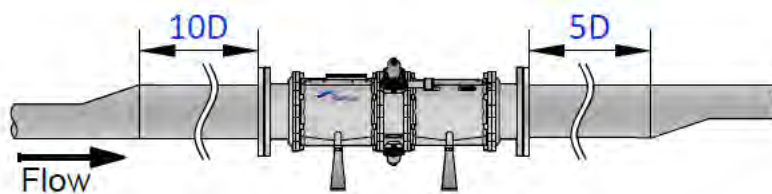


Figure 3-1: Configuration for eccentric expanders in horizontal installation

■ Mechanics.

- Prepare the supports needed for the Atlantium Unit, ensuring that the minimum height above the floor is 750mm / 29.5 inch (between the floor and the bottom of the provided triangle support brackets).
- Check that the piping on the inlet and outlet sides has adequate support structures so that there are no mechanical stresses that may endanger the Unit.
- Check the upstream and downstream piping for vibrations. Make sure that the piping to be attached to the Atlantium Unit is anchored and reinforced to protect the Unit against vibration. A high vibration level can cause damage to any hydraulic system, including the Atlantium unit. High vibration can be caused by an unbalanced pump or by sudden valve close.

■ Hydrodynamics

- Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. Make sure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check your facility system procedures, including:
 - ◆ When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
 - ◆ Sudden valve closures can cause water hammer. Make sure that valves close gradually, enough to avoid this problem.
 - ◆ Ensure that all pumps that impact the Atlantium system employ soft start-up procedures.
 - ◆ The Atlantium Unit must be totally filled with water. Assure that no air can be trapped in the unit and that air can be released. Verify through the Unit's viewport after installation that there are no bubbles.

3.1.3 Flow Data

- Evaluate your flow trends to determine whether you require a flow meter or flow switch. If the flow varies, a **flow meter** is required to detect and report on the flow coming into the unit. Make sure that there are to be **no** branches on or off the pipes between the flow meter and the Unit inlet. If the flow rate is static, a **flow switch** can be used instead.

3.1.4 Application

- Verify that the dose-related parameters, such as the UVT and the flow, are the same as those defined in the Atlantium application scope of supply.

- Determine what a low flow situation means to your process and whether you need to shut down the Unit (low flow indicates upstream problems) or recirculate (low flow normal) when the flow goes below the normal range.

3.1.5 Feed Water Quality

If the water entering the Atlantium Unit contains debris or particles in suspension, pre-treatment (e.g. filtration) may be required to maximize the effectiveness of UV disinfection and protect internal parts from blockage or damage. Consult with your Atlantium representative.

3.2 Typical Installation Set-Up

Use the diagram below as a basis for your installation drawings. Vertical installation with upward flow is preferred where it is possible and accessible, as it prevents the accumulation of air pockets. If horizontal installation is required, be sure to include a 5 degree slope (recommended/minimum) in the direction of flow to reduce the potential for air pockets. Be sure to include service access as you fit the Atlantium system into your facility (i.e., nearby pipes, electrical fixtures, support beams, etc.)

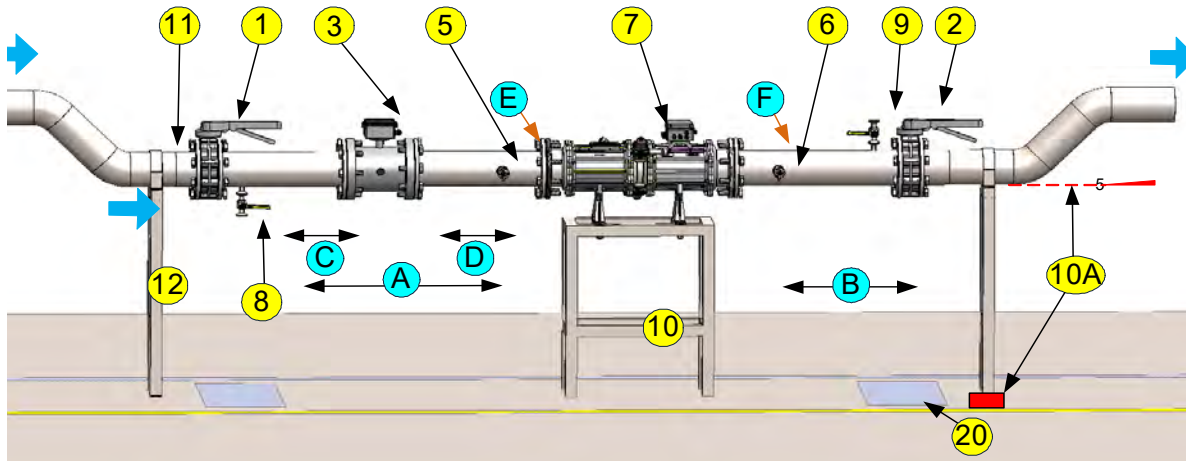


Figure 3-2: Generalized Atlantium system setup, horizontal (no bypass)

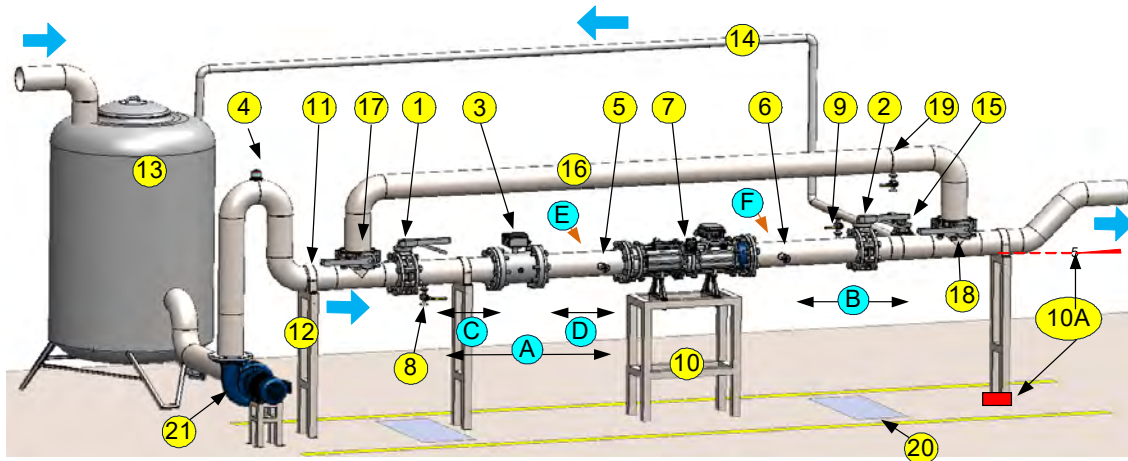


Figure 3-3: Generalized Atlantium system setup, horizontal (with bypass)

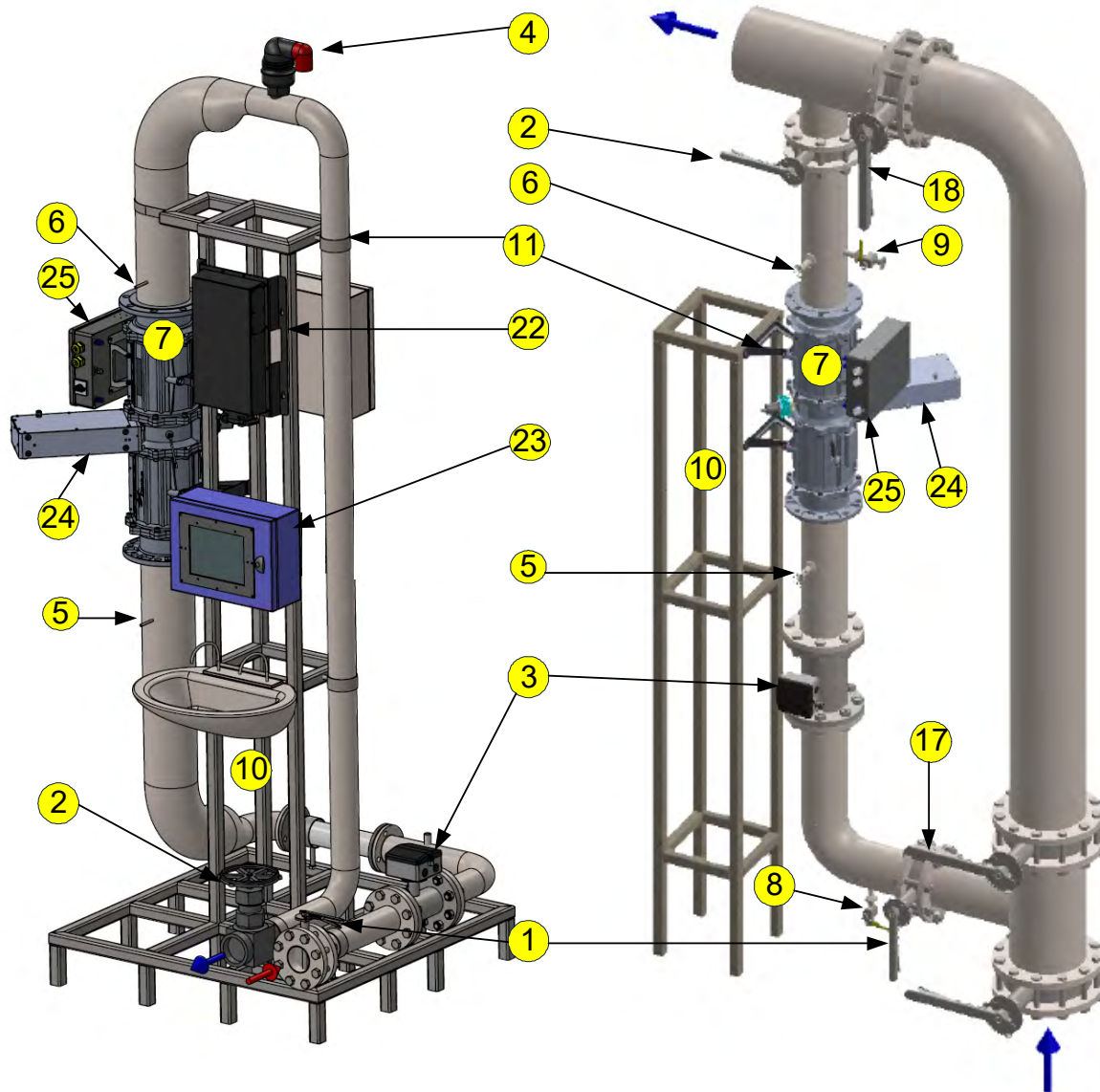


Figure 3-4: Generalized Atlantium system setups, vertical - Left: no bypass, Right: with bypass

Table 3-1: Typical Atlantium System Component Setup Key

#	Description		Atlantium P/N or recommended and available at Atlantium
	Direction of water flow in the pipes		-
1	Inlet isolation Valve ■ Isolating Valve on the Inlet side		-
2	Outlet isolation Valve ■ Isolating Valve on the Outlet side		-
3	■ Flow Meter flow measurement device with 4-20mA analogue output signal ■ To be installed on the same line as the Atlantium system, with no branching before or after the flow meter ■ Follow the manufacturer's Installation instructions		Available Flow Meters: KROHNE OPTIFLUX 6000 KROHNE OPTIFLUX 2300
4	Air Release valve - To prevent air bubbles where necessary ■ Automatic Air Release valve is used to release air bubbles trapped within the line ■ Install at a high position to allow air bubbles to accumulate and release from this point		-
5/6	Inlet/Outlet Sampling Points ■ Position aseptic sampling valves horizontally (90°) on the pipe center, within 50cm of the Unit ■ Aseptic sampling valve must be a dedicated valve for sampling only and resistant to alcohol and flame		■ Aseptic Sampling Valve Kit for Stainless Steel Pipe - KTB005700 ■ Aseptic Sampling Valve Kit for Plastic Pipe - KTB0011500
7	Generic Atlantium Unit		-
8	Draining/CIP Port - For performing CIP on the Atlantium Unit ■ The Draining port should be positioned close to the Inlet valve and between that and the Unit, on the bottom of the pipe ■ It is also used as one of the CIP ports and connects to a hose		■ CIP Kit - SAB012900 ■ CIP Valve Kit for Stainless Steel - KT0011600 ■ CIP Valve Kit for Plastic - KT0011700
9	CIP Port - For performing CIP on the Atlantium Unit ■ The CIP port should be positioned between the Unit and the Outlet valve, on top of the pipe, at the highest point before the isolation valve, in order to allow the pipes to be completely filled with solution		(the valve itself is not part of the kits) See CIP Ports and CIP Kits on page 31., for information on CIP equipment.

Table 3-1: Typical Atlantium System Component Setup Key

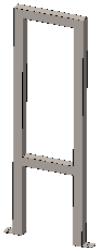
#	Description		Atlantium P/N or recommended and available at Atlantium
10	System Supports Supports are required to assure no mechanical stress on the Atlantium Unit		-
10A	Shim, block or higher downstream support, to achieve 5° slope in horizontal installation		supplied by customer
11	Support brackets - Brackets are used to hold the Atlantium Unit and pipes attached to the support mounts - Is to include internal flexible material to absorb vibrations and assure that the pipes and Atlantium Unit are mounted tightly and securely		-
12	Pipe Supports Pipe Supports are required to assure no vibrations in the pipeline		-
13	Water tank		-
14	Recirculation Line >5m³/h (>20gpm) (Optional) A Recirculation line is highly recommended to keep water flowing constantly, thereby assure a high disinfection level		-
15	Recirculation Line Isolation valve (Optional)		-
16	By-Pass (Optional) If the facility uses quartz corrosive CIP chemicals, such as caustic soda, use a By-Pass line to prevent damage to the Atlantium Unit		-

Table 3-1: Typical Atlantium System Component Setup Key

#	Description		Atlantium P/N or recommended and available at Atlantium
17	By-Pass Inlet isolation Valve (Optional)		-
18	By-Pass Outlet isolation Valve (Optional)		-
19	By-pass draining port (Optional) The Draining port should be positioned on the bypass line, at the lowest point of the pipe		-
20	Drain It is recommended to install a drain in the floor, close to the draining valve		-
21	Pump To prevent water hammer, the pump must be able to start up gradually.		-
22	Ballast Module (certain Atlantium systems)		
23	Control Module		
24	DPM cleaning system (certain Atlantium systems)		
25	Connection Box		

Table 3-2: Typical Atlantium System Distances Setup Key

#	Description
A	- The pipe between the Inlet isolation valve and the Atlantium Unit should be straight and horizontal. - For most applications, a minimum 10D (mm/ inch) of straight horizontal pipe is needed between the Inlet isolation valve and the Atlantium Unit. - For Power Plants, and other applications with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.
B	- The pipe between the Atlantium Unit and the Outlet isolation valve should be straight and horizontal. - For most application, a minimum 5D (mm/inch) of straight horizontal pipe is needed between the Atlantium Unit and the Outlet isolation valve. - For Power Plants, and other applications with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.
C	Some manufacturers require positioning the Flow Meter at a minimum of 5D (1500mm/59inch) of straight horizontal pipe immediately preceding it on the Inlet side. Refer to the manufacturer's instructions.
D	- Some manufacturers require positioning the Flow Meter at a minimum of 2D (mm/inch) of straight horizontal pipe immediately following it before the Atlantium Unit. - Refer to the manufacturer's instructions.
E	Position the Inlet Sampling Point at a maximum distance of 0.5 meters/1.5ft. from the Atlantium System Inlet.
F	Position the Outlet Sampling Point at a maximum distance of 0.5 meters/1.5ft. from the Atlantium System Outlet.

3.3 Electrical Requirements

The electrical infrastructure is to be installed and in place prior to the system's delivery.

For information on the footprint of the components of the Atlantium system, see the section, **About the RZB300 System Systems** on page 12. Use the Electrical ICD provided by Atlantium to help you plan the electrical infrastructure. See also **Power, Voltage and Grounding Requirements** on page 18 and **Connecting the Ballast Module Cables** on page 51.

Take into account:



For locations with 440Y/254 VAC 3~ +PE 50/60HZ
or 480Y/277 VAC 3~ +PE 50/60HZ

The Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).

- Evaluate the quality of your power and the likelihood of blackout/brownouts, electrical surges or interruptions. Determine if you need power surge protection and or uninterrupted power supply if any.
- To verify that you order the correct lengths for cables (i.e., Cable Harness, power cables, data cable, etc.), measure the distance between the proposed location of the relevant Atlantium components and between them and the Power source.
 - Planning the placement of the Ballast Modules, **Control Module** and the path of the various cables must take into consideration the length of the **Cable Harness** and the other cables. Cables must be placed in a path that does not interfere with passage ways.
 - The power mains connections must be in the vicinity of the Ballast Modules, Control Module.
 - The power cable and connecting plug are to be obtained locally.

- If the location experiences periodic power outages, consider including uninterrupted surge protection devices.
- Prepare cables of sufficient length to connect from the auxiliary equipment (i.e., the flow meter/flow switch, etc.) to the Atlantium system.

3.3.1 Preparing for the Electrical Wiring to the Mains

You must prepare the elements listed below for connecting the Atlantium system to the mains.



- For preparing all electrical connections, be sure to use the *Electrical ICD* drawing provided to you by Atlantium with this manual. If no electrical drawings were provided please contact your Atlantium representative.
- Atlantium systems require connection to mains power of 400 to 480 VAC. See *Mains power supply up to 480VAC* on page 18.
- Mains system area circuit breaker, according to the total systems' power consumption to be connected to this line. See *Electrical Usage* on page 17.
- Install the Required Circuit Breakers, as defined in the Electrical ICD.
- If you intend to lengthen or replace the mains power cable connected to the Atlantium Unit with a longer cable, prepare the cables according to your location electrical distribution system requirements.
 - For locations with 400 VAC: 5-wire system that contains 3 phases, Neutral and Ground (The Neutral and Ground are mandatory)
 - For locations with 440Y/254 VAC 3~ +PE 50/60HZ
or 480Y/277 VAC 3~ +PE 50/60HZThe Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).
- For sites with 400VAC 3 phase + Neutral, the Control Module's Mains feeding voltage is 230 VAC (L1 to N or L2 to N or L3 to N).
- For sites with 440/480VAC 3 phase, the Control Module **cannot** be fed from one phase of the 3 phase supply. The Control Module **must** be connected to a separate mains feeding, which is in the voltage range of 100-240 VAC.

Voltage option are:

- 1 phase x 120 - 240VAC +N +PE
- 2 phase x 120 - 240VAC +PE

3.4 The Piping Infrastructure

To review how to prepare the infrastructure, read through the typical installation section before you begin and refer to the section, *About the RZB300 System Systems* on page 12.

Install the piping infrastructure before the Atlantium system arrives, but to avoid built-in mechanical stresses perform the final pipe adaptations and welds when the unit is actually situated in its permanent position.

ATTENTION!

- When tightening bolts to connect piping, use a torque wrench to apply correct and uniform torque, according to accepted standards and practices. Guidance for bolt tightening techniques may be found at:
http://www.wermac.org/flanges/flanges_torque-tightening_torque-wrenches.html

- When sealing the flanges/ferrules and gaskets, **do not** use sealing material that contains graphite as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- Be sure to align the piping on the inlet and outlet sides of the designated location for the Atlantium Unit with the unit to avoid mechanical stress on the Unit.

Be sure to secure the piping around the designated location for the Atlantium Unit so that there is no vibration.

- Be sure to position adequate supports on the piping line needed to hold up and stabilize the process piping on the inlet and outlet side of the proposed location of the Atlantium unit.
- Determination of support requirements and design should be made by an on-site mechanical engineer.

3.5 Flow Measurement/Detection

The Atlantium unit requires a flow signal to calculate the dose and operate properly. The flow signal is a measurement of the actual flow or simply a flow/no-flow indicator. The flow signal is collected from either a flow meter or a flow switch.

- A flow meter is required in facilities where the flow rate is variable for any reason.
- A Flow switch can be used in facilities where the flow is at a fixed, steady rate. In this situation the value of the known steady rate is entered manually into the system controller (see **Configuring the Flow Settings** on page 87).

The Flow meter or flow switch does not necessarily have to be installed in close proximity to the Atlantium Unit. It may be more remotely positioned, provided that it is located between the water inlet and the Atlantium Unit and there is **no** branching or splitting of the water flow.

ATTENTION!

The Atlantium system cannot be operated when there is no water or no water flow. Any damage caused due to operating the system without water or without water flow invalidates the Atlantium warranty.

3.5.1 Flow Meter

In most cases, a flow meter with a 4-20 mA output is required to measure the flow and automatically adjust the output of the UV system. The flow meter must be able to measure the expected maximum water flow of the specified Atlantium unit. The power source for the flow meter must come from an external source according to the manufacturer's instructions.

ATTENTION!

Be sure to provide the power source for the flow meter as part of the preparations for installing the Atlantium Unit.

Install the flow meter in strict accordance with the instructions supplied by its manufacturer.

Examples of Flow Meters

Flow meters are available from Atlantium, such as:

- **KROHNE OPTIFLUX 6000** flow meter for high-purity water applications
- **KROHNE OPTIFLUX 2300** flow meter – OPTIFLUX 2000 flow meter, includes **KROHNE IFC 300** signal converter

Consult your Atlantium representative.

OPTIFLUX 6000



OPTIFLUX 2300



Figure 3-5: KROHNE OPTIFLUX Flow Meters

3.5.2 Flow Switch

The flow switch tells the Atlantium Unit when water is or is not flowing. The recommended flow switch type features a paddle connected to a Dry Contact micro-switch, with minimum flow sensing of 0.2m³/hr (1gpm), see Figure 3-6.

Install it in a vertical (12:00 o'clock) orientation on top of the pipe. Follow the manufacturer's instructions.



Figure 3-6: Example of Flow Switch

3.6 Sampling Valve Components and Kits

Sampling valve kits are available from Atlantium.

One Accessory kit is required for each Sampling point.

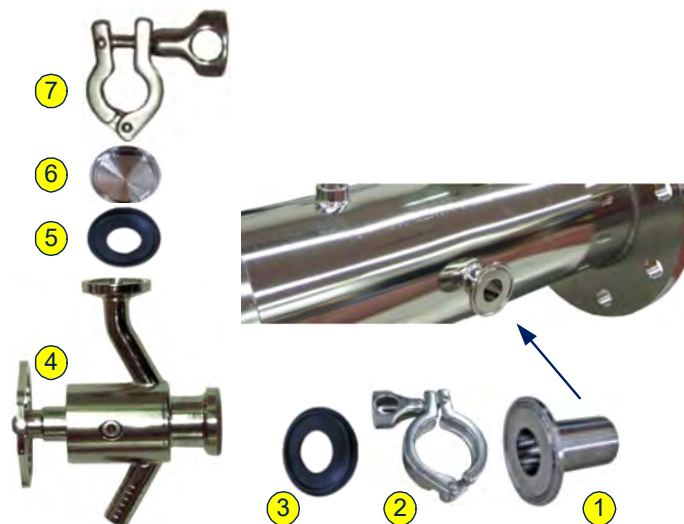


Figure 3-7: Sampling Valve Assembly

The table below details them. For information on installing the Sampling valves, see *Installing the Inlet/Outlet Sampling Points* on page 42.

Table 3-3: Sampling Valve Components and Kits

	Catalog number	Description		Image	Kit for Stainless Steel Pipe KTB005700	Kit for Plastic Pipe KTB005750
1	FAC003700	Long welding ferrule 1" Adapter	For welding the tri-clamp port to the pipe		2 Pcs	-
1	FAB008100	Threaded Male Adapter 1/2" BSP	For connecting the tri clamp port to the plastic pipe		-	2 Pcs
2	FA0000045	Clamp 1.5", ST.ST 304	For connecting the valve to the welded port		2 Pcs	2 Pcs
3	FA0000046	Gasket 1", FJ31.05-E	For sealing the valve connected to the pipe		2 Pcs	2 Pcs
4	COB000500	Aseptic sampling valve, ST.ST 316L	For taking water samples		2 Pcs	2 Pcs
5	FA0000048	Gasket 1/2", FJ31/25-E	For sealing the valve's flushing port		2 Pcs	2 Pcs
6	FA0000049	Solid End Cap, 1/2", ST.ST 304	For covering the valve's flushing port		2 Pcs	2 Pcs
7	FA00000047	Clamp 1/2", ST.ST 304	For sealing the flushing port of the valve		2 Pcs	2 Pcs
	FA0008300	Round Brush - 15/60/150	For cleaning the valve during the process		2 Pcs	2 Pcs

3.7 Draining Valves

It is important to plan for draining the system. Install one draining valve at the lowest point of the pipeline, before or after the Atlantium unit. If you plan to install CIP ports, one of them can double as a draining valve. If there is a Bypass line used occasionally to bypass the Unit in any circumstance, add an additional draining valve to assure that the bypass line does become a potential source of standing water. A bypass line must be empty when not in use.

3.8 CIP Ports and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve. The CIP Cart includes a reservoir, a pump, a supply hose, a return hose, and a pocket to hold these hoses. See the diagram below. Refer to **CIP Cart ICD drawing ICB012900**.

3.8.1 CIP Cart

For systems with 400VAC: catalog number: SAB012900

For systems with 440/480VAC: catalog number: SAB012950

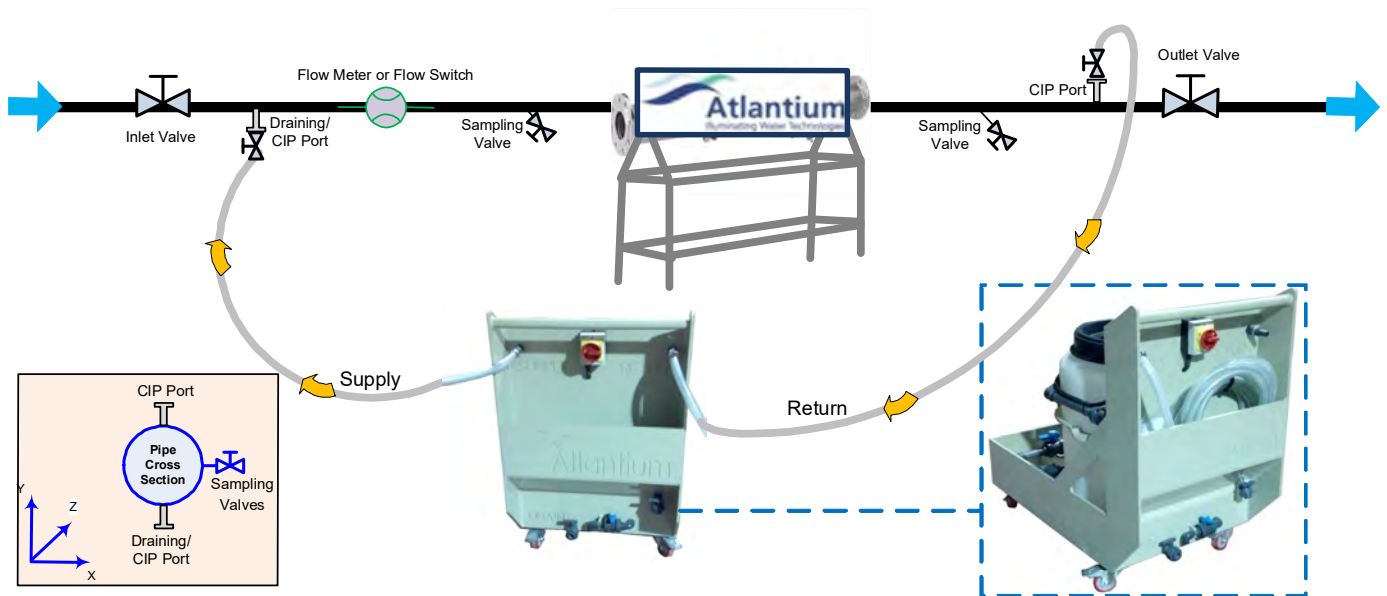


Figure 3-8: CIP Recirculation System Connections

3.9 CIP Accessories and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve, see Figure 3-8.

Two Accessory kits are required, one for each CIP port.

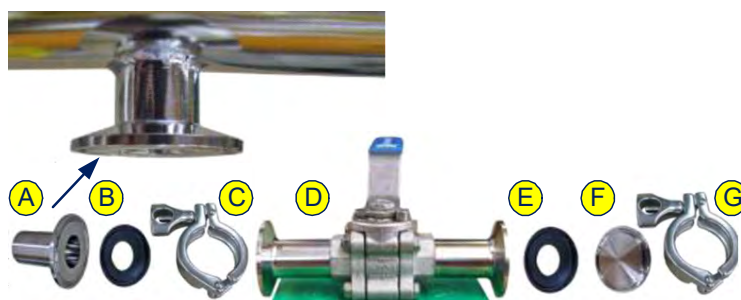


Figure 3-9: CIP Valve Kit

Table 3-4: CIP Accessory Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011500
A	FAC003700	Long welding ferrule 1" Adapter	For welding the tri-clamp port to the pipe		2 Pcs	-
A	FAB008100	Threaded Male Adapter 1/2" BSP	For connecting the tri clamp port to the plastic pipe		-	2 Pcs
B,E	FA0000046	Ferrule Gasket 1", FJ31.05-E	For sealing the valve connected to the pipe		2 Pcs	2 Pcs
C,G	FA0000045	Clamp 1.5", ST.ST 304	For connecting the valve to the welded port		2 Pcs	2 Pcs
F	FAC004100	Solid End Cap, 1 1/2", ST.ST 304, EG16A3A 1.54L	For covering the CIP port while not in use		2 Pcs	2 Pcs
D		Valves	Two high-purity chemical resistant brass ball valve (aseptic ball valves), 1 inch diameter, with tri-clamp fitting connection mechanism (A3 Standard) on both sides.		2 Pcs Not Included Purchased by Customer	2 Pcs Not Included Purchased by Customer

Table 3-4: CIP Accessory Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011500
	FAB005700	Hose Adaptor 1" to 3/4"	To attach the CIP hose		2 Pcs	2 Pcs

3.9.1 CIP Safety Equipment

Required but not included in Atlantium's CIP Kits: protective equipment such as goggles, clothing and (chemical-safe) gloves, according to the safety standards specified at your facility.

3.10 Connection to Facility Control

Options are available to set up control of the Atlantium system via your facility control system. Control setup can be made via any of the following:

- **Signal control** - utilizing System Ready, External On/Off, and General Alarm functions, etc. (See **Connection Box** on page 52.)

With any of the above control options, you can still utilize the function of the **Control Module** of the Atlantium system. Whichever control facility is utilized, the system communicates to the other control systems such that the last command issued determines the current status.



4 Safety Overview

The Atlantium System has been designed according to the highest safety standards, assuring the safety of operating personnel, the environment and the treated water.

DANGER!

Improper use of controls or adjustments or performance of procedures other than those specified herein could result in significant hazards.

Therefore, personnel operating or servicing this system must be thoroughly familiar with all safety requirements and operating procedures and are to adhere to them during use of this system.



To guard against injury, observe basic safety precautions, including the following:

All persons operating the Atlantium system must be aware of proper use and the potential hazards of violating safeguards. Be certain that all personnel carefully review the safety information and the procedures specified in this manual. Only authorized individuals with appropriate training and knowledge, including local regulations, should operate, assist in the operation of or provide maintenance to the system.

All service and repair of the Atlantium system must be performed by Atlantium field service technicians or other factory-authorized personnel **ONLY.**

4.1 UV Lamp Safety



UV Exposure: The system generates ultraviolet (UV) light within the Unit, which can cause serious eye damage or blindness if you stare at it directly when it is operating. Use the Viewport only. Do not look directly into the lamp enclosure during system operation, examining, or servicing the system's internal components during operation, or when lamp's breather caps are open, or if energized testing is required.

The Viewport is specifically designed with a UV filter to prevent UV light from escaping so it is safe to look inside. Nevertheless, personnel should not stare at it or any other brilliant light source for any but short intervals to check on Unit operations. The Viewport cover should also be preserved intact.

If the viewport is broken, DO NOT look directly into it. Be sure to request a replacement part immediately. Do not open the viewport's metal cover until the replacement is installed.

Persons potentially subject to UV light exposure – technicians who are servicing the Unit – must wear appropriate eye protection whenever the system is operating.

During maintenance procedures that are conducted with the UV lamps turned on:

- Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection, such as containing polycarbonate lens that meet EN1ultrasonicultrasonic, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**



Protect Hands: Do not touch the UV lamp with bare hands.

- Wait at least 10 minutes until the lamp is cooled down before touching the lamp or starting the replacement procedure.
- Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.
- Lamps can reach a temperature of 1,000C° under operating conditions.
- Wait ten minutes to allow the lamp to cool down fully before replacing it.
- Keep materials that are sensitive to heat, or which contain solvents, a safe distance away from the lamp, Unit, and electrical connections.



Electric Shock: Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, make sure the switches on the Connection Boxes of the relevant lamps are turned OFF and place a sign on the Atlantium Unit and Control Module alerting others NOT to touch the switches or the screen during maintenance so that no one can inadvertently turn it on while maintenance is in progress.

ATTENTION!

Prevent Damage:

Handle the UV lamp assembly by holding only the contact housings (white ceramic casing).

When handling the UV lamp, always place it on a flat surface or table, so that it is not accidentally damaged.

If the body of the UV lamp is touched accidentally, clean the fingerprints off with alcohol (at least 70%) and wipe dry with a soft, clean, lint-free cloth (usually provided with the lamp). Do not use cleaning rags or materials that can leave a residue.

Do not use a UV lamp that shows any scratches, cracks, or other damage.



Mercury Poisoning:

Lamps contain mercury, which is a hazardous substance. Always wear gloves and safety glasses when handling or replacing a UV lamp.

Inhalation of vaporized mercury compounds can be harmful to the lungs, kidneys, and nervous system. Mercury that penetrates the skin or is ingested can also be harmful.

If mercury is inhaled, penetrates the skin, or is ingested accidentally, seek emergency medical treatment immediately.

If a UV lamp breaks during handling and releases mercury, the following precautions should be observed to minimize the risk of exposure to mercury:

- Always wear gloves and safety glasses when handling or replacing a UV lamp.
- Leave the area immediately to avoid inhalation of the mercury vapor.
- Thoroughly ventilate the area for at least 30 minutes or until the mercury vapor concentration is in compliance with applicable federal and local health and safety regulations.
- After handling a broken UV lamp, carefully remove gloves, and then wash hands thoroughly with soap and water. Follow all applicable federal and local health, safety, and environmental regulations.
- Do not turn on or operate the system until all mercury contamination has been cleaned up and removed.
- Establish a Hazard Response Plan to deal with all related hazards and provide it to all personnel associated with the Atlantium system.

Refer the Chemical Safety information at: <http://www.inchem.org/documents/icsc/icsc/eics005ultrasonic.htm>

ATTENTION!

Lamp Compatibility:

Use original Atlantium UV lamps ONLY. Other lamps are incompatible. They void the warranty and may damage the system.

Proper Disposal of Lamps

Expired lamps are classified as hazardous waste and must be disposed of appropriately.

- A non-ruptured UV lamp that is no longer used must be disposed of according to specific local or federal environmental and hazardous waste regulations.
- Broken lamps and parts contaminated by the mercury have to be vacuum-packed and disposed of. Broken lamps, as well as contaminated packing material and other parts have to be considered as special waste, which may only be removed by authorized waste disposal companies.

4.2 Electrical Hazards and Safety Considerations

DANGER!

UNDER NO CIRCUMSTANCES

SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE **OFF** POSITION AND THAT **NO** ELECTRICAL FEED IS LIVE.



DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL **ALL** ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.



- **DO NOT DRILL INTO THE CONNECTION BOXES OR ALTER THEM IN ANY WAY!!!!**
- The Connection Box already contains entry glands for **ALL** of the cabling needs.
- Making holes in a Connection Box will destroy its waterproof seal, endangering the components inside.
- Do Not draw electricity from the Atlantium Unit's Connection Box as it will damage the Unit's electrical configuration.



The Atlantium Hydro-Optic Water Disinfection system incorporates high-voltage internal components, which can cause serious injury or fatal electrical shock if not used or serviced properly. High voltage components can retain a charge for some period of time even after the system has been turned off.



- Never attempt to remove any system covers or to dismantle any parts, except when performing the maintenance procedures detailed in **System Maintenance** on page 97.
- No portion of the Atlantium system is to be opened or removed by anyone other than a trained and authorized technician.
- Do not spray or pour any type of fluid directly into the Unit or electrical components. Moisture causes damage to the equipment and electrical shock may result.
- Do not operate the system if the power cables or harnesses are frayed or otherwise damaged.

CAUTION!



DO NOT enter **Technician Mode** in the **Control Module** unless instructed to do so by Atlantium certified personnel or by an Atlantium System User Manual, for a specific task.

If you are in Technician Mode, **DO NOT WALK AWAY!**



NEVER LEAVE THE ATLANTIUM SYSTEM WHEN IT IS OPERATING IN TECHNICIAN MODE!!

A SYSTEM LEFT IN TECHNICIAN MODE WITHOUT SUPERVISION CAN RESULT IN PERMANENT DAMAGE!!



Utilize best practices regarding electrostatic discharge (ESD).

Avoid direct contact with the electronic circuit boards of the **Control Module**, which is sensitive to electrostatic discharge (ESD).

Wear an electrostatic discharge (ESD) Wrist Strap when performing procedures where contact with a circuit board is possible.

4.3 Keep the Unit Full of Water

ATTENTION!

The Atlantium system cannot be operated when there is no water. Any damage caused due to operating the system without water invalidates the Atlantium warranty.

4.4 Chemical Use for Cleaning in Place (CIP)



ATTENTION!

- The chemicals selected and the concentrations used must be approved by Atlantium's Application staff, and the ship's quality assurance and safety officials. The CIP process must comply with the written protocols and procedures set in place by those departments. Refer to the section, **Selecting the Correct Chemicals for CIP** on page 98
- Use and handle all chemicals in strict accordance with their manufacturer's instructions, product information sheets, and Safety Data Sheets (SDS).
- NaOH (Caustic Soda) as it causes irreversible damage to quartz. It is strictly forbidden to use it without a specific analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

Refer the Chemical Safety information at: <http://www.inchem.org/documents/icsc/icsc/eics005ultrasonic.htm>

4.5 System Safety Features

The Atlantium Hydro-Optic Water Disinfection system includes the following safety features:

4.5.1 UV Protective Seal



Each lamp is enclosed within a sealed lamp chamber that prevents the UV light from escaping. This means that with regular operation there is no ambient UV escaping from the Unit.

While servicing the UV Unit or checking the Intensity sensor, UV blocking safety glasses should be worn by service staff or others working in the area to prevent exposing the eyes to the UV light.

4.5.2 Lamp On/Off Switch



CAUTION!

An On/Off switch is located on the Connection Box on top of the Atlantium Unit.

In the event of an emergency, as well as for maintenance purposes, each lamp can be shut down immediately by turning this switch to the Off position.

To restore operation, rotate the button clockwise and then the system restarts.

4.5.3 Software Safety Controls on Control Module



CAUTION!

When the system is started, the Atlantium software performs a series of internal self-testing routines prior to starting system operation.

If the system **does not** pass these tests satisfactorily, UV lamp operation is disabled until the problem is resolved.

If problems arise during normal operation, appropriate error messages with varying levels of severity are generated on the touch-screen of the Control Module.

Control Module screens that provide setup options and compliance reports are password protected.

ATTENTION!

DO NOT leave the Control Module unattended, even for a short time, while setting change screens are accessible. An untrained person could damage the operation of the Atlantium system by making unauthorized setting changes.

4.6 Atlantium Training

As part of the Commissioning and Hand-Off process, Atlantium provides hands-on training on all aspects of the Atlantium system for your designated personnel. Should any component be replaced with upgraded features, Atlantium offers training as needed and is ready to provide additional training for new personnel or any other needs related to its systems.



5 Installing the Atlantium System

This section provides details about Installation the Atlantium system.



For preparing all electrical connections and mechanical installation, be sure to review the electrical and mechanical Interface Control Drawings (ICD) provided to you by Atlantium with this manual. If no electrical drawings were provided, please contact your Atlantium representative.

5.1 Installation Overview

Hardware setup to be implemented by your facility personnel:

- Phase 1** Unpacking and checking the contents of the Atlantium system packaging (See below)
- Phase 2** Installing Sampling valves and CIP ports (See page 41)
- Phase 3** Positioning the Atlantium Unit and connecting to the Piping Infrastructure (See page 45)
- Phase 4** Mounting the **Control Module(s)**, Ballast Module(s), etc. (See page 48)
- Phase 5** Cabling the **Control Module(s)**, Ballast Module(s), etc. (See page 51)

Commissioning to be implemented by Atlantium personnel:

- Phase 6** Starting up the system (See page 61)
- Phase 7** Configuring the **Control Module** (See page 69)
- Phase 8** Tuning the System (See page 120)
- Phase 9** Completing the System Setup and Testing (See page 82)

The Installation Process Flow diagram below maps out the installation process

5.2 Installation Process Flow Diagram

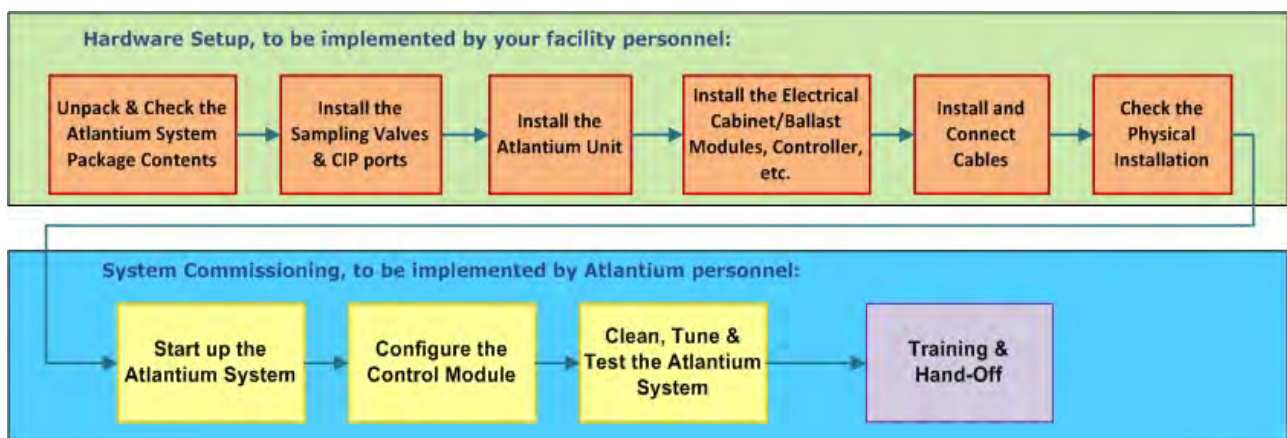


Figure 5-1: Installation Flow Diagram

5.3 Setting the Main Circuit Breakers to OFF Position

Before beginning work to connect the wires/cables of the Atlantium system components, turn the main circuit breakers of the electrical lines feeding

the Atlantium system components to the **OFF** position. **Verify that no electrical feed to the Atlantium system components is live.**



DANGER!

UNDER NO CIRCUMSTANCES

SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE OFF POSITION AND THAT NO ELECTRICAL FEED IS LIVE.

DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL ALL ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

5.4 Unpacking and Checking Package Contents

The Atlantium system is packaged and shipped in specially packed crates that contain:

- The AtlantiumUV Unit
- Ballast Module(s)
- Control Module
- Documentation
- Peripheral accessories as per your order and listed on your delivery documents are crated according to their size and weight. Small accessories may be crated together.

Each of the crates is connected to a forklift-friendly wooden pallet. The Atlantium crates must be lifted off the transport vehicle using a forklift or crane controlled by a qualified operator.

➡ **To unload and check the system:**

- Step 1.** Using the forklift or crane, lift each of the crates off the transport vehicle and lower them to a level ground surface.
- Step 2.** Walk around and inspect each crate for damage. Note if there is any separation in the joints or any other indication of any damage in transit. If there is any indication of damage, take a photograph of it and use extra caution when proceeding to the next step.
- Step 3.** Unclamp and remove the panels of the crates.
- Step 4.** As you remove the wrapping and shipping protections, verify that the contents of the crates are correct according to the delivery documents. The Atlantium Unit contains a plate with the model number and other product information. Verify that the correct product version is supplied.



- Immediately report to Atlantium and to the insurance carrier any damage to the crates or to the Atlantium system found before opening the crates or during every step of unpacking, installing or testing of the system.
- If you ordered more than one Atlantium system, check the serial numbers against the delivery documents to verify which components comprise each individual system.

5.5 Installing Sampling Points and CIP/Draining Ports

Both the Sampling Points and CIP/Draining Ports must be installed on the inlet and outlet sides of the Atlantium Unit.

5.5.1 Installing the Inlet/Outlet Sampling Points

Aseptic sampling valves are to be located indoors in clean environment without away from direct contact with fan or A/C air flow. They each must be a dedicated valve for sampling only and resistant to alcohol and flame. For information on the Sampling valve's components, see *Installing Sampling Points and CIP/Draining Ports* on page 41. See *Taking Microbial Samples* on page 94. for information on using the Sampling valves.

On the inlet and outlet sides of the water line, connect the aseptic sampling valves.

➔ To install an inlet/outlet sampling point:

Step 1. On the Inlet side, install the **Long welding ferrule** or **threaded male adapter (1)**.

- Position aseptic sampling valves (1 & 2 in Figure 5-6, on page 44) horizontally (90°) on the pipe center, within 50cm of the Unit.

■ For Stainless Steel. Pipes:

- Weld the 1 inch **ferrule (1)** clamp fitting at a horizontal orientation (3 or 9 o'clock).

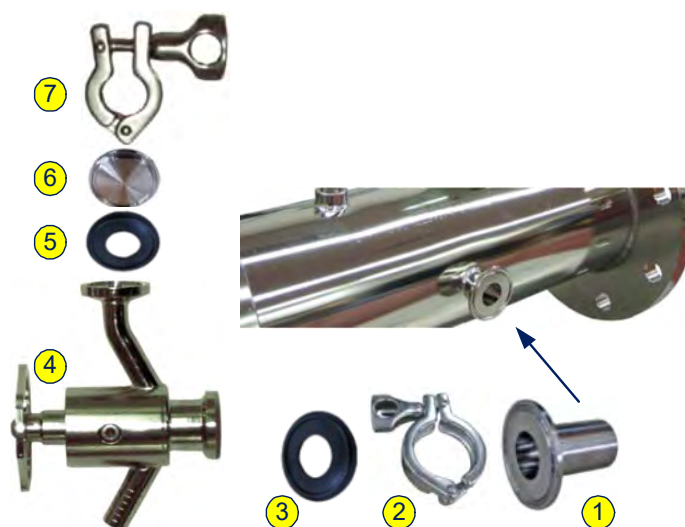


Figure 5-2: Sampling Valve Assembly

- Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
 - Passivate the pipes to avoid corrosion.
 - **For Plastic Pipes:**
 - Thread the **male adapter (1)** clamp fitting to the port.
 - Repeat for the Outlet side sampling point.
- Step 2.** Fit the gasket **(3)** into the **welding ferrule or threaded male adapter (1)**.
- Step 3.** Fit the aseptic **sampling valve (4)** on over the gasket **(3)**.
- Step 4.** Fit the clamp **(2)** around the aseptic **sampling valve (4)**/gasket **(2)**/welding ferrule or threaded male adapter **(1)** and close and lock it.



Figure 5-3: The clamp is placed around the aseptic **sampling valve** /gasket /welding ferrule or threaded male adapter

Closing Off the Aseptic Sampling Valve When Not in Use

- Step 5.** Into the spout of the sampling valve, fit the 1/2" gasket **(6)** and fit the cap **(7)** over it.
- Step 6.** Fit the 1/2" clamp **(5)** around the sampling valve spout/gasket **(6)** /cap **(7)** and close and lock it.



Figure 5-4: The clamp is closed



Figure 5-5: The clamp is closed

5.5.2 Installing the CIP/Draining Ports

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve. For more information on performing CIP, see **Cleaning In Place (CIP)** on page 95.

A CIP KIT includes the CIP Cart with water reservoir and a pump, capable of pumping up to 4m³/hr (17.6gpm). It includes a supply hose and a return hose, both with ID 19mm (0.75") and length 5m (15 feet) stored in a convenient pocket. See the diagram below.

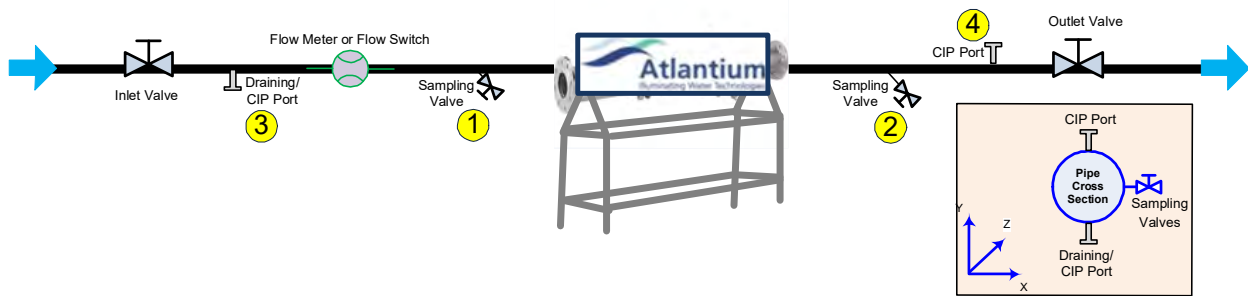


Figure 5-6: CIP Ports and Sampling Valves Orientation

Position the CIP/draining valves (**3 & 4**) on both the inlet and outlet sides of the Atlantium unit. To enable connection of the CIP kit, use high-quality, sanitary equipment, including a 1 inch tri-clamp/ferrule on both sides. A CIP/Draining valve is to be located 1D after the Inlet valve and 1D before the Outlet valve. When not using the CIP ports, close the valve outlet using the tri-clamp cover.

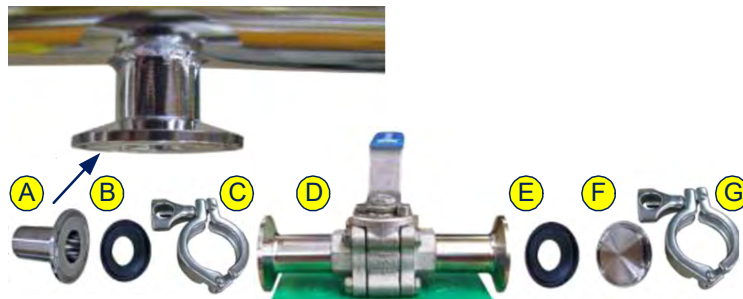


Figure 5-7: CIP Valve Kit



For preparing all electrical connections and mechanical installation, be sure to review the electrical and mechanical ICD drawings provided to you by Atlantium with this manual. If no electrical drawings were provided, please contact your Atlantium representative.

➡ To install the CIP ports:

Step 1. Weld or thread the 1" ferrule at a vertical orientation:

For municipal applications, prepare 1" screw ports for the valves and insert the valves:

- **Draining Valve:** at a 6:00 o'clock orientation on one side of the Atlantium unit.
- **CIP Valve:** at a 12:00 o'clock orientation on other side of the Unit.

For all application except municipal:

■ For Stainless Steel. Pipes:

Weld the 1" clamp fittings:

- **Draining port:** at a 6:00 o'clock orientation on one side of the Atlantium unit.
- **CIP port:** at a 12:00 o'clock orientation on other side of the Unit.
- Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
- Passivate the pipes to avoid corrosion.

■ For Plastic Pipes:

- Thread the clamp the valve fittings to the port..

Attaching the Valve to the ferrule

Step 2. Fit the gasket (**B**) into the **welding ferrule** or **threaded male adapter (A)**.

Installing the Atlantium System

Step 3. Fit the aseptic **valve (D)** on over the gasket **(B)**.

Step 4. Fit the clamp **(C)** around the aseptic **valve (D)**/gasket **(B)**/welding ferrule or **threaded male adapter (A)** and close and lock it.

Closing Off the CIP/draining Ports When Not in Use

Step 5. Into the ferrule, fit the gasket **(E)** and fit the cap **(F)** over it.

Step 6. Fit the clamp **(G)** around the CIP port/gasket **(E)** /cap **(F)** and close and lock it.

Preparing the CIP Hoses

Step 7. On one end of a hose **(A)**, using an appropriate screwdriver, loosen the band **(C)**.

Step 8. Connect the hose to one of the adapters on the cart **(B)**. Push the hose up as far as you can.

Step 9. Using the screwdriver, tighten the band **(C)**.

Step 10. On the other end of a hose **(D)**, using an appropriate screwdriver, loosen the band **(F)**.

Step 11. Connect the hose to one of the hose adapters **(E)**. Push the hose up as far as you can.

Step 12. Using the screwdriver, tighten the band **(F)**.

Step 13. Repeat with the second hose to the adapter on the other side of the cart.

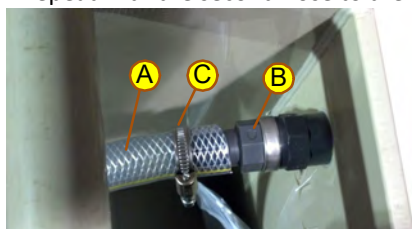


Figure 5-8: Hose Connection to One Side of Cart

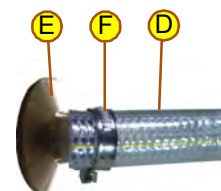


Figure 5-9: Hose Connection to Hose Adapter

5.6 Positioning the Atlantium Unit

Read through this typical installation section **before** you begin.

Make sure to use your installation drawing during this procedure. The Unit's triangle support brackets must rest on a strong, stable, level surface and anchor the Unit. Trace the path to be taken to position the Unit to its planned location and clear it of all obstacles.

In general, Atlantium systems can be installed horizontally or vertically in your piping system. Vertical installation of the Atlantium unit and piping, with an upwards direction of flow, is the preferred configuration for installation, where this is possible.

If mounting the unit in a horizontal configuration, a slope of at least 5 % is recommended to minimize the risk of air pockets developing or remaining trapped in the Atlantium unit. The Customer is to supply the mechanical means for ensuring this slope (see Figure 3-2 on page 21). For a vertical installation, the Atlantium Unit can be positioned against a wall or on skids.

5.6.1 Required Equipment

Supply the following tools and equipment:

- Lift or crane
- Lifting strap
- Bolts, washers & lock washers
- Supports for inlet/outlet piping and for the Unit as shown on your installation plan



The Atlantium unit is heavy - refer to Table 2-3, **Ballast Module Physical Details**, on page 17. Two people are required to get the Atlantium Unit to the designated installation location - a qualified lift or crane operator, at least one other person to guide the Unit as it is transported, and others as needed.

➤ To position the Atlantium unit:

- Step 1.** Note the direction of the water flow that is to occur in the piping process into which you intend to place the Atlantium Unit and the Flow indicator marked on the unit. Use this information to decide at which angle to lift the Unit for transport. This is to help you to prepare for placing the Unit at the location.
- Step 2.** To lift the Unit from its packing crate, loop a lifting strap through the lifting support rings **(1)** at the end fittings on each side of the Unit.
- Step 3.** Join the straps together with a third central strap that is being connected to the lift or crane.--

Figure 5-10: Lifting Strap Locations

- Step 4.** Ensure that all straps are connected securely to the Unit, to each other and to the lift/crane.
- Step 5.** Using the lift/crane, lift the Unit by the straps, out of the crate and position it for safe transport (near-ground level is preferred).
- Step 6.** Using the lift/crane and the designated person to guide, carry the Unit to its intended location.
- Step 7.** Install the Atlantium Unit:

Horizontal Installation

- At the designated location, check the installation plan, including the distances between the triangle support brackets.
- Position the customer-provided supports /surface so that the Unit's triangle support brackets are at least 750mm / 29.5 inch above the floor. (Refer to the **RZB300 System Interface Control Drawings** provided by Atlantium.)
- Carefully lower the Unit to its position.
- Using the appropriate bolts, washers and lock washers, bolt the triangle support brackets to the customer-provided corresponding supports.

For Vertical Installation

Position the Atlantium Unit vertically so that the Flow label is pointing upwards



Figure 5-11: Flow direction arrow label

- At the designated location, check the installation plan.
- Position the vertical supports / skids so that the Unit is at least 750mm / 29.5 inch away from the wall, to enable maintenance from all sides (Refer to the **Interface Control Drawings** provided by Atlantium.)
- Carefully lower the Unit to its position.

- d Using the appropriate bolts, washers and lock washers, bolt the triangle support brackets to the customer-provided corresponding supports.

5.6.2 Connecting to facility piping

Step 1. To connect the fitting on one side of the Unit to the fittings of the facility's pipes, insert a gasket between the Unit's and the facility's pipes. In some cases, a reducer or expander may be necessary.

q`

- Step 2.** Repeat the step above for the other side of the Unit.
- Step 3.** Tighten the fastening hardware so that the torque pressure around the fittings is even, and the Unit is firmly connected to the facility's water line.
- Step 4.** For installations with multiple units, repeat this procedure for the second unit.
- Step 5.** To check for any leakage, open the Inlet and Outlet valves to allow water to flow through the newly installed Unit and its surrounding process piping.
- Step 6.** If you find any leakage, tighten the connections.
- Step 7.** Close the Inlet and Outlet valves and use the Draining valve, drain the Atlantium system piping from Inlet valve to Outlet valve.



Figure 5-12: Generic Flange fitting to the Water Line

5.7 Installing Surge Protection

Verify that you have a protective device that prevents electrical surges and fluctuations from causing damage. You must install at least one of these two types of surge protection:

- UPS (Uninterrupted Power Supply) has an internal battery that allows the equipment plugged into it to continue to run in the event of a power outage until power is restored. It also has the benefit of performing the same functionality of surge protectors, protecting against spikes and surges. If needed, the UPS allows the system to be shut down cleanly if power is out for a lengthy period.
- Surge protection device that protects your equipment from variations in electrical current, such as surges and spikes.

The diagram to the right shows the generic wiring for a surge protection device. See the manufacturer's installation instructions with the surge protection device to be installed. For more information about surge protection devices available via Atlantium, see **Surge Protector Devices** on page 21.

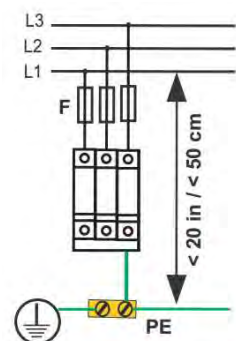


Figure 5-13: Surge protection device wiring - 440/480VAC

5.8 The Control Module

#	Item
1	Main circuit breaker
2	Power supply (100-120/200-240 VAC)
3	Controller + I/O Cards
4	Ethernet Module
5	Connection to Screen Display
6	Various Component Connectors (See Digital Outputs on page 55)
7	Holes for Peripheral Component Connections
8	RJ-45 Connector to Connection Box (Figure 5-22 on page 54)

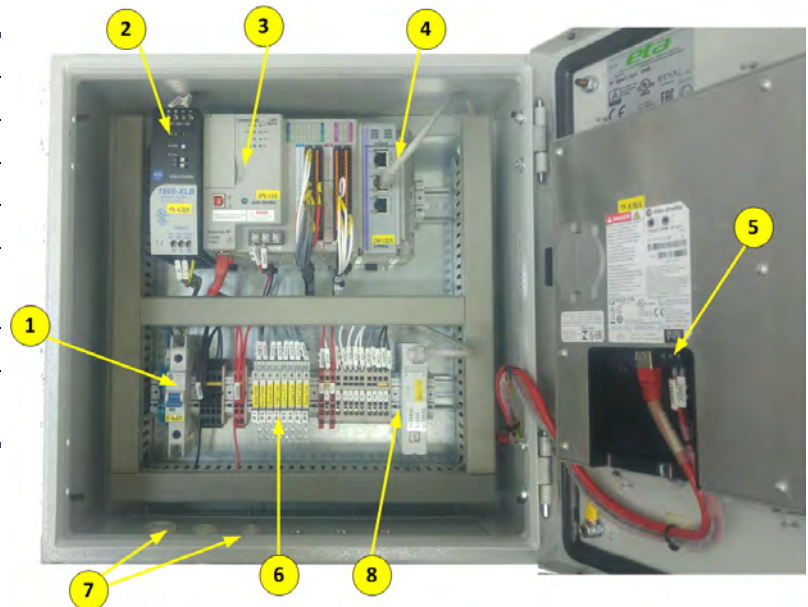


Figure 5-14: Control Module Interior

5.8.1 Mounting the Control Module

For exact physical dimensions of all units, be sure to consult the Mechanical ICD drawings provided to you by Atlantium with this manual.

The **Control Module** can be mounted on a wall or suitable anchoring point. The **Control Module** comes with an electrical 1.6M/ 5.25ft. long (1.5m/4.92ft. from the PG gland) and a Data cable (for length options, see the table below). However, it can be placed up to 50M/ 164.04 ft. from the Atlantium Unit, in a place that it is easy to reach the touch screen and read the monitoring information. A longer Data Cable can be ordered for this purpose.

ATTENTION!

- When mounting the **Control Module**, take into consideration the length of the cables.
- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**



For preparing all electrical connections and mechanical installation, be sure to review the electrical and mechanical ICD drawings provided to you by Atlantium with this manual. If no electrical drawings were provided please contact your Atlantium representative.

➤ To mount the Control Module on the wall:

- Step 1.** On the wall space designated for the **Control Module**, make four marks according to the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.
- Step 2.** The mounting holes **(1)** are 8.5mm (0.33") in diameter. Using the appropriate drill bit, drill the four holes.
- Step 3.** In each hole, insert a plastic anchor.
- Step 4.** Uncrate the **Control Module** package and carefully lift it out.
- Step 5.** Open the door of the **Control Module** and remove the plastic covers on the four holes.
- Step 6.** Position the **Control Module** so that its four holes fit over the drilled holes.
- Step 7.** Using appropriate screws and with a screwdriver, screw in the four screws in a diagonal pattern (one corner, then the opposite corner).
- Step 8.** To secure the **Control Module** firmly, return to each screw in the diagonal pattern and tighten the screws securely.

➤ To mount the Control Module on skids:

- Step 1.** At the location designated for the **Control Module** set up the skids with spacing that matches the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.
- Step 2.** The mounting hole **(1)** are 8.5mm (0.33") in diameter. Using the appropriate bolts and nuts, with a screwdriver, screw in the four bolts and add the nuts leaving around 5.6mm (0.20-0.24") sticking out.
- Step 3.** Uncrate the **Control Module** package and carefully lift it out.
- Step 4.** Open the door of the **Control Module** and remove the plastic covers on the four holes.
- Step 5.** Position the **Control Module** so that its four holes fit over the corresponding holes on the skids.
- Step 6.** Using appropriate bolts and nuts and with a screwdriver, screw in the four bolts adding the nuts in a diagonal pattern (one corner, then the opposite corner).
- Step 7.** To secure the **Control Module** firmly, return to each bolt in the diagonal pattern and tighten them securely.

5.9 Installing the Ballast Module(s)*

Each UV lamp requires a corresponding Ballast Module. The location for mounting the Ballast Module(s) must take into account the length of the cables to be attached from the Atlantium Unit. If you need to calculate space-cooling requirements for the location in which you are installing the Ballast Module(s), refer to the table for heat load per Ballast Module, shown below:

Heat Load per Ballast Module: 210W

ATTENTION!



- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits or cable trays and must not be stretched.**
- **Install the Ballast Modules of the Atlantium system in an area where the air is clean and free of dirt particles in suspension (e.g. sand, dust, oil, food powders, etc.). This precaution protects the BM cooling fans/vents from blockage by dirt.**

*For maintenance procedure and diagram of Ballast Module Types B/C, see **Checking & Cleaning the BM Cooling Fans** on page 124

Ballast Modules must be installed vertically and positioned at least one meter above ground level. Ballast Modules must be at least 300mm (12") from each other or from any other object around them to allow proper ventilation. Ballast Modules must not be installed within closed cabinets.

You can mount them next to each other horizontally or one above another vertically. Atlantium recommends you arrange the Ballast Modules in consecutive order from left to right according to the number marked on the bottom of the Ballast Module.

Consult the **Ballast Module's ICD** accompanying this manual.

BALLAST MODULE C (Power Unit version).

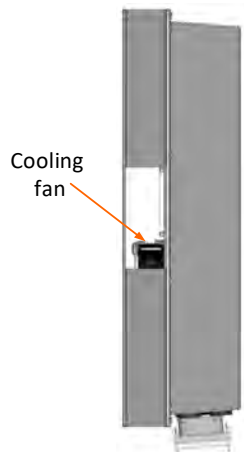


Figure 5-15: Ballast Module C side view

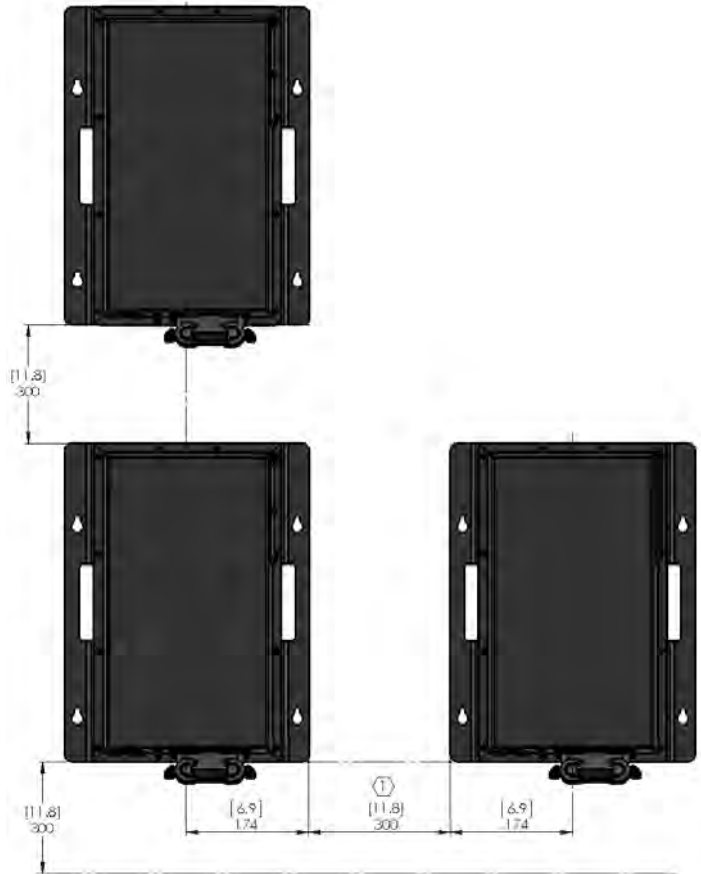


Figure 5-16: Multiple Ballast Module C Installation Diagram

➡ To mount a Ballast Module on the wall:



The Ballast Module is heavy! Be sure that the wall on which you intend to mount the Ballast Module is strong enough to carry its weight.

- Step 1.** On the wall space designated for the Ballast Modules, make four marks according to the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.
- Step 2.** The keyhole slots for mounting are 7mm (0.28") in diameter for screws size M6 OR 1/4" bolts. Using the appropriate drill bit, drill the four holes.
- Step 3.** In each hole, insert a plastic anchor and with the appropriate screwdriver, screw in the screw leaving around 5.6mm (0.20-0.24") sticking out of the wall.
- Step 4.** Uncrate the Ballast Module package and carefully lift out the Unit.
- Step 5.** Fit the keyhole slots onto the protruding screw heads and slide the Unit downward until the top of the keyhole slots rest on the screws.
- Step 6.** To secure the Unit firmly, tighten the four screws.

Installing the Atlantium System

➔ To mount a Ballast Module on skids:

- Step 1.** At the location designated for the Ballast Modules, set up the skids with spacing that matches the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.
- Step 2.** The keyhole slot for mounting is 7mm (0.28") in diameter for screws size M6 OR 1/4" bolts.
- Step 3.** Using the appropriate screwdriver, screw in the four bolts and add the nuts leaving around 5.6mm (0.20-0.24") sticking out.
- Step 4.** Uncrate the Ballast Module package and carefully lift out the Unit.
- Step 5.** Fit the keyhole slots onto the protruding bolt heads and slide the Unit downward until the top of the keyhole slots rest on the bolts.
- Step 6.** To secure the Unit firmly, tighten the four bolts and their nuts.

5.9.1 Connecting the Ballast Module Cables

One side of the **Cable Harness** is attached to the connection box of each lamp of the Atlantium Unit. The other end of the **Cable Harness** connects to the corresponding Ballast Module via a single connector. Refer to the **Electrical ICD** drawing provided to you by Atlantium with this manual

➔ To connect the cable:

Table 5-1: Ballast Module Details

#	Item
A	Single Connector with lock latch
B	48VDC for Fan - 440/480VAC only
C	Pressure Relief
D	Fuse Housing (for Atlantium Technicians only)
E	Connector width 13.11cm / 5.159inch.



Figure 5-19: Connector Detail

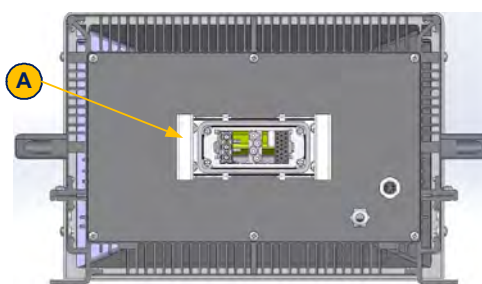


Figure 5-17: Ballast Module A-Waterproof Box

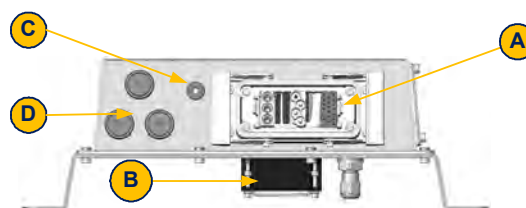


Figure 5-18: Ballast Module C-Waterproof Box (bottom view)

- Step 1.** According to the electrical drawing of the Installation plan, bring the **Cable Harness** attached to the Connection Box of Atlantium Unit up to the Ballast Module.
- Step 2.** On the Ballast Module, pull the lock latch of the connector to the open position and remove the protective cover.



Be sure to save the protective cover of the Ballast Module connector in an accessible location for the future in case you need to dismantle the Ballast Module for any reason.

- Step 3.** Connect the lamp cable lead to the Lamp connection port (**A** in the figures above) (the connector is keyed so it only matches in one position) and pull the lock latch closed to secure the connection.

5.10 Connection Box

WARNING!



DO NOT DRILL INTO THE CONNECTION BOXES OR ALTER THEM IN ANY WAY!!!!

The Connection Box already contains entry glands for **ALL** of the cabling needs.



Making holes in a Connection Box will destroy its waterproof seal, endangering the components inside. **Do Not** draw electricity from the Atlantium Unit's Connection Box as it will damage the Unit's electrical configuration.



Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Atlantium uses the following types of Connection Box (LM) cards:

Table 5-2: Connection Box (LM) cards

Connection Box Card Type	P/N	For use in systems....:
700/750	AP0000700/ AP0000750	systems with Ultrasonic Cleaner
1300	AP0001300	RZ systems without Ultrasonic Cleaner
1350	AP0001350	RZB systems
1400	AP0001400	together with Card 1300 or 1350 for systems with DPM cleaner
1500	AP0001500	together with Card 1300 or 1350 for systems without DPM cleaner



Do not alter any cable connection that is supplied from the factory.

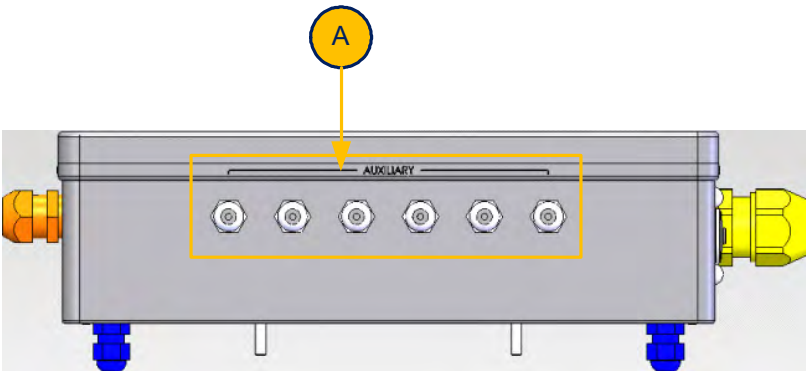


Figure 5-20: Connection box with cable insertion holes [A]

5.10.1 Connecting the Connection (LM) Card

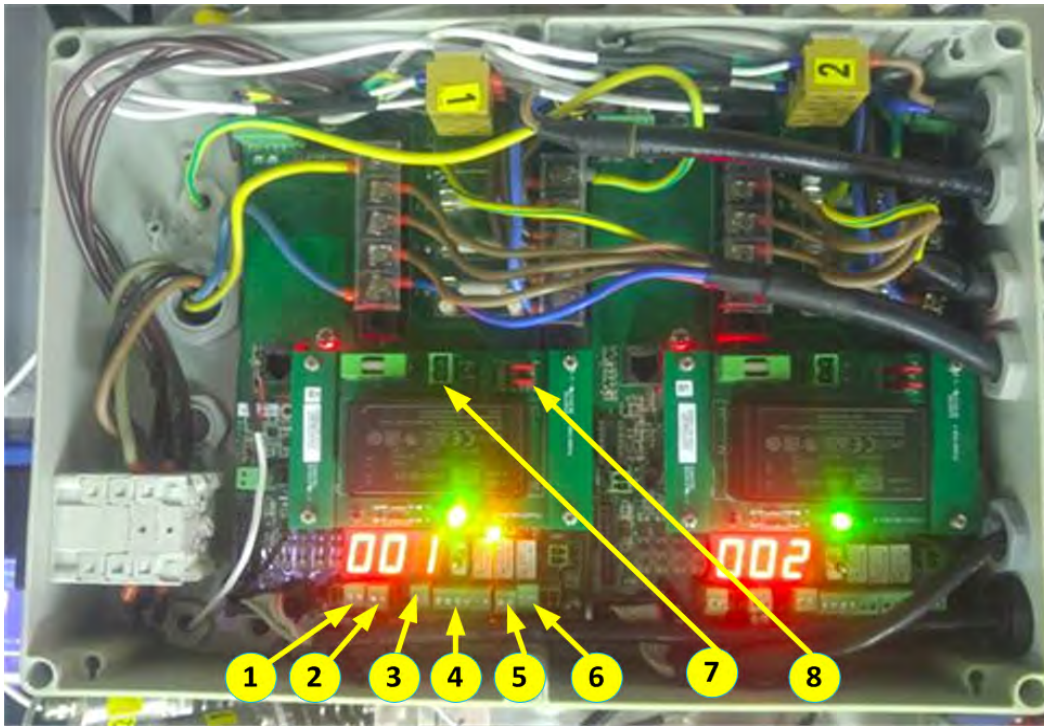


Figure 5-21: Connection Box Card 1350+1500



Atlantium recommends connecting the signal components to the **Control Module**. However, connection to the LM card may be used as an alternative. You should consult your Atlantium technician about this.

Table 5-3: 1350+1500 Card Connections*

#	Connector*
1	UVTS
2	UVIS
7	100 - 240VAC Power Connection
8	Jumpers to define connection type

*Connectors 3-6 not used

- Step 1.** On the first **Connection Box**, remove the covers of holes (marked **A** in the diagram below) as needed for the auxiliary components in your system configuration.
- Step 2.** Add plastic glands to each hole to be used. Make the connections to the terminals listed in Table 5-6 as appropriate*.
- Step 3.** Fasten the plastic nut of each of the cable

➡ To connect the Data cable:

- Step 1.** Extend the Data cable connected to Connection Box #1, out to the **Control Module** and connect it to the following terminals on the RJ45-7.7 port of the Control Module (Figure 5-22).

- Step 2.** Open the lid of the RJ-45 connector on the AB Control Module (Figure 5-22)
- Step 3.** Connect the Data cable wires to the RJ-45 connector as follows (see label in Figure 5-22):
- TXRXD+ to pin #1
 - GND to pin #5
 - TXRXD- to pin #8
- Step 4.** Secure Data cable in place and close lid of the RJ-45 connector

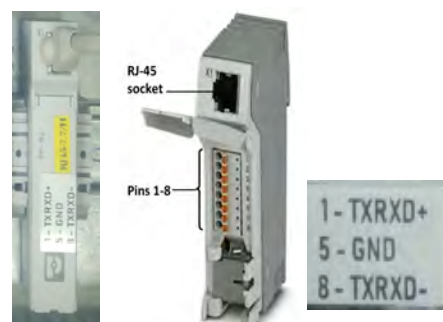


Figure 5-22: RJ-45 connector - L closed, M opened, R label



Do not make alterations to or drill into the Ballast Modules, Control Module, or any other component of the Atlantium System. Such alternations void the Atlantium Warranty.

5.11 Connecting Peripheral Equipment

5.11.1 Digital Outputs

The following relay output connections are available on the AB Control Module:

- R1 – System Ready / Outlet Valve - D00.0
- R2 – General Alarm - D00.1
- R3 – Shutdown alarm - D00.2
- R4 – High Temperature - D00.3
- R5 – Low UVT - D00.4
- R6 – Low Status - D00.5
- R7 – Inlet Valve - D00.6
- R8 – Lamp/Ballast Failure - D00.7

See **Configuring the Signal Settings** on page 69 and **Other Settings** on page 70

5.11.2 Analog I/O and External On/Off

The following relay I/O connections are available on the AB Control Module:

- 1 - +24VDC = External On/Off
- 2 - DI 0.0 = External On/Off
- 3 - +24VDC = Flow Switch
- 4 - DI 0.1 = Flow Switch
- 5 - AI 0+ = Flow Meter
- 6 - AI 0- = Flow Meter
- 7 - AI 1+ = Conductivity Meter
- 8 - AI 1- = Conductivity Meter
- 9 - AO 0+ = Dose Out +
- 10 - AO 1+ = UVT Out +
- 11 - AO 0- = Dose Out -
- 12 - AO 1- = UVT Out +

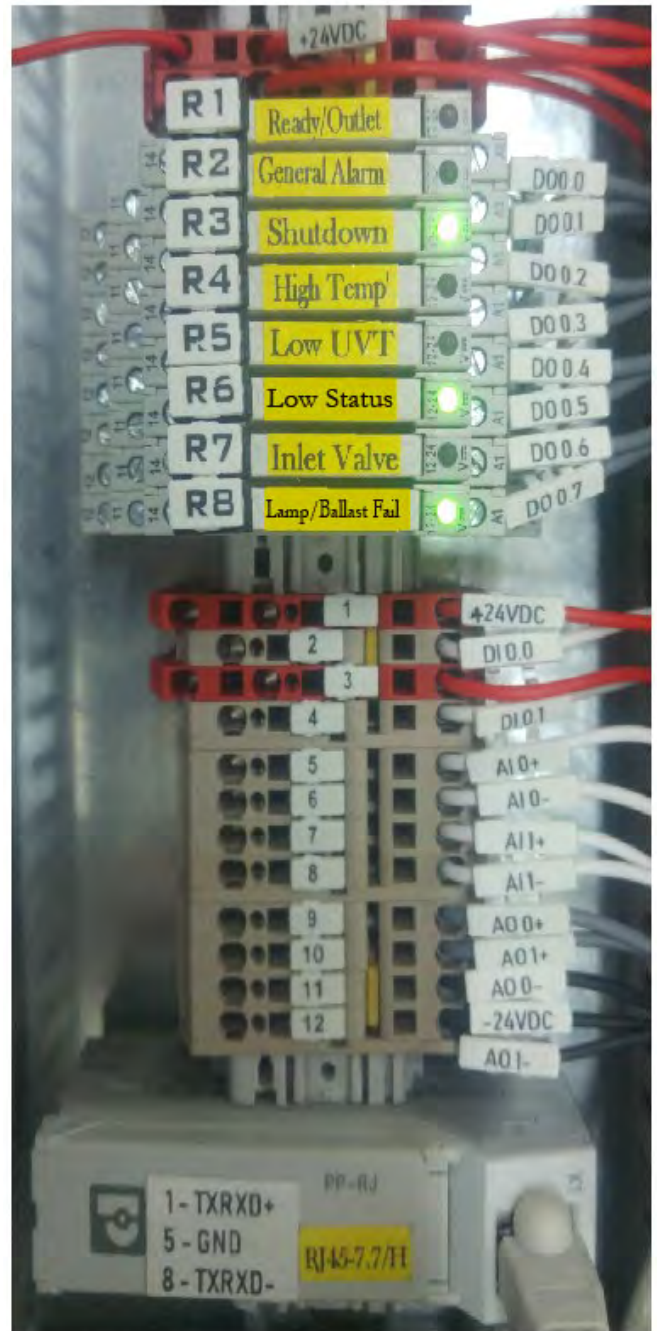


Figure 5-23: AB module connections

5.12 Connecting the Atlantium System to the Mains



DANGER!

ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENT MUST BE SET TO THE OFF POSITION AND LOCKED WITH AN APPROPRIATE LOCK TO PREVENT ACCIDENTAL CHANGE TO THE ON POSITION.

DO NOT REMOVE THE LOCK OR SET ANY MAIN CIRCUIT BREAKER TO THE ON POSITION UNTIL ALL ELECTRICAL WIRING IS PROPERLY CONNECTED.

UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

The infrastructure of the electrical connections is to have been prepared according to the *Electrical ICD* drawing provided to you by Atlantium with this manual.

For installations that require 440-480VAC only, the customer must provide the Power cable to the mains.

5.12.1 Mains Connection Phases

Phase 1 Connect the **Control Module** to the mains.

Phase 2 Connect the Atlantium Unit's Power cable to the mains.

ATTENTION!

- Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.

5.12.2 Connecting the Control Module's Power Cable

The **Control Module** comes with a power cable of 2m/6.5ft. length. For the electrical specifications, See *The Control Module* on page 22. Prepare the appropriate plug for the electrical standards of your region. Considering the distance between the **Control Module** and your electrical box, you may want to replace the electrical cable with a longer one.

➔ To connect the Control Module's power cable:

- Connect the appropriate plug for the electrical standards of your region.
- **Do not** plug in the **Control Module** to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See *Initializing the Atlantium System* on page 61.)

5.12.2.1 Replacing the Control Module's Power Cable

You can order a longer cable from Atlantium or acquire one that conforms to the specifications, from a local source.



Avoid direct contact with the electronic circuit board which is sensitive to electrostatic discharge (ESD). Utilize best practices regarding ESD when replacing the power cable.

5.12.3 Connecting the Atlantium Unit's Power Cable

The Atlantium Unit comes with a power cable of (5m/15ft.) or 10m/30ft (according to you order.) Prepare the appropriate cables and plugs for the facility mains connection according to the electrical standards of your region. Considering the distance between the Atlantium Unit and your electrical box, you may want to replace the electrical cable with a longer one.



- **A certified electrician is required to perform these procedures. Follow your local regulations.**
- **Do not** connect the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See *Initializing the Atlantium System* on page 61.)
- **Do not** connect the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See *Initializing the Atlantium System* on page 61.)

➔ To connect the Atlantium Unit's power cable:

- Connect the appropriate plug for the electrical standards of your region and connect it to a mains outlet.



- The power cables supplied with the unit has 5 wires: 3 Phases (L1, L2, L3), Neutral (N) and Ground (PE). However, for the 440/480V installation, the Neutral wire is redundant and is not to be connected on either side of the cable.

- Step 5.** Loosen the plastic nut of the Power cable's gland (**C**) from the underneath and unthread the power cable.
- Step 6.** Thread the replacement power cable into the gland (**C**).
- Step 7.** Using the appropriate screwdriver, connect the replacement Power cable leads (**G, H, I**).
- Step 8.** Return the switch to its place and refasten the Switch holder (**F**).
- Step 9.** Return the Switch place (**E**) to the correct position and reattach the Switch knob (**D**).
- Step 10.** Using the appropriate screwdriver, on the terminal block, connect the replacement Power cable leads (**G, H, I**) and return the cover (**J**).
- Step 11.** Tighten the plastic nut of the gland (**C**) from the underneath.
- Step 12.** Using the appropriate Allen wrench, close the Connection Box cover.

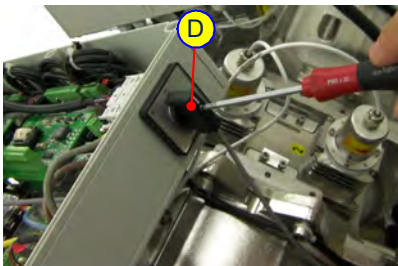


Figure 5-24: Replacing the Power Cable 1

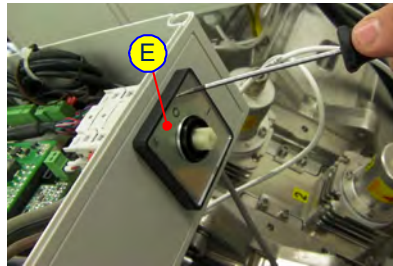


Figure 5-25: Replacing the Power Cable 2

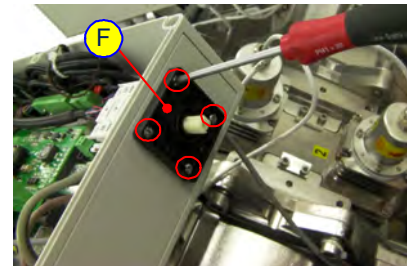


Figure 5-26: Replacing the Power Cable 3

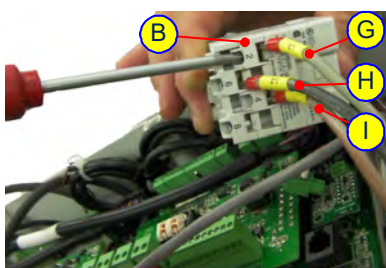


Figure 5-27: Replacing the Power Cable 4

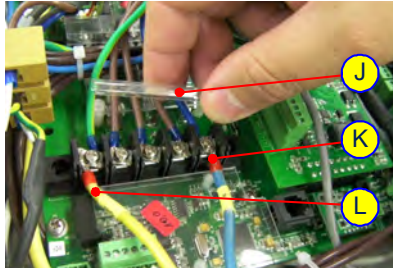


Figure 5-28: Replacing the Power Cable 5

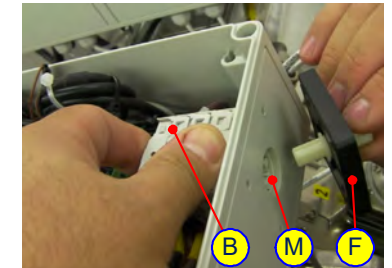


Figure 5-29: Replacing the Power Cable 6

5.13 Installation Check List

When you are ready to schedule the Atlantium technician to commission the Atlantium system, complete the check list below, based on Atlantium's **Commissioning Requirement Specifications**.

Send this check list together with digital photos of every angle of the installed system to your Atlantium representative. It generally requires at least 2 weeks after the check list has been completed to schedule the Atlantium commissioning.

Table 5-4: Atlantium System Installation Check List

Item	
System Serial Number	S.N: _____
System Orientation and Positioning The Atlantium Unit is installed: ☐ Vertically - Water is flowing upward ☐ Horizontally, with slope The following Modules are correctly and securely mounted and located in their final position: ☐ Ballast Module ☐ Control Module ☐ Ultrasound Module (if purchased) ☐ Transformer Module ☐ All other purchased components are securely installed and have sufficient service clearance on all sides (according to the ICDs). ☐ The Atlantium Unit and all of its components are not located where they may be in contact with direct water or chemical dripping. ☐ Cable from system is connected to the Electrical Cabinet(s) Ballast Module(s) according to installation manual. The electrical cables between components are not stretched.	
Power Connections ☐ Power cable is connected to the Electrical Cabinet(s) Ballast Module(s) Voltage rate _____ Number of phases _____ ☐ Correct voltage was measured at entrance to Ballast Module Connection boxes You are scheduling an electrician to be on site and available on day and time of System commissioning - a MUST or the installation cannot be completed!.	
Water Flow ☐ The Atlantium Unit is connected to the water pipes, according to the Installation plan developed by facility engineers and Atlantium personnel. ☐ Proper pipe lengths (Inlet - 10D, Outlet - 5D) are installed. ☐ The Unit has been checked while full of water and no bubbles or leaks exist in the system. The system is balanced, supported and pressure stable (no vibration, etc.). ☐ No detected water hammers or pressure above 7 bars are on the line.	

Table 5-4: Atlantium System Installation Check List

Item
Flow Measurement Device (only one option is required) A flow meter or flow switch is installed and the electrical and data connection are complete. ☐ If flow meter: 4 - 20 mA signal and Com wires are connected as instructed in this guide. ☐ If flow switch: 2 wires from the flow switch are connected as instructed in this guide.
Accessories The following accessories are installed on the line and in accordance to Atlantium instructions ☐ Sampling Valves ☐ Draining Valves ☐ CIP Ports ☐ Air Release Valves
Inputs/Outputs required by the customer (require hard wiring): ☐ Dose out ☐ Ready Signal ☐ General Alarm ☐ Shutdown Warning ☐ High Temperature ☐ UVT Out ☐ Low UVT ☐ Low Status ☐ Remote On/Off ☐ Outlet Valve Output ☐ Inlet Valve Output ☐ Bypass Valve Output ☐ Lamp/Ballast Failure
To be sent to Atlantium, attach photos of the following with this Checklist ☐ Atlantium Unit ☐ Ballast Module(s), Control Module and other components ☐ Accessories (Flow meter or flow switch, CIP ports, Sampling valves, etc.)
Customer Approval <i>By signing, I hereby declare that the above filled information is correct and nothing prevents the arrival of Atlantium engineer on the designated/agreed date to perform the activation and commissioning.</i> <i>I am aware that lack of water flow or electricity, or any delay providing it and/or major deviation from above filled information, shall incur additional costs and a fully paid additional visit.</i> Name of Customer's authorized representative: _____ Position: _____ Signature of Customer's authorized representative: _____ Date: _____
Atlantium Commissioning Document Atlantium Commissioning Document ☐ Received



6 Initializing the Atlantium System

As indicated above in Figure 5-1 on page 40, Phases 6-9 are considered commissioning and are to be implemented by Atlantium personnel. The Atlantium system starts up with the initialization of the **Control Module**.

➔ To start up the Atlantium system:

- Step 1.** Check that the power cables of the Ballast Modules are all connected to the mains power source.
- Step 2.** Connect the power cable of the **Control Module** to the mains power source.
- Step 3.** Using the attached key, open the screen panel of the **Control Module**.
- Step 4.** Locate the circuit breaker (1) and turn it to the **On** position.

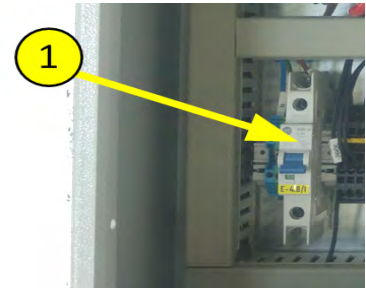


Figure 6-1: Control Module CB

The system powers up and the **Control Module's** touch-screen monitor is activated, displaying the Atlantium software's opening screen, for a short time while the system performs its internal self-testing routines. When the self-testing routines are complete, the **Main Operation** screen appears.

The **Lamp Power** button  on shows that UV lamps are off



Figure 6-2: Main Screen of AB Control Module (lamps on)

- Step 5.** Press the Lamp Power button  to ignite the lamps.



7 Configuring the AB Control Module

7.1 The Allen Bradley (AB) Control Module HMI

The **AB Control Module** provides the Human Machine Interface (HMI) for control of operations and dose measurements of the Atlantium system.

The regulations applying to the type of application determine sizing as well as software defaults and operational control options. This is set up at the original manufacturing facility and completed on site by the Atlantium technician.

The **AB Control Module** provides a real-time display of operational data. A Ready indicator turns green when the system is ready for operation.

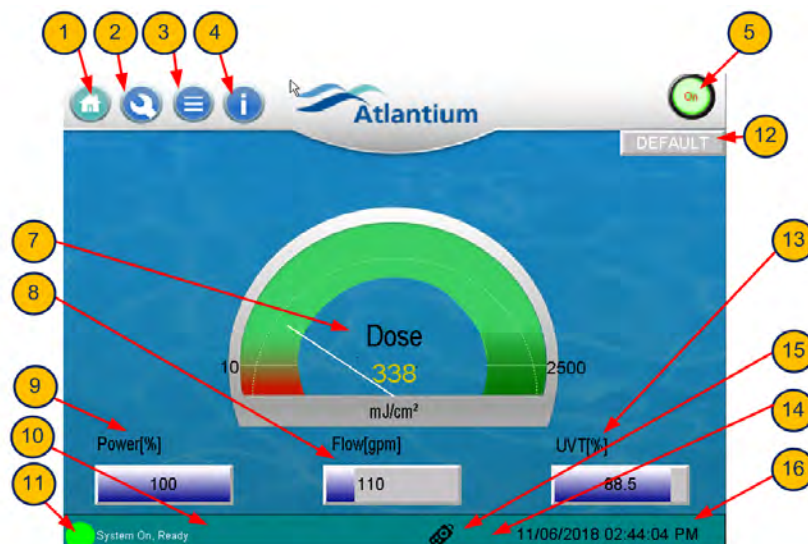


Figure 7-1: Initial Main Operation Screen*

Table 7-1: AB Control Module Main Screen Details

#	Item	Icon	Description
1	Return to Main screen		To return to the Main screen.
2	Settings Screen Access		To access the Settings screen. and other screens according to permissions associated with the usernames and password. Enter by default to monitor screen and not to configuration screen
3	Event Log Access		To the Events screen. Access the list of events that have occurred during system operation. No need user name & password.
4	System information Screen Access		Includes the Generic Info tab that displays general system information and lamp status and the Info Monitor tab that displays a summary of system status information.

Table 7-1: AB Control Module Main Screen Details

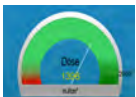
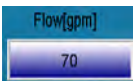
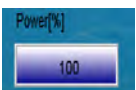
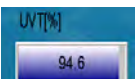
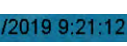
#	Item	Icon	Description
5	Main Unit Operation Button		ON/Off toggle button turns on/off the Atlantium unit components (lamp(s), UV power, etc.) in an orderly sequence.
6	N.A.		
7	Dose Gauge		Represents the system dose range. The red area of the gauge represents off-spec low dose. The dark green area represents the user's set maximum allowable dose. The current dose (mJ/cm ²) is displayed in the middle of the gauge. The minimum and maximum allowable dosage levels are user-defined.
8	Flow		Scale shows the current rate of water flowing through the Unit The scale color changes to red when the flow is either less than minimum flow setting or above the maximum flow setting. When there is no flow the scale goes gray.
9	Power		Scale shows the current Power level.
10	Bottom Footer		Displays certain system events for a short period of time. A relevant icon appears when a critical event occurs. It disappears after the event has been acknowledged in the log.
11	Ready LED		Displays the basic system status. When the system is ready for operation (all the lamps are ignited and the dose is above the minimum settings, the system displays the green Ready icon. If the system startup fails or the system is no longer working at or above the minimum dose, the system displays the red icon.
12	User Login		The Login button enables access to the settings.
13	UVT (Transmittance)		Scale shows the amount of UV light transmitted through the water. If the UVT percentage factor is high, the transparency of the water to UV light is high, and vice versa. When there are several UVT sensors in one system, this Scale displays the average of the UVT values measured by each sensor. The Scale color becomes red when the UVT is lower than minimum UVT setting. The color changes even if the UVT alarm is set to None.

Table 7-1: AB Control Module Main Screen Details

#	Item	Icon	Description
14	Space for Event Icons	 Communication  Emergency shutdown	A relevant icon appears when a critical event occurs. It disappears after the event has been acknowledged in the log.
15	Remote On/Off indicator		The Remote-On/Off indicator appears if marked in the Signals tab settings.
16	Date and Time		Date and time are displayed according USA or European conventions



The specific configuration for your Atlantium system is based on the Application type set for your industry and according to the parameters required for your facility's process.

Configuring the **AB Control Module** begins with logging in and accessing the Settings screen and follows the general order of the screen components.

7.2 Log In

Most screens of the AB Control Module are accessible without any password.

User must log in as Administrator with the password 9999 to change configurations.

Table 7-2: AB Control Module User Privileges

Purpose	Screen	No login needed	Administrator
Password			9999
View Main Screen	Main Screen	√	√
Configuration	Tech Settings Screens		√
Signal Setting	Signals Screens	√	√
Flow Setting	Flow Screen	√	√
Dose Setting	Dose Setting Screen	√	√
General Settings	General Settings Screen	√	√
Tech Monitor	Tech Monitor Screen	√	√

Table 7-2: AB Control Module User Privileges

Purpose	Screen	No login needed	Administrator
Password			9999
Tuning	Tuning	√	√
View System Information	View System Information	√	√
View Monitor Screen	View Monitor Screen	√	√
View Event Screen	View Event Screen	√	√

7.3 Settings Screens

The Settings Screens include the following :

- Monitor
- Tuning
- Signals
- Flow Settings
- Dose Settings
- General
- Configuration



WARNING!

Take care to ensure that the system configuration made via the Settings screen is not tampered with! Unauthorized changes may have a serious impact on the function of the Atlantium system. **Only trained and authorized** facility technicians are to access the Settings screen.



AB screen display values are shown in US units.

➤ To access the Settings screen:

- Step 1.** Tap  at the top of the screen.
- Step 2.** The System Info screen is displayed with the Monitor tab open:



Figure 7-2: Settings Screen, showing Monitor Tab (generic example with 12 lamps)

- Step 3.** Tap the **System Info** tab. The **System Info** screen appears



Figure 7-3: System Info Screen

Step 4. Tap the **Configuration** tab. The **Configuration** screen appears.

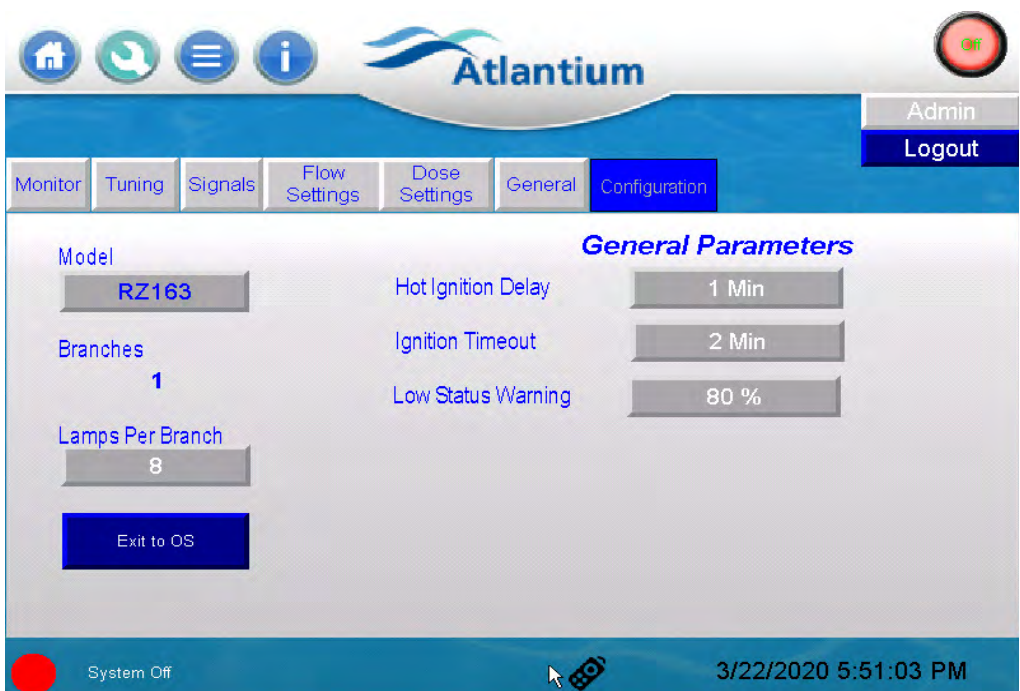
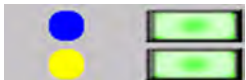


Figure 7-4: Configuration Screen (sample screen)

The configuration screen is important for use when replacing the LM card in the connection box

Table 7-3: Configuration Screen settings

Feature	Description
Hot Ignition Delay	Minimum time interval [1-10 min] between switch off to re-ignition of lamp. Enables lamp to cool down before re-ignition.
Hot Ignition Delay test	1 Re-ignite lamps immediately after switch off. Lamps should reignite only after set interval. 2 Re-ignite lamps after set interval from switch off. Lamps should reignite immediately.
Ignition Timeout	Time interval [1-30 min] during which system attempts to ignite lamps following switch on. This interval starts at end of Hot Ignition Delay. 1 If one or more lamp(s) fail(s) to ignite during Ignition Timeout interval, ignition attempt stops and lamp ignition fault appears: 21 - ignition failed - lamp not ignited/gone out lamp number 3 2 The faulty lamp is disabled and appears in red on Monitor screen (see Figure 7-10 on page 76).  3 The Lamp/Ballast fault goes active - green outline (see Figure 7-5) Lamp / Ballast  On successful ignition of all lamps the lamp marker appears in blue and the Lamp/ Ballast fault de-activates.
Low Status Warning	When the system is on and the lamp status is lower than this Low Status Warning value for more than 30 sec the "Low Status Lamp #" warning is raised, the lamp marker goes yellow and the Low Status output is triggered. 

Step 5. Tap the **Signals tab**. The **Signals Tab** appears.

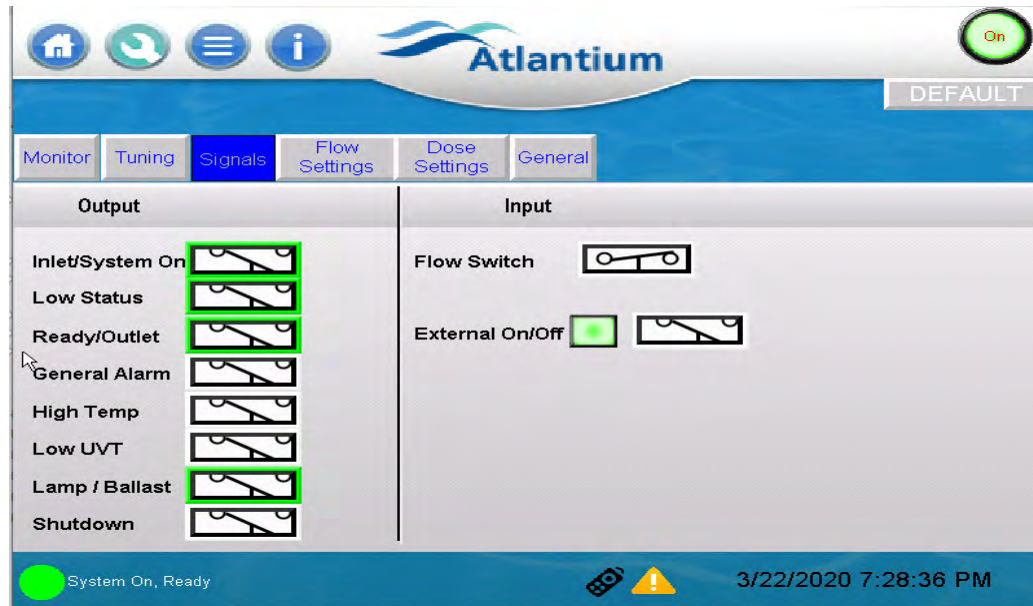


Figure 7-5: Signals Tab on Settings Screen

7.4 Configuring the Signal Settings

The Signals tab displays the settings for the system’s dry-contact Input and Output connections: either **Normally Open (NO)** () or **Normally Closed (NC)** () see Figure 7-5 on page 69. The **green** border around the NO/NC symbol shows that the connection is active (logic = 1).

The following is an explanation of Input and Output signals. See **Digital Outputs** on page 55 to view physical output connections.

7.4.1 Signal Outputs

Inlet valve/System on - The facility’s electronically-controlled **Inlet** and **Outlet** valves, that control the water supply to the Atlantium unit, as well as the **Bypass valve** (if a bypass is implemented), must each be configured to receive control signals from the **AB Control Module**. These valves affect various facility processes relevant to the Atlantium system, such as:

- Managing high flow conditions
- Bypassing and streaming the water to a different place if the water quality has degraded
- Bypassing for cleaning chemicals that are not compatible with the quartz chambers
- When the system is activated (either manually via pressing the **Operation** button on the **AB Control Module**’s main screen, or via an external signal utilizing the **External On/Off**), the **AB Control Module** sends a signal, within 10 seconds, that opens the **Inlet valve**. This signal can also be used to provide feedback that the system is activated.
- When the all the lamps are ignited and ready to use, the **AB Control Module** sends a signal to the **Outlet valve** to open. This signal can also be used to notify that the system is **ready**. The **AB Control Module** also sends a signal to the **Bypass valve** to close.
- Similarly, when the system is deactivated, (either manually via pressing the **Operation** button on the **AB Control Module**’s, or via an external signal utilizing the **External On/Off**), the **AB Control Module** sends a signal that closes the **Inlet** and **Outlet** valves, and also opens the **Bypass valve**.

Configuring the AB Control Module

- **Signals to Alarms** - To set off a General Alarm mechanism, such as a flashing light, alarm, bell, or a call or message to a telephone, a system alarm signal must be connected to a SCADA control system. See **Viewing the Alarm Log** on page 87.

General Alarm - requires connection of a General Alarm mechanism to receive a signal from the **AB Control Module**. Once configured, the alarm is triggered by any of following signals (connection required in each case):

- **High Temperature** - signal output when system temperature has exceeded the configured maximum.
- **Low UVT** - signal output when the system's UVT has fallen below the configured minimum.
- **Lamp Ballast** - signal output - green outline - when one of the lamps has failed to ignite.
- **Shutdown** - signal output when an event causes the system to shutdown.

7.4.2 Signal Inputs

- **Flow Switch** - If your system utilizes a Flow Switch, you must configure its electrical polarity in the **AB Control Module**. For the Flow Meter, see **Configuring the Flow Settings** on page 71.
- **External On/Off** - A switching mechanism to synchronize the Unit with the facility processes. This remote PLC turns the Atlantium Unit on/off. For details about the External On/Off signal wiring, see **Viewing the Alarm Log** on page 87.

For example, the facility's SCADA (Supervisory Control And Data Acquisition) control system can send a signal to the Unit to start or stop its operation as needed. Using this setting, the Atlantium system recognizes this signal and turn the Unit on or off, as appropriate, without the need for manual intervention.



When the **External On/Off** sends a signal to activate the system, the Atlantium **AB Control Module** sends a response within **10 seconds** that the command was received. If no response is received, it is likely that the **External Operation Button** on the **AB Control Module** main screen has been manually set to the **Off** position. Check this button and if needed manually tap it to reinstate the **On** position, so that the **External On/Off** can control the system activation.

For regular wiring, set according to the table in **External On/Off - Additional Notes** on page 59:

- For **IN+** select **Normally Closed**
- For **IN-** select **Normally Open**
- **Low/Normal Mode** - for systems with an external signal for switching to Low Flow mode.

7.4.3 Other Settings

- **Maximum Temperature Warning** - The System shuts down automatically if the water temperature exceeds the maximum set value. You can set the water temperature threshold for sending a warning notification, which alerts you to take action to prevent the water from reaching the maximum temperature that triggers shutdown. It is recommended to set the warning notification threshold temperature to 5% below the shutdown threshold temperature.
- **System Auto Power On** - You can also set the Atlantium system to restart automatically when power is restored after a facility-wide power failure, so that it reactivates according to the configured **Auto Power On** settings.

➔ To configure the Signals settings:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

The states of the **Outputs** and **Inputs** are set via toggle switches that by default are set to **NC (Normally Closed)**. To change any of them to **NO (Normally Open)**, tap the corresponding toggle switch. It changes to **NO**. Tap again to change it back to **NC**.

Step 2. Under **Output**, set the electrical polarity for:

- **Inlet valve/System On**
- **Low Status**

- Outlet valve/Ready
- General Alarm
- High Temperature
- Low UVT
- Lamp/Ballast Failure
- Shutdown

Step 3. Under **Input**, set the electrical polarity for:

- **Flow Switch** (if installed)
- **External On/Off**

Step 4. To enable the External On/Off, mark the **External On/Of Enabled** checkbox.

7.5 Configuring the Flow Settings

If a flow meter is utilized, the **AB Control Module** data must be configured to match the flow meter at the time the system is originally commissioned, but you may need to change this configuration if a flow meter is changed, etc. The Max Flow setting provides a threshold to compare to the actual flow rate. If the actual flow rate exceeds this set Max Flow value, the system sends notification and triggers an alarm.

➔ To configure the flow settings:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **Flow tab**. The **Flow Settings** tab appears. The **Current Flow** indicator displays the current rate of water flowing through the Atlantium system in **GPM** (gal/min) units. The **Main Flow Input** indicator displays the **Flow Switch (FS)** or the **Flow Meter (FM)** in **GPM** units.

- For Flow Meter:
 - **Flow Meter** appears as the **Main flow input** source,
 - The **Current flow** is displayed on the right.



Figure 7-6: Flow Setting Screen - Flow Meter

7.5.1 Low Flow Mode

Set the Atlantium system's response to a low water flow situation via the Low Flow mode. This unique Atlantium feature minimizes the on/off cycles by allowing the system to operate at an extremely low flow value. The Atlantium Application Engineer defines this value according to the specific application. In some cases, this value can be zero flow for a predefined period of time. The **Low Flow Mode** options are:

- ◆ Restart
- ◆ Shutdown
- **Restart Mode** – (Default) This option assumes that low flow is not necessarily a sign of a fault and ensures that the system is ready to come up to full dose or power immediately when the flow rises.

Atlantium allows the system to continue operating at a low or even zero flow rate up to a max time interval, the **Flow Delay**, as defined by the Atlantium application engineer. With this setting, we reduce the number of times the UV lamps are switched on and off, thus extending their lifetime, while preventing the flow of non-disinfected water into the line.

- As long as the flow rate stays below the **Min Flow** for less than the amount of time set in **Flow Delay**, the Unit continues to operate in standby mode, generally at the lowest possible power.
- However, if the water flow rate drops below the set **Min Flow** rate for a time that is **longer** than the **Flow Delay**, then:
 - ◆ The **Low Water Flow** warning is triggered
 - ◆ The lamps are automatically shut down
 - ◆ An event notification is written to the event screen and log file
- The **Restart** mode causes the system automatically restart when the water flow rate returns to a level above the **Min Flow**. The Unit automatically and instantly returns to full operation, without any operator intervention or connection to a higher control level, and without any delay or cool down/ warm up period.

The default values for the minimum flow rate and maximum flow delay appropriate for **Restart** mode are preset by Atlantium. These values may be changed based on local site requirements and conditions, but **be sure to coordinate** with the Atlantium representative.

- **Shutdown Mode** – In this mode, in the event of low flow, which may be a sign of an upstream malfunction, the system shuts itself down in an orderly fashion, but operator intervention is required to start it up again.
 - As long as the flow rate stays below the **Min Flow** for less than the amount of time set in **Flow Delay**, the Unit continues to operate in standby mode, generally at the lowest possible power.
 - However, if the water flow rate drops below the set **Min Flow** rate for a time that is **longer** than the **Flow Delay**, then:
 - ◆ The **Low Water Flow** warning is triggered
 - ◆ The lamps are automatically shut down
 - ◆ An event notification is written to the event screen and log file
- In the **Shutdown mode**, operator action is required to turn on the Atlantium system manually, once the water is back to the proper Flow rate.

⇒ Flow Settings

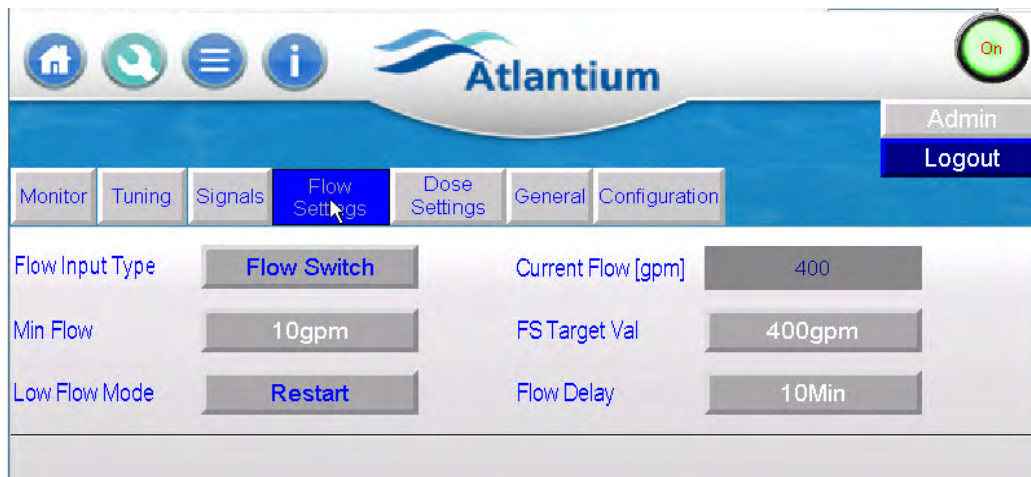
- Step 1.** For **Min Flow**, using the **-/+** buttons, enter a value for the minimum water flow rate threshold. Below this threshold, the system enters **Low Flow Mode**.
- Step 2.** For **Low Flow Mode**, tap the arrow and from the drop-down list, select one of these options:
 - ◆ Restart - When the flow returns after shut down the system will start again automatically.
 - ◆ Shutdown - when the flow returns after low flow shutdown the system remains off until the user starts it up manually.
 - ◆ Restart OK

- ◆ Shutdown Fail - Need to add Shutdown message and to add general and the shutdown outputs signals.
- Step 3.** The default value for maximum **Flow Delay** appropriate for **Restart** mode are preset by Atlantium. These values may be changed based on local site requirements and conditions, but be sure to coordinate with the Atlantium representative. If you need to change this value, set the desired value using the **-/+** buttons. The rate can be between 1 and 45 minutes, default 1 min
- Step 4.** For **mA at 0 Flow (Flow Factor)**, to ensure that the system does not operate on the basis of a discrepancy in the Current Flow readings, turn off the flow and then check the value for Current Flow. If it is not zero, adjust the value, by using the **-/+** buttons, until the Current Flow value is zero. Then resume the water flow. The range 2.00 to 6.00 the default is 4.00 with 2 decimal point resolution. .
- Step 5.** For **Flow x 100 at 20mA (Offset)** on the flow meter data sheet, find the meter's maximum-allowed flow rate.
- Step 6.** In the **Flow x 100 at 20mA** field, using the **-/+** buttons, enter the meter's maximum-allowed flow rate divided by 100. (If the max flow rate is 100 m³/h, enter the value **1**. Convert GPM values into m³/h values to make this calculation.)
- Step 7.** For **Max Flow**, using the **-/+** buttons, enter a value for the maximum water flow rate threshold. If the actual flow rate exceeds this threshold, the system delivers a notification message and triggers an alarm.
- Step 8.** To set the action to be performed if the water flow rate exceeds the value set in the **mA at 0 Flow** field. The options are:
 - ◆ **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
 - ◆ **Caution** - One or more caution messages are issued.
 - ◆ **Warning** - One or more warning messages are issued.
 - ◆ **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm.

For Flow Switch:

Flow Switch appears as the **Flow Input Type**.

The current flow is displayed on the right.



The screenshot shows the Atlantium control interface. At the top, there are navigation icons (Home, Back, Menu, Info) and the Atlantium logo. A status indicator on the top right shows 'On' with a green light. Below the navigation bar is a menu with tabs: Monitor, Tuning, Signals, Flow Settings (selected), Dose Settings, General, and Configuration. On the right side of the menu, there are 'Admin' and 'Logout' buttons. The main configuration area for 'Flow Settings' contains the following fields:

Flow Input Type	Flow Switch	Current Flow [gpm]	400
Min Flow	10gpm	FS Target Val	400gpm
Low Flow Mode	Restart	Flow Delay	10Min

Figure 7-7: Flow Setting Screen - Flow Switch

- Step 1.** For **FS Target Val**, using the **-/+** buttons, enter the maximum value expected for the flow in gpm

For **Min Flow**, using the **-/+** buttons, enter a value for the minimum water flow rate threshold. Below this threshold, the system enters **Low Flow Mode**. The Min Flow value rate shall be between 1 to (FS Target Val - 1)gpm as integer. The default number 100gpm. The Min Flow value rate shall be between 1 to (FS Target Val - 1)gpm as integer. The default number 100gpm.

Step 2. For **Low Flow Mode**, tap the arrow and from the dropdown list, select either **Restart** or **Shutdown**.

Step 3. The default value for maximum flow delay appropriate for **Restart** mode are preset by Atlantium. These values may be changed based on local site requirements and conditions, but be sure to coordinate with the Atlantium representative. If you require changing this value, for **Flow Delay**, using the **-/+** buttons, set the desired value.

7.6 Power Settings

The AB Control Module controls the operation of the Atlantium System according to Power Settings. The Operator sets the system at a particular value for UV Lamp power, as a percentage (%) of maximum lamp output, and the system operates at the prescribed power level. Most operational settings either provide information or shut-down the system under specified conditions.

➔ To configure the Power settings:

Step 1. At the top of the screen, tap . The **Monitor Tab** is displayed.

Step 2. Tap the **Dose Settings tab**. The **Dose Settings** screen appears.

The Operation Mode is preset as Power Mode.



Figure 7-8: Dose Settings Screen - Power Mode

Step 3. In **Power**, set the desired power level for UV lamp output. The system is set to operate at this power level as a constant.

7.7 Configuring the General & Connection Settings

You can set a number of general settings, as well as the connection settings, via the General tab.

The following General Settings are to be configured:

- Date and Time

- **Maximum Temperature** - This is the maximum allowed water temperature, in °F, which triggers a warning notification. You should set this value so that you can take action to prevent the water from reaching the maximum temperature that triggers automatic shutdown. It is recommended to set this threshold to 5% below the automatic shutdown temperature allowed for your system.
- **System Auto Power On** setting, which enables automatic power-on of the system, in the event of restored power after power failure.

➔ To configure the General & Connection settings:

- Step 1.** At the top of the screen, tap . The **Monitor Tab** is displayed.
- Step 2.** Tap the **General tab**. The **General Settings** screen appears.

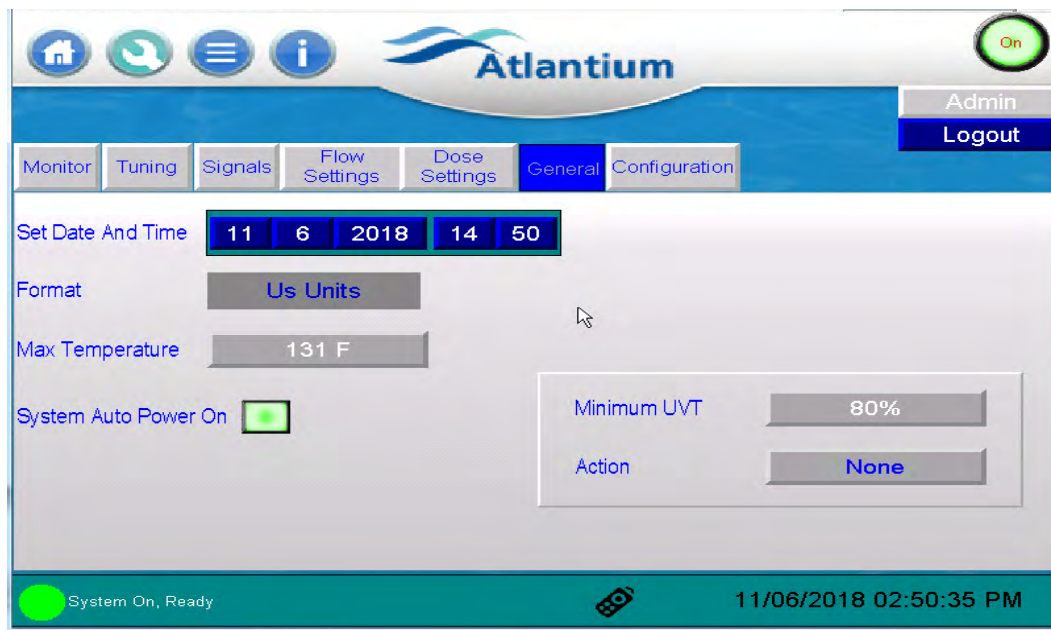


Figure 7-9: General tab

General Settings

- Step 3.** **Set Date And Time**, (US format: MM/DD/YYYY 23:59)
- Step 4.** Tap **Set**. The correct time appears on the **Output/Input** Tab and is set for the system.
- Step 5.** Set **Max Temperature**, (°F) range: 34 to 300 as integer number, the default value is 100. To set **Max Temperature**:
- a Set the system water flow to the minimum possible.
 - b Wait a few minutes for the system to stabilize.
 - c Check the on-screen temperature reading.
 - d Under **Max Temperature**, using the +/- buttons, set the maximum temperature threshold to the temperature reading in previous step, + 9°F. When this threshold is reached, a warning notification is sent.
- Step 6.** To enable the system to restart automatically when power is restored, mark **System Auto Power On** checkbox.

Afterwards, during system operation:

- When system temperature exceeds (**Max Temp** - 10)°F for more than 30 seconds, the system outputs a High Temp Warning signal. You must take action to prevent the water from reaching the temperature that triggers shutdown!
- When system temperature exceeds **Max Temp** for more than 30 seconds, the AB Control Module outputs a High Temp Shutdown Warning signal and shuts down the system automatically.

Minimum UVT

Step 7. For **Minimum UVT**, using the **-/+** buttons, set the threshold value for the UV transmission below which the system is to issue an alarm, warning of below-spec UVT.

Step 8. For **Action**, from the drop-down list, select the action to be performed if the factor falls below the Min UVT and becomes off-spec. The options are:

- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
- **Warning** - One or more warning messages are issued.
- **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a General Alarm.

7.8 Lamp Monitor Screen

Tap Monitor to open the Monitor tab of the AB Control Module (Figure 7-10).

Note that in this Figure all lamps are in Enabled state  and that lamps 1 & 2 are on .



Figure 7-10: Monitor Screen

Table 7-4: Lamp color indicators in Monitor Screen

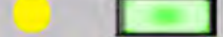
Color	Status	Output Signal
	Lamp On solid blue and Enabled	Ready signal after all lamps are ON
	Lamp In Ignition flashing blue and Enabled	Inlet Valve when Ignition Starts
	Lamp On Low Status solid yellow and Enabled	Low Status
	Lamp/Ballast Fail solid red and Disabled	Ballast/Fail
	Lamp Off solid black and Disabled	NA

Table 7-5: Lamp Monitor screen items

#	Item	Range	Description
1	Lamp ID	1-24	ID number for lamp
2	Power %	0-100%	Power of system. 0% when off, min power at Min Flow, otherwise at 100%
3	Status %	0-100%	% status of lamp, according to calculation
4	UVT %	0.0 - 100.0%	% UVT of lamp. Applicable when UVT sensor signal > 3.00mA
5	UVIS mA	0.00 - 30.00mA	Current generated by UVI sensor
6	UVTS mA	0.00 - 30.00mA	Current generated by UVT sensor
7	Lamp kW	0.0 - 30.0kW	Lamp power
8	Lamp V	0 - 2000V	Lamp voltage
9	Lamp A	0.0 - 30.0A	Lamp current
10	Temp °F	0 - 200°F	System Temperature
11	On/Off Indicator	 Off  On	Lamp Status
12	Lamp Enable Checkbox	 Disabled  Enabled	Check to enable ignition of that lamp

7.9 System tuning screen

Tap Tuning to open the Tuning tab of the AB Control Module:



Figure 7-11: Tuning Screen

Table 7-6: Tuning Screen Items

#	Item	Range
1	Lamp ID	1-24
2	Status %	0-100%
3	UVT %	0.0 - 100.0%
4	UVIS mA	0.00 - 30.00mA
5	UVTS mA	0.00 - 30.00mA
6	UVIS-T0 mA	0.00 - 30.00mA
7	UVTS-T0 mA	0.00 - 30.00mA
8	UVT-T0	0.0 -99.0
9		Scroll up/down lamp table

To perform tuning, see **System Tuning** on page 120.

7.10 System Information Screens

These 2 AB screens, System Info and Monitor, are accessible without password.

7.10.1 System Info Screen

Tap System Info to open the System Info tab of the AB Control Module:



Figure 7-12: System Info Screen

7.10.2 Operator Monitor Screen

Tap System Info to open the System Info tab of the AB Control Module.

Tap Monitor to open the Monitor tab of the AB Control Module.

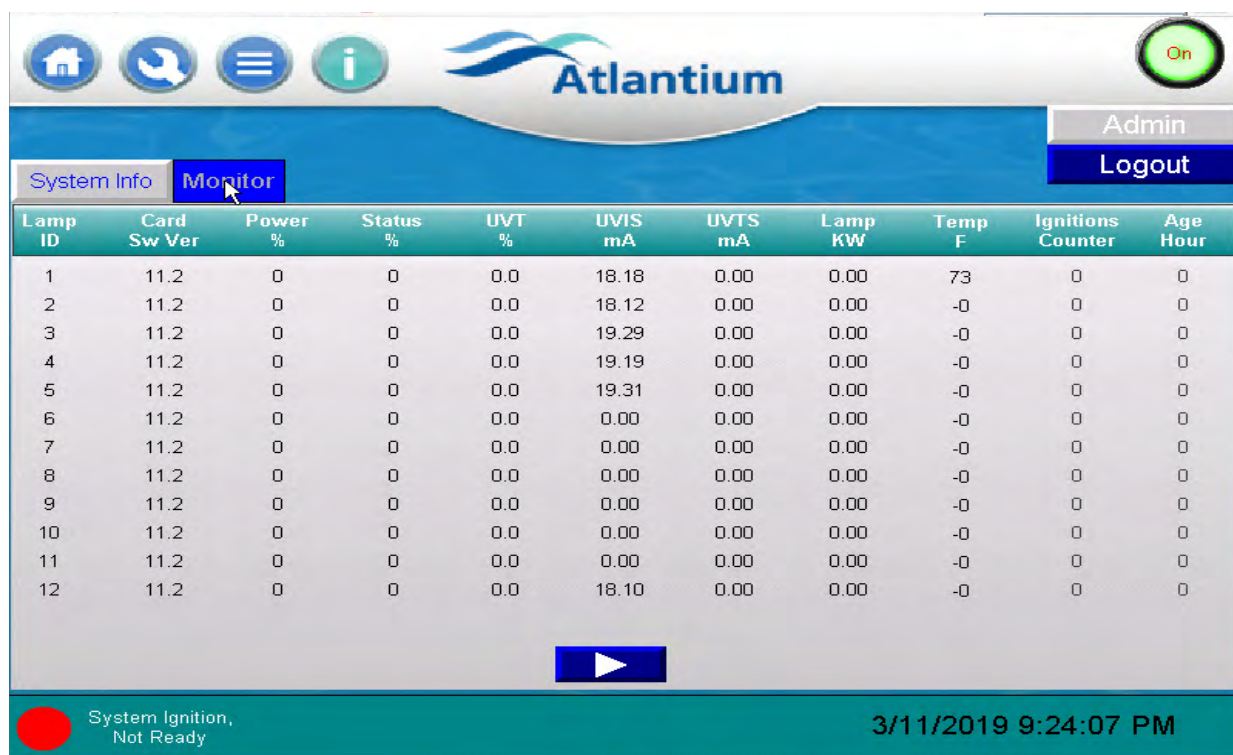


Figure 7-13: Operator Monitor Screen

Table 7-7: Operator Monitor Screen Items

#	Item	Range	Description
1	Lamp ID	1-24	ID number for lamp
2	Card SW ver	0.00 - 9.99	Software version of each lamp card
3	Power %	0-100%	Power of system.
4	Status %	0-100%	% status of lamp
5	UVT %	0.0 - 100.0%	% UVT of lamp
6	UVIS mA	0.00 - 30.00mA	Current generated by UVI sensor
7	UVTS mA	0.00 - 30.00mA	Current generated by UVT sensor
8	Lamp kW	0.0 - 30.0kW	Lamp power
9	Temp °F	0 - 200°F	System Temperature
10	Ignitions Counter	0 - 99999	Increased by 1 for every successful ignition. When a lamp is replaced, touch value to reset.
11	Age Hour	0 - 9999 hours	Lamp age, +1 for every hour that lamp is lit. When a lamp is replaced, touch value to reset.
12			Scroll up/down lamp table

7.11 AB Control Module Event Screen

See *Viewing the Alarm Log* on page [87](#)



8 First Time System Activation

Following **Control Module** configuration, completing the system commissioning includes the following major steps:

- Cleaning the Atlantium Unit (See below)
- Filling unit with water (See below)
- Igniting the UV Lamps (See below)
- Initiating the Water Flow (See page 83)
- Adjusting the Auxiliary Equipment (See page 83)
- System Tuning (See page 83)
- Installation QA (See page 84)

8.1 Cleaning the Atlantium Unit

The inner surfaces of the Atlantium Unit are to be cleaned via the CIP process at installation, before starting up and proceeding with the commissioning phase. Follow the directions in **Cleaning In Place (CIP)** on page 98.

8.2 Filling unit with water

The unit must be filled with water before full activation.

Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. **Make sure that water hammer is NOT present in your facility's water line.**



To prevent water hammer, consult your system engineer and check your facility system procedures, including:

- When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
 - Sudden valve closures can cause water hammer. Make sure that valves close gradually enough to avoid this problem.
 - Ensure that all pumps used with the Atlantium system employ soft start-up procedures.
-

➞ To introduce water:

- Open the **Inlet valve** and, using the **Viewport** on the Unit, check that the quartz chamber fills with water completely.

8.3 Igniting the Lamps

The lamps of the Atlantium Unit are ignited to commence live system operation.



- **Temporarily, the breather caps on the lamps must be left open to allow any condensation water to evaporate. During this time, set up Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
 - Wear white cotton gloves.
 - Wear long sleeves rolled down to your wrist.
 - Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
 - **DO NOT look directly into openings that emit UV light.**
-

➤ To ignite the lamps:

- Step 1.** On the **Control Module** Monitor screen, tap  and wait approximately 3 minutes for the **Ready Lamps On** signal to transmit.
- Step 2.** Check that the Main screen shows that system is working properly according to spec (see Figure 8-2)



Figure 8-1: Monitor Screen*

*For explanation of the Monitor screen, refer to Table 7-4 and Table 7-5 on page 77.

- Step 3.** Check lamp performance parameters. Tap  to scroll up/down lamp list.
- Step 4.** Let the system operate for at least one hour.
- Step 5.** Continue with Section 8.4 and Section 8.5 below while the lamps complete the first hour of operation.
- Step 6.** Close all of the lamps' breather caps and remove the Caution floor signs.

Figure 8-2: Main screen showing on-spec system (generic)

8.4 Initiating the Water Flow

When initiating the water flow, the water may be set to drain out or collect in a tank until the system is fully operational. Once the entire installation and QA testing is complete, the water is set to flow normally as part of the facility operation.

➤ To initiate the water flow:

- Step 1.** Open the relevant valve that allows water to flow through the Atlantium unit. This may be the draining valve or bypass valve.
- Step 2.** Open the viewport and verify that the Unit is completely filled with water and that no air gaps and no air bubbles exist.

8.5 Adjusting the Auxiliary Equipment

All relevant auxiliary equipment, i.e., flow meter, automatic valves, PLC, etc., must be adjusted at this time. Refer to the manufacturers' user documentation of the relevant equipment.

If your configuration includes a flow meter, check that the value of the flow displayed on the **Control Module** matches the reading on the flow meter. For flow meter calibration, see **Configuring the Flow Settings** on page 87.

8.6 System Tuning

The Atlantium system must be tuned via the **Control Module**. Follow the instructions in **Replacing/Checking a UVT Analyzer or UVIS Sensor** on page 127.

8.7 Installation QA

Atlantium personnel performs a quality assurance procedure (see **Installation Check List** on page 59) following which the Atlantium system is ready for regular facility operation.



9 Operating the Atlantium System

This section details how to operate the system and handle basic and periodic routines to be carried out by Atlantium trained local staff.

For system without **Cleaning system**, disregard the irrelevant sections.



WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

9.1 Operational Guidelines

Follow these guidelines for smooth Atlantium System operations:

- **Incoming Water Supply** - Be alert about incoming water supply. If you do not have pre-treatment, be alert to reports of contamination by upstream slugs of mud, particles, oil, etc.
If the water source was not previously in operation or the line is new, - flush the line prior to introducing water into the Atlantium system
- **Avoid chemicals that are harmful to UV systems** - Avoid caustic soda and other corrosive chemicals that can degrade gaskets and O-rings. For information on approved chemicals for CIP use, see **Selecting the Correct Chemicals for CIP** on page 116
- **Prevent Water Hammer** - Water hammer can cause damage to any hydraulic system, including the Atlantium unit. Take steps to ensure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check you facility system procedures for following:
 - Avoid filling empty lines with water without keeping valves open for releasing the air from the pipes and the system.
 - Avoid rapid closure of water lines by sudden closure of valves.
 - Before introducing the water, ensure that all pumps that impact the Atlantium system employ soft start-up procedures.
- **Avoid vibrations on upstream and downstream piping** - Make sure that the piping to be attached to the Atlantium Unit is reinforced to protect the Unit against vibration.
 - A high vibration level can cause damage to any hydraulic system, including the Atlantium unit.
 - High vibration can be caused by an unbalanced pump or by sudden valve close.

9.2 Basic Operational Tasks

- Shutting down the Atlantium system - See below
- Starting up the Atlantium system - See page 86
- Taking microbial samples - See page 93

9.2.1 Shutting Down the Atlantium System

You can shut down the Atlantium system via the **Control Module**.

➤ To shut down the Atlantium system:

- Step 1.** In the **Control Module**, on the top of the screen, tap . The button turns pink. Wait approximately three minutes for the **Lamps Off** signal to transmit.
- Step 2.** Check the **Main Operations** screen and verify that the system parameter for **Power** is **0%**.
- Step 3.** Using the viewport on the Atlantium Unit, check that the lamps are indeed turned off.

9.2.2 Starting up the Atlantium System

You can start up the Atlantium system via the **Control Module** and then check the water and system parameters.

➤ To start up the Atlantium system:

- Step 1.** Check that the power cable of the Ballast Modules is connected to the main power source.
- Step 2.** Connect the power cable of the **Control Module** to the main power source.
- Step 3.** Using the attached key, open the screen panel of the **Control Module**.
- Step 4.** Locate the circuit breaker and turn it to the **On** position.
- Step 5.** In the **Control Module**, on the top of the screen, tap . The button turns green. Wait 10 minutes for lamps to turn on and the system to stabilize.
- Step 6.** Open the viewport and check that there are no air bubbles in the water flow. If necessary, release all air bubbles from within the system.
- Step 7.** Check the **Control Module** Screens (See **Configuring the Control Module** on page 78.) and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within wanted acceptable range (see the table below to enter your expected values).

Notify the executive maintenance engineer in case the values are out of range.

Table 9-1: Abbreviation of Table in Appendix A. Checking the System Parameters

	Parameter	Expected Value	
		Minimum	Maximum
1	Power		
2	Flow Rate		
3	UVT		
4	UV Dose		
5	Lamp Status		



For periodic checks on Atlantium system parameters, a convenient check list is supplied in Appendix A, **Checking the System Parameters** on page 142.

9.2.3 Viewing System Information

➔ To view information about the Lamps:

Step 1. At the top of the screen, tap . The **System Info** screen appears (see Figure 7-3 on page 66).

Step 2. Tap the **Monitor** tab. The following information is displayed,:

- ◆ Lamp ID
- ◆ Power in %
- ◆ UVT %
- ◆ UVIS mA
- ◆ UVTS mA
- ◆ Lamp KW
- ◆ Lamp V
- ◆ Lamp A
- ◆ Temp F
- ◆ On/Off
- ◆ Enabled LED



Lamp ID	Power %	Status %	UVT %	UVIS mA	UVTS mA	Lamp KW	Lamp V	Lamp A	Temp F	On Off	Lamp Enabled
1	100	100	95.0	17.80	18.00	1.70	220	7.7	77	Blue	Green
2	100	100	0.0	17.80	0.00	1.70	220	7.7	-0	Blue	Green
3	100	100	95.0	17.80	18.00	1.70	220	7.7	-0	Blue	Green
4	0	0	0.0	4.00	0.00	0.00	508	0.0	-0	Red	Green
5	100	100	95.0	17.80	18.00	1.70	220	7.7	-0	Blue	Green
6	0	0	0.0	4.00	0.00	0.00	504	0.0	-0	Black	Green
7	100	100	0.0	17.80	0.00	1.70	220	7.7	-0	Yellow	Green
8	100	100	0.0	17.80	0.00	1.70	220	7.7	-0	Blue	Green
9	0	0	0.0	0.00	0.00	0.00	0	0.0	0	Black	Green
10	0	0	0.0	0.00	0.00	0.00	0	0.0	0	Black	Green
11	0	0	0.0	0.00	0.00	0.00	0	0.0	0	Black	Green
12	0	0	0.0	0.00	0.00	0.00	0	0.0	0	Black	Green

Figure 9-1: AB Monitor Screen*

*For explanation of the Monitor screen, refer to Table 7-4 and Table 7-5 on page 77.

9.3 Viewing the Alarm Log

Every defined event /alarm is written to the Alarm Log of the **Control Module**. alarms with alerts also trigger an alert icon at the bottom of the screen. You can acknowledge each alarm or acknowledge all together.

➔ To view the alarm log and acknowledging alarms:

- Step 1.** At the top of the screen, tap  the **Alarm Log** screen is displayed.
- Step 2.** To navigate through the log, tap the up/down arrows ▲▼

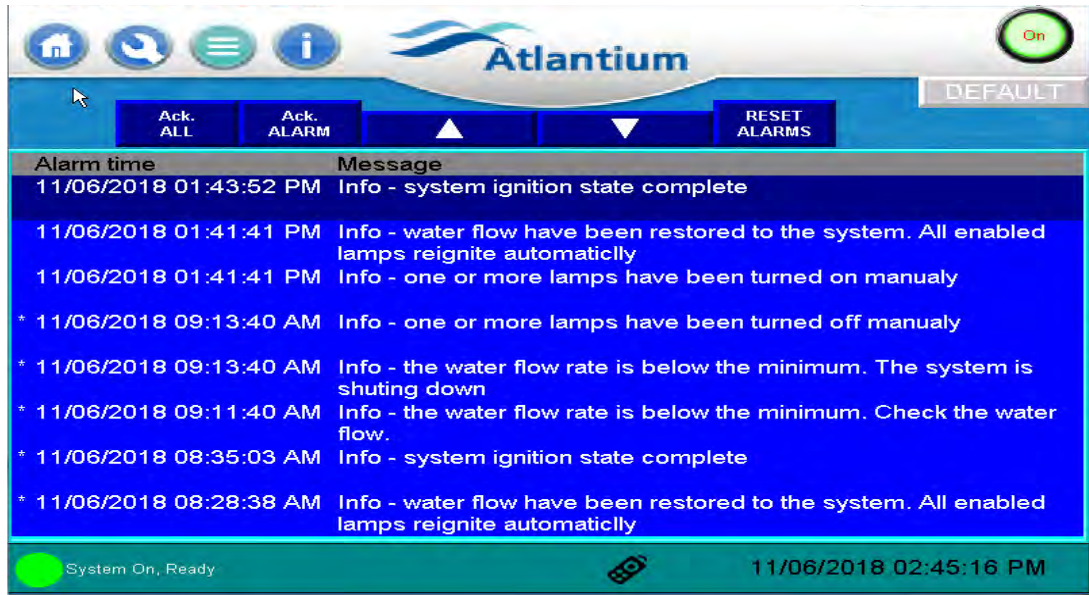


Figure 9-2: AB event log screen

- Step 3.** To view more information about a single alarm, tap the alarm. The alarm item expands to display the full information, including reading for System Power, Dose, Flow, Status, UVT, and Temperature at the time the alarm occurred. When expanding an alarm with an alert, the alert is acknowledged.
- Step 4.** To acknowledge an alarm without opening it, tap Ack Alarm.
- Step 5.** To quickly acknowledge all open alarms, tap Ack All
- Step 6.** To reset all alarms, tap Reset Alarms.

Table 9-2: AB alarms Screen Items

#	Item	Description
1	Alarm time	Alarm date and time
2	Message	Brief description of alarm
3	Ack. ALL	Acknowledge all alarms
4	Ack. ALARM	Acknowledge current highlighted alarm
5		Scroll up/down
6	RESET ALARMS	Reset all alarms

9.3.1 AB Control Module Alarm List

The following is a list of alarms/warnings displayed on the AB Control Module:

Table 9-3: List of AB Control Module Alarms

#	Alarm Title	Alarm Severity	Alarm Info Text	Data	Remarks
1	Dose is below the minimum!	CONFIGURABLE	The dose too low! It is below the minimum defined in the system settings.	System	When the calculated dose is lower than the minimum dose settings for at least of 30 second.
2	Temperature is too high!	CAUTION	The system's temperature is too high -- check the water flow.	System	When the system temperatures is above the max temp settings - for at least 30 seconds.
3	Temperature is critically high!	SHUTDOWN	The System temperature is above the maximum defined in the system settings. The system is shutting down.	System	When the system temperature is above the max temp settings for at least 30 seconds.
4	Temperature sensor fault	CAUTION	One of the temperature sensors has malfunctioned or the communication with the Connection Box card was lost.	Lamp	
5	Flow rate is below the minimum!	INFO	The water flow rate is lower than the minimum flow rate defined in the system settings. Check the water flow.	System	
6	UVT is below the minimum	CONFIGURABLE	The UVT is lower than the minimum UVT defined in the system settings.	System	
7	Lamp ignition failed	CAUTION	A UV lamp has failed to ignite. Check if it needs to be replaced, or if there is a faulty connection or ballast malfunction.	Lamp	
8	System turned on manually	INFO	One or more lamps have been turned on manually.	System	Manually: via the main screen button
9	System turned off manually	INFO	One or more lamps have been turned off manually.	System	Manually: via the main screen button

Table 9-3: List of AB Control Module Alarms

#	Alarm Title	Alarm Severity	Alarm Info Text	Data	Remarks
10	System tuned on automatically - power restored	INFO	Power was restored to the system. All enabled lamps reignite automatically.	System	When the system is on and suddenly the Control Module loses power. When power to CM is resumed it will turn lamps back on automatically
11	System turned off automatically - water flow too low	INFO	The flow rate is below the minimum defined in the system settings. The system is shutting down.	System	When the flow rate is below the minimum flow setting for interval $\geq$ the set Flow Delay, the system will shut down. The Flow Delay counter starts when all lamps are ignited.
12	Lamp ignition stage complete	INFO	System ignition stage complete.	System	When all lamps are ignited.
13	System Temperature sensors fault	SHUTDOWN	All temperature sensors in a system branch have malfunctioned or communication with Connection Box cards is lost. The system shuts down.	Lamp	
14	Unexpected lamp shutdown	CAUTION	A lamp shutdown unexpectedly.	Lamp	When a lamp shuts down a warning message is issued. Try to reignite the lamps after 3 minutes and try after 3 minutes to turn on the lamp(s). Via the Lamp communication card includes a bit for on/off indication of lamp.
15	Ballast or lamp malfunction	CAUTION	A ballast or lamp has malfunctioned. Check for faulty connection, ballast or lamp.	Lamp	See the lamp ignitions for more details of lamp/ballast fault
16	Communication error - Single Connection Box/Control Module	CAUTION	Communication lost between one of the Connection Boxes and the Control Module.	Lamp	

Table 9-3: List of AB Control Module Alarms

#	Alarm Title	Alarm Severity	Alarm Info Text	Data	Remarks
17	Control Module turned on	INFO	Power is restored to the Control Module.	System	This signal is sent every time the Control Module is powered up.
18	System turned on externally	INFO	One or more lamps have been turned on via external signal.	System	Or via the mod bus communication
19	System turned off externally	INFO	One or more lamps have been turned off via external signal.	System	Or via the mod bus communication
20	System turned on automatically - Water flow restored	INFO	Water flow was restored to the system. All enabled lamps reignite automatically.	System	In restart mode only
21	Communication error - All Connection Boxes/ Control Module	SHUTDOWN	There is a communication loss between all of the Connection Boxes and the Control Module. The system is shutting down.	System	.
22	System turned on automatically - communication restored	INFO	Communication was restored to the system. All enabled lamps reignite automatically.	System	When the system is on, and suddenly the Control Module loses communication with one or all lamp cards. When communication returns, it will turn re-ignite lamps automatically

9.4 Taking Microbial Samples

The designated aseptic sampling valves (i.e., EGMO ESV valves) are the **Inlet** and **Outlet Sampling Points** located on the sides of the Atlantium Unit. Before taking a microbial sample, a valve is sterilized internally with Ethanol 70%, as well as with flame.

Required Equipment

- Sampling containers that are clean, laboratory-grade, sampling bottles with a volume no less than 100ml (3.38oz).
- Ethanol 100-70%
- Bunsen Burner and lighter
- Permanent marker
- Sterile gloves
- Proper Sampler attire according to local laboratory protocol



CAUTION!



- Use gloves during sampling and treat them with Ethanol 70%.
- Use **sterile disposable** sampling containers.
- Prevent contact between the sampling containers and the valves' surfaces.
- Be sure that no foreign matter or liquids fall into the sampling containers.
- Perform the sample collection as fast as possible to minimize environmental contamination of the sample.
- **Use open flame to treat the surrounding air while sampling.**

➤ To take a microbial sampling:

- Step 1.** Open the Sampling valve and allow water to flow for 2-3 minutes. Then close the valve.
- Step 2.** If the sampling valve contains a cover at the top, open its cover and place it in a clean place.
- Step 3.** Spray all over the Sampling valve and inside it thoroughly with Ethanol 70%.
- Step 4.** Using a Bunsen burner, apply the flame comprehensively over the entire Sampling valve surface from its connection to the main pipe to its spout.
- Step 5.** Spray the cap thoroughly with Ethanol 70% and place it on the Sampling valve.
- Step 6.** Open the valve and allow the water to flow for 2 minutes. If you have applied the flame properly, vapor emerges from the spout. (If there is no vapor, application of the flame was insufficient. Repeat the previous step.)



Figure 9-3: Cleaning the Sampling Valve with Ethanol

ATTENTION!

Make sure never to touch the opening at the Sampling container's neck.
While the Sampling container is open, do not turn the cap upside down.

- Step 7.** On the Sampling container, with one hand, remove its cap carefully, avoiding touching the inside of the cap with fingers or any other object. **Do not turn the cap upside down.**
- Step 8.** With your other hand position the Sampling valve under the water stream and take a water sample. Make sure never to touch the valve's opening or the opening at the container's neck. Fill the sampling container, leaving ample air space in bottle (at least 2.5 cm).
- Step 9.** Carefully replace the container's cap and tighten it to avoid cross contamination.



Figure 9-4: Taking a Sample

- Step 10.** Label the Sampling container with the following data:
- Sampling site
 - Sampling point
 - Sampler name
- Step 11.** Close the Sampling valve.
- Step 12.** Spray the sampling container thoroughly with Ethanol 70%.
- Step 13.** Store the water samples in a dark, refrigeration unit at a temperature of 2-8°C (35.6-46.4°F) until they are examined.

9.4.1 Microbial Analysis Guidelines

The microbial analysis is to be conducted according to the company guidelines which are used in the local microbiology laboratory.

- Atlantium's recommendation is to use the microbial filtration method for the testing.
- If a number of consecutive anomalous microbiological results are obtained in one of the sampling points, take the samples only after performing CIP with a disinfection chemical (peracetic acid is recommended). It is important to let the chemicals clean and sanitize the aseptic valves, by letting the cleaning solution flow through them. See **Cleaning In Place (CIP)** on page 116.
- In the event that system outlet counts are higher than the inlet counts, the results are to be considered as a sampling error.
- In the event that the microbial counts are not proportional in different dilutions, the results are to be considered as a laboratory error.
- For further guidance regarding microbial sampling and analysis, contact the Atlantium Application department.



10 System Maintenance

To keep the Atlantium system in peak form, preventive and periodic routine maintenance is essential. During commissioning, Atlantium trains your local staff how to do the tasks on a prescribed schedule or as needed.



WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.



Before you begin a procedure, carefully read through it so that you can anticipate the steps efficiently.

The periodic routine maintenance schedule is:

- Weekly Maintenance Tasks:
 - Walk-Around Inspection Tour - see below
- Monthly Maintenance Tasks:
 - Checking the Ballast Module Cooling Fans (See page 103)
- General Lamps, Sleeves Maintenance Tips (See page 103)
- Yearly Maintenance Tasks (or as needed):
 - Cleaning the Quartz Sleeve (or Replacing as needed) and Replacing the Quartz Sleeve O-rings (See page 111)
- Maintenance Tasks Performed as Needed:
 - Cleaning In Place (CIP) (See page 95)
 - Replacing a Lamp (See page 103)
 - Checking/Replacing a Sensor (See page 119)
 - Tuning the Atlantium System (See page 120)
 - Checking/Replacing the Quartz Sleeve (See page 111)
 - Cleaning/Replacing a UVIS Wave Guide (See page 108)
 - Cleaning/Replacing a UVIS Mirror (See page 110)
 - Temperature Sensor Maintenance
 - ◆ Replacing the Temperature Sensor (See page 122)
 - ◆ Replacing the Temperature Converter Card (See page 123)

10.1 Walk-Around Inspection Tour



Walk-Around and inspect the Atlantium system once a month to check for water leaks, visible damage to the cables and harnesses, or any other part of the system. Report any problem you find to an authorized technician.

➔ To perform a Walk-Around inspections:

- Step 1.** Walk around the Atlantium unit and inspect it. Look for leakage at the incoming and outgoing pipes.
- Step 2.** Inspect the **Cable Harness** and cables running between the Unit and the Ballast Module(s) and **Control Module** for signs of fraying and any other visible damage.
- Step 3.** Inspect the Ballast Module(s) and **Control Module** for any signs of damage.
- Step 4.** Report any leakage or damage to an authorized technician immediately.
- Step 5.** Inspect the Atlantium Unit. If you see any corrosion, clean the systems' parts by washing the parts with soap water and scrubbing it with an abrasive cloth, such as Scotch Bright. Spray the Unit with anti-corrosive material. See on page 121.

10.2 Cleaning In Place (CIP)



The CIP procedure consists of two parts:

- Cleaning to remove the accumulated deposit (particles) that collect on the quartz sleeve over time
- Disinfectant cleaning to sterilize the quartz sleeve (not applicable to some Municipal installations)

Clean the Atlantium Unit's inner surfaces and quartz sleeve generally on a periodic basis that is determined by observing that the performance is on target and the UVT Sensor readings are stable. You can clean using this process less frequently, depending on the specific chemistry of your water.

A CIP recirculation system is available from Atlantium as an optional CIP Kit. Connect the recirculation system to the CIP ports to create a closed-loop flow line that pumps a diluted solution through the Unit.



- In facilities where CIP is performed on the plant level, circulating through all piping, the CIP process is the same for the Atlantium System. The inlet and outlet valves referred to in these sections are irrelevant.
- If the Atlantium Unit(s) is/are integrated into the facility's CIP loop, ensure that the water flow rate through the Unit is at least 60 m3 per hour.
- In facilities with plant-wide CIP, perform CIP with the Atlantium System running.

The estimated time for a CIP process is 1.5 hours.

Prepare the following:

- Protective chemical-safe goggles, clothing and gloves
- An external pump-driven recirculation system with a built-in 100 liter reservoir (26.5 US gallons) or Atlantium's CIP Kit including its accessories (See **Sampling Valve Components and Kits** on page 11.)
- Scalant cleaning solution
- Disinfectant cleaning solution (if disinfection is planned)

10.2.1 Selecting the Correct Chemicals for CIP

It is **critically important** to select the correct cleaning and disinfecting chemical for the CIP process. It must be powerful enough to clean the inner surfaces of the Atlantium unit, while at the same time:

- Does not harm the components of the Unit that it comes into contact with (stainless steel 316, quartz and Viton® o-rings).
- Does not contaminate the product for which the water being treated by the system is used.



WARNING!



- The chemicals selected and the concentrations used must be approved for use by the facility's quality assurance and safety officials, and must comply with the written protocols and procedures set in place by those departments.
- Use and handle in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- Use of NaOH (Caustic Soda) may cause irreversible damage to the quartz. Use of this material requires an analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.



- All chemicals must be obtained locally.
- Atlantium does not supply chemicals - get them locally from commercial chemical suppliers.

Recommended Scalant Cleaning Solutions

The following chemicals have been tested and found to be very effective for cleaning and are recommended for use when cleaning the Atlantium system:

- Phosphoric acid (H_3PO_4) 2% diluted solution
- Nitric acid (HNO_3) 2% diluted solution
- HCl 2% diluted solution - recommended for metal based (rust-colored) deposits
- Sulfamic acid (H_3NSO_3) 1.5% diluted solution - recommended for calcium based deposits

Recommended Disinfectant Solutions

(not applicable to some Municipal installations)

(The Disinfectant CIP process is not applicable to some Municipal installations)

The following have been tested and found to be very effective for disinfection and are recommended for use when disinfecting the Atlantium system:

- Peracetic acid-based chemicals - 30 min. contact time, for example:
 - P-3 Oxonia Active 150 - 0.5% diluted solution according to manufacturer's recommendation
 - Divosan Forte - 0.5% diluted solution according to manufacturer's recommendation
- Hot water at 85°C (185°F) for 30 min.



WARNING!

- It is the responsibility of the Atlantium system's owner/operator to use and apply the correct chemicals.
- The use of any other cleaning and disinfection chemicals requires Atlantium's approval.



10.2.2 Preparing the Chemical Solution

For both cleaning and disinfecting, the liquid in the reservoir must contain the desired percentage of the chemical additive (see above). Therefore, the purchased chemical solution must be diluted and added to the reservoir. Calculate the correct amount of chemical solution to be used for the CIP process.



Protect yourself from the ill-effects of chemicals with protective chemical-safe goggles, clothing and gloves.



- The concentrations refer to room temperature.
- The diluted solution values denote the required dilution of the chemical circulating through the Atlantium Unit.
- While calculating the diluted volumes, consider all volumes including the Unit, pipes and the recirculation reservoir.

➔ To calculate the correct amount of chemical solution:

Step 1. Calculate the total volume you need for the CIP* by adding the water volumes in:

- The Atlantium Unit volume (measurements in liters/gallons) - Refer to the **Physical Specifications** Table in **Atlantium System Footprint** on page 15
- The pipes between the Atlantium Unit and the **Inlet** and **Outlet Valves**
- The CIP recirculation system
- The hoses between the Unit and the recirculation system

Step 2. Calculate the amount of chemical additive to be added to the reservoir to achieve the desired dilution. Use this formula:

$$C = \frac{A * V_t}{B}$$

A - The target concentration for the liquid (chemical substance + water) in %
B - The concentration of the chemical substance to be added in %
C - Volume of the chemical substance to be added in liters/gallons
V_t - Volume of the water in the Unit + piping + CIP Reservoir (as described above*) in liters/gallons

OR:

Assume that you need a 2% diluted solution in the CIP process and assume that your purchased solution is of a 35% concentration in a 4-liter container.

Calculate according to this calculation: [200 liter X 2%] / [35% X 4 liter] = [400] / [140] = 2.9 = 3 containers.



CAUTION!

* If you intend to collect the solution into the CIP reservoir or other container, the total volume of the cleaning solution must not exceed the volume of the CIP reservoir or container.

Be sure to obtain from an engineer the correct calculation for the amount of chemicals to be used for the process you intend to use.

10.2.3 Performing CIP



CAUTION!

The anti-scaling procedure is performed only if necessary. Atlantium recommends that anti-scaling always be followed by disinfection.

The CIP process is performed utilizing fresh water. For systems with seawater, drain the system and rinse before performing CIP.



- In facilities where CIP is performed on the plant level, circulating through all piping, the CIP process is the same for the Atlantium System. The inlet and outlet valves referred to in these sections are irrelevant.
- If the Atlantium Unit(s) is/are integrated into the facility's CIP loop, ensure that the water flow rate through the Unit is at least 60 m³ per hour.
- In facilities with plant-wide CIP, perform CIP with the Atlantium System running.

This section details the CIP process with recommended steps for carrying it out. The procedure is to be adapted according to the specific circumstances of your installation.

The CIP process is to follow these major steps:

(Refer to the next section for the detailed description on the CIP process.)

- Step 1.** Close the inlet and outlet valves and immediately shut down the Atlantium system
- Step 2.** For systems with seawater, if needed, drain the system
- Step 3.** Remove the caps covering the CIP ports
- Step 4.** Connect the supply hose to the bottom CIP port and connect the return hose to the top CIP port
- Step 5.** Fill the CIP reservoir with fresh water or seawater
- Step 6.** Starting the circulation
- Step 7.** Adding the cleaning solution (See **Selecting the Correct Chemicals for CIP** on page 95.)
- Step 8.** Running the CIP circulation for at least 30 minutes or as specified according to the chemical manufacturer
- Step 9.** Rinsing the system and the CIP cart
- Step 10.** Performing CIP Disinfection with the disinfection solution
- Step 11.** Ending CIP

The CIP diagram below is followed by the full recommended CIP procedure.

CIP Setup

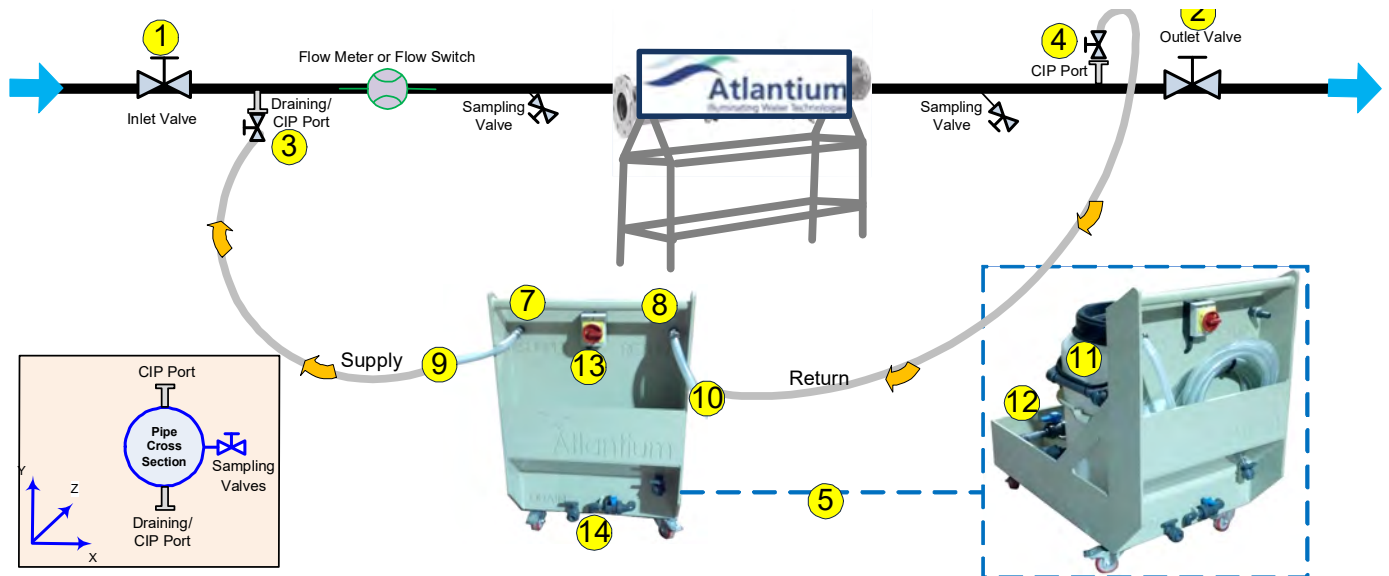


Figure 10-1: Recommended CIP Setup



If the installation is vertical, the water flow is from bottom to top. The Draining/CIP port (3) is on the bottom, under the Atlantium Unit and the CIP port (4) is above the Atlantium Unit.

➔ To Perform CIP:

ATTENTION!

If you are only performing disinfection: Follow Steps 1 through 20. Then resume with Step 22.

If you are performing both procedures: Follow all of the steps below.

Shutting Down the System

- Step 1.** Close the Unit's inlet isolating valve (1) and outlet isolating valve (2). Immediately proceed to the next step.
- Step 2.** Shut down the Atlantium system, on the **Control Module**, by tapping the **Main Unit Operation Button** **Operation** button in the upper-right corner.
- Step 3.** On each CIP port, remove the cap's clamp and remove the caps covering the CIP ports (3 & 4). Place the caps in a safe place for reuse later. Verify that the gasket is still inserted into the ferrule.

ATTENTION!

Make sure that the hose fittings are secure and tight.

Water must fill the entire path through the piping on both sides of the Atlantium Unit and the Unit itself, as well as the Supply and Return hoses and the CIP Kit's reservoir.

Connecting the CIP Hoses

Step 4. Remove the hoses from their pocket on the Cart.

Step 5. Using the CIP cap's clamp, connect the CIP Kit's Supply hose to the bottom CIP port **(3)**. Verify that the gasket is present.

Step 6. Using the CIP cap's clamp, connect the CIP Kit's Return hose **(10)** to the top CIP port **(4)**. Verify that the gasket is present.

Filling the CIP Reservoir

Step 7. To fill the CIP reservoir **(A)** with water:

■ Filling option 1:

- Slightly open the Unit's inlet isolating valve **(1)** and open the top CIP port on the outlet side **(4)** and allow the facility's water to fill the CIP Kit's reservoir **(11)**.
- There is a full line indicator on the reservoir **(11)**. Make sure the water in the reservoir is up to the 100 liter (26.5 US gallons) level.

◆ Close the Unit's inlet isolating valve **(1)**.

◆ Close the CIP port **(4)** valve.

■ Filling option 2:

- Open the top cover of the CIP Kit's reservoir **(11)**.
- Using a hose of an external water source, fill the reservoir with 100 liter reservoir (26.5 US gallons of room-temperature tap water ($\pm 25^{\circ}\text{C}/75^{\circ}\text{F}$)).
- There is a full line indicator of the reservoir. Make sure the water in the CIP Kit's reservoir **(11)** is at the 100 liter reservoir (26.5 US gallons) level.
- Close the cover.

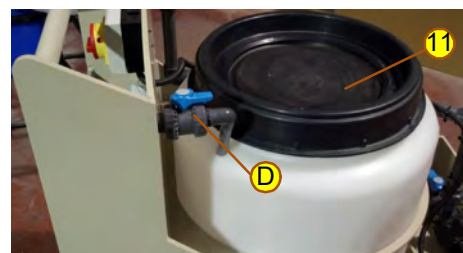


Figure 10-2: CIP Kit's Reservoir



Figure 10-3: CIP Cart's Inlet and Drain Port



WARNING!

Observe chemical safe rules! Follow all safety recommendations in the chemical manufacturer's product information and MSDS publications!

Adding the Chemical Solution



Protect yourself from the ill-effects of chemicals with protective chemical-safe goggles, clothing and gloves.

Step 8. Open the top cover of the CIP Kit's reservoir **(11)** and add the prepared chemical solution to the water in the reservoir. (See **Selecting the Correct Chemicals for CIP** on page 95.)

➔ Starting the CIP Circulation Process

ATTENTION!

The Atlantium Unit must be absolutely full for the solution to reach all of the inner surfaces.

Step 9. If the bottom and top CIP ports (**3 & 4**) are valves, open them.

Step 10. To start the CIP recirculation pump, on the CIP cart, turn the switch (**13**) to the **On** position and allow the solution to circulate through the Unit for **the period of time specified in the chemical's Use instructions, but not less than 30 minutes**.

Step 11. Verify that the Atlantium Unit is full of water and that the water is circulating correctly.

Step 12. During the circulation period open and close the sampling valve several times to assure chemical contact with the CIP chemicals.

If the draining valve is separate from the CIP ports, open and close the draining valve.

Rinsing the System and CIP Kit

After the chemical solution has circulated through the Unit for a sufficient duration, the chemicals solution must be rinsed from the system.

Step 13. Connect the CIP recirculation system's drain valve (**B**) to the facility's sewage system. See **Alternatives for Draining the Water with Cleaning Solution** on page **102**.

Step 14. To turn on the Atlantium system, on the **Control Module**, by tapping the **Main ON/OFF Operation** button in the upper-right corner.

ATTENTION!

Before performing the next step, verify that the lamp(s) are on and the system has stabilized and indicates it is ready. If not, restart the lamp(s) again and wait until the system has stabilized and indicates it is ready.

Step 15. Close the inlet CIP port (**3**), stop the recirculation pump (**12**) and open the CIP recirculation system's drain valve (**C**). Immediately slightly open the inlet valve (**1**) to rinse the Atlantium system thoroughly.

Allow this rinsing process to operate until at least three cycles of water volume have gone through the Atlantium Unit to remove all of the scalant cleaning/disinfectant solution.

Step 16. To assure that all the chemicals are flushed out, use a measurable indicators, such as measuring the pH, or other indicators depending on the chemical being used (consult your chemical supplier).

Step 17. Slightly open the outlet valve (**2**) and close the CIP port on the outlet side (**4**). To avoid hammer effect if there is air trapped in the system, carefully open the inlet valve (**1**) completely. Resume normal operation.

Rinsing the CIP Reservoir

After the chemical solution has circulated through the Unit for a sufficient duration as specified by the chemical supplier (typically 30 minutes), the chemicals solution must be rinsed from the system by the steps below.

Step 18. Open the CIP cart's drain valve (**C**) and completely drain the CIP reservoir (**11**).

Step 19. Using an external water source, rinse the CIP cart's reservoir, letting the rinse water drain out of the CIP cart's drain valve (**C**).

Step 20. Close the CIP cart's drain valve (**C**).

Performing CIP Disinfection

Step 21. If you are planning to perform an additional CIP process using the disinfectant solution, refer to **Preparing the Chemical Solution** on page **97** and follow the CIP process again from **Filling the CIP Reservoir** in **Step 7**. above.

Or,

To return to normal operating status, proceed to the next step.

➤ Ending CIP

Step 22. To drain out any water inadvertently collected in the CIP cart's storage pocket or base, use the CIP cart's drain valves **(A and B)**.

Step 23. Roll up the CIP Kit's hoses and return them to their storage pocket **(F)**.

Refilling the System

ATTENTION!

Make sure that the Atlantium Unit is filled with water before powering it up and turning on the lamps.

➤ Returning Power to the Lamps

Step 24. On the Connection Box next to each lamp, turn the switch to the **On** position.

Step 25. Turn on the Atlantium system, on the **Control Module**, by tapping the **Main ON/OFF Operation** button in the upper-right corner.

Step 26. Turn on the Atlantium system at 100% power and wait until the system has stabilized and indicates that it is ready.

Alternatives for Draining the Water with Cleaning Solution

■ **If local regulations permit, and a physical drain setup exists**, drain the cleaning solution to the facility's sewage system:

- a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.
- b With the Atlantium System operating, slightly open the Inlet valve.
- c Close the inlet bottom CIP port **(3)**.
- d To stop the CIP recirculation, on the CIP cart, turn the switch **(13)** to the **Off** position.
- e Open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.
- f Connect the CIP cart's drain valve **(C)** to the facility's sewage system.

Or,

■ **According to local regulations, use the Draining/CIP port to drain the solution in the system into a external container.**

Or,

■ **Use the CIP reservoir (11) to collect the chemical solution.**

- a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.
- b With the Atlantium System operating, close the inlet bottom CIP port **(3)**.
- c To stop the CIP recirculation pump, on the CIP cart, turn the switch **(13)** to the **Off** position.
- d Allow the chemical solution to drain entirely into the CIP reservoir **(11)**.
- e Dispose of the contents according to regulations. Connect a drain pipe to the CIP cart's drain pipe and open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.
- f Disconnect and close the CIP cart's drain valve **(B)**.

10.3 Checking the Cooling Fans



For Ballast Module B, once a month the cooling fans of the Ballast Modules must be checked to verify that the air is flowing out.

➔ To check the cooling fans:

- Check that air is being blown out of the fan on top of the Ballast Module.

10.4 General Lamps Sleeves Maintenance Tips

The following are some tips that are meant to help you understand maintenance requirements:

- Lamps and quartz sleeves traverse the width of the Unit and are held in place by holders and connectors at both sides. For maintenance, you can access the lamps and quartz sleeves from either side. Depending on the constraints of the installation (sometimes one side is less accessible) you can usually decide from which side to access the them.
- To access the quartz sleeve, you must remove the lamp housed within it.
- On rare occasions, quartz sleeves and their o-rings may need to be replaced because of a maintenance issue that arises.

10.5 Replacing a UV Lamp



The UV lamp can be used as long as it provides the required dose and efficient service. (Generally, a lamp service status of 80% or more is considered efficient service.) A lamp should be replaced **ONLY** when necessary.

Each lamp's performance is measured directly by a dedicated UV Intensity Sensor. Each lamp's performance is measured directly by a dedicated UV Intensity Sensor. Replace a lamp when:

- Its status or performance declines and the lamp no longer provides the required performance at the appropriate electrical cost
- Once its rated hours expire, be alert to any changes in efficiency that could indicate the need to change the lamp

In the **Control Module**, you can set a margin to assure notification as status declines and a warning is sent to the operator at both the user setting level and the internally coded minimum.



Generally, a lamp is replaced after shutting down the entire Atlantium system (See **Shutting Down the Entire System** on page 105.). For systems with multiple lamps, in a facility where there is no redundant UV purification, and you require minimizing system downtime, except for a need to replace lamp 1 or 2, the system may continue to operate while any one of the other lamps is being replaced. **However, in any dual lamp chamber, both lamps in the lamp chamber must be shut down before replacing one of them** (See **Shutting Down a Pair of Lamps** on page 105.)

For systems with dual lamps, in a dual lamp chamber, both lamps in the lamp chamber must be shut down before replacing one of them.



CAUTION!

- Allow the old lamp at least 10 minutes to cool down before starting procedure.
- Read carefully the instructions provided in the new Lamp package.
- Wear white cotton gloves.
- Before installing a new lamp, verify that it is clean and free of grease. If not, rub it gently with the provided lens tissue. Never touch the lamp with your bare hands - hold it only by the end-connectors. Oily residues from fingerprints can damage the lamp.



WARNING!

- The UV lamps are designed with high internal positive pressure. Wear protective eye-wear while replacing a lamp. Pay strict attention to the safety warnings and precautions in the **Safety Overview** on page 34 in the front of this manual.
- In case of lamp breakage, refer to the Safety instructions in **Safety Overview** on page 34.



CAUTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

10.5.1 Lamp Replacement Options

- Replacing a Lamp after System Shutdown (See page 105)
- Replacing a Lamp while System is Operating (See page 105)

Generally, a lamp is replaced after shutting down the entire Atlantium system. However, in a facility where there is no redundant UV purification, and you require minimizing system downtime, you can replace a lamp while other lamps continue operation provided you follow these safety requirements:

Replacing a Lamp after System Shutdown

Prepare the following:

- Replacement lamp
- Lamp cleaning cloth (comes with the lamp)
- White cotton gloves
- Eye protection
- #3 Allen wrench
- #4 Allen wrench
- Caution signs

10.5.2 Shutting Down the Entire System

Generally, a lamp is replaced after shutting down the entire Atlantium system.

➞ To shut down the entire system:



A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp's switch before you proceed to replace the lamp.

Step 1. For the lamp that is to be replaced, on the connection box, turn the lamp switch to the **Off** position.

Step 2. Continue with the procedure, **Replacing the Lamp** on page 105.

10.5.3 Shutting Down a Pair of Lamps

When replacing a lamp while the rest of the system is still in operation, you must follow these safety requirements:



- Shutdown the electricity supplied to the lamp to be replaced.
- **During the time when the lamp's caps are removed, use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards. The UV lamps are designed with high internal positive pressure.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

When replacing a lamp, you must shut down power to both lamps of a pair associated with single Connection Box.

➞ To shut down electricity to a pair of lamps:



- A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp's switch before you proceed to replace the lamp.

Step 3. For the lamp that is to be replaced, on the connection box, turn the lamp switch to the **Off** position. Both lamps in the corresponding lamp chamber are shut down.

Step 4. Continue with the procedure, **Replacing the Lamp**, below.

10.5.4 Replacing the Lamp

Once the electricity is turned off for the system or for the specific pair of lamps (at least one of which is to be replaced) and preparations are made (see above) you can replace the lamp.

➔ To replace a lamp:

Step 1. If you are replacing a lamp while other lamps are still in operations, follow these safety requirements:



- During the time when the lamp's caps are removed, use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into the lamp enclosure during the entire lamp replacement procedure.**

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

ATTENTION!

For systems that contain wave guides, **be sure to access the lamp from the side that contains the sensors.** The wave guide is fragile and may be damaged easily if the lamp is inserted from the opposite site.

Removing the Faulty Lamp



The lamp is hot! DO NOT touch the lamp with bare hands, particularly the quartz portion.

Wait at least ten minutes until the lamp has cooled down or use an appropriate pair of long-nose pliers to carefully pull out the lamp.

Cleaning/Replacing the UV Intensity Wave Guide or Mirror

Step 2. If you need to clean or replace the **UV Intensity Wave Guide** or **Mirror**, do so now before installing the new lamp.

- For wave guide, refer to *Cleaning/Replacing the UV Intensity Wave Guide* (on page 108)
- For mirror, refer to *Cleaning/Replacing the UV Intensity Sensor's Mirror* (on page 110)

Installing the Replacement Lamp

Step 3. Unpack the new UV lamp from its vacuumed sealed package.

Step 4. Make a note of the serial number of the new lamp.

Step 5. Clean the new lamp using the cleaning pad provided with the lamp. Verify that no visible fingerprint or dust is on the lamp.

Step 6. Insert the new lamp **(F)** into the quartz sleeve, all the way until it stops. Ensure that the male contact point **(G)** on the end of the lamp has meshed with the female contact point **(G)** in the lamp's housing on the other side of the Unit.

Step 7. Replace the lamp housing's O-ring.

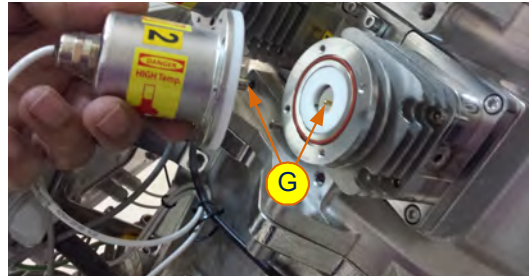


Figure 10-4: Lamp, Connector Assembly and connectors

Assembly

Step 8. Reinstall the lamp's housing and, using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Step 9. Reinstall the lamp's connector assembly pushing it straight in. Ensure that the male contact point **(G)** on the end of the lamp has meshed with the female contact point **(G)** in the lamp's housing on the other side of the Unit.

Step 10. Using the # 3 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Allowing the Atlantium Unit to Run with Caps Open

Step 11. Temporarily place Caution floor signs at a distance of 1m/ 40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.

Step 12. Unscrew and remove the two caps **(H)** on each side of the lamp. (See the figure to the right.)

Step 13. For the lamp that was to be replaced, on the connection box, turn the lamp switch to the **On** position.

If the Entire System is Off

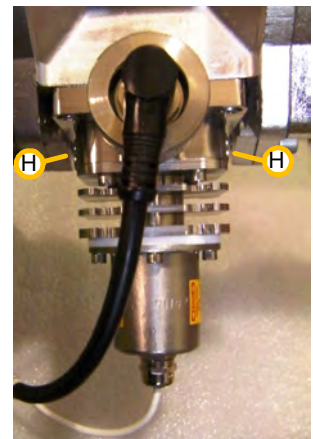


Figure 10-5: Lamp's Caps (Generic View)

If Only the Pair of Lamps are Off - Reactivating the Lamps

For All

Step 14. Check that the lamp ignites properly.

Step 15. Operate the system without the lamp's caps for one hour.

Step 16. Return the lamp's caps **(G)**.

System Tuning

Step 17. Perform system tuning according the instructions on page **120**.

Recording Lamp's Serial Number

Observing Lamp Efficiency

Step 18. Observe the system's efficiency for **100 hours**. If after this period there is a performance degradation of more than 5%, remove the UV intensity sensor and check for humidity. Clean if necessary.

Step 19. Perform **System Tuning** according the instructions on page **120**.

10.6 Cleaning/Replacing the UV Intensity Wave Guide



Performed
as needed

If your system includes wave guides:

For each UVIS sensor in the Atlantium Unit, there is an internal wave guide for directing UV light into the sensor. The UV Intensity Sensor's wave guide must be cleaned every time the lamps are replaced.

Prepare the following:

- Replacement UV Intensity Sensor's Wave Guide
- Wave Guide tool
- White cotton gloves
- Alcohol/isopropanol and lint-free lens cleaning paper
- Appropriate Phillips screwdrivers



CAUTION!

- Allow the old lamp at least 10 minutes to cool down before starting.



- Be very careful when handling the wave guide. The wave guide is made of quartz and is fragile and easily broken.

➔ To clean/replace the wave guide:

Accessing the Wave Guide to be Replaced

Step 1. On the side of the Unit where the UVIS sensors are located, identify the lamp whose wave guide is to be cleaned/replaced.

Follow the directions for removing a lamp. See **Replacing a UV Lamp on page 103**.

Removing the Wave Guide

Step 2. Screw the Wave Guide tool (**L**) onto the wave guide socket (**J**).

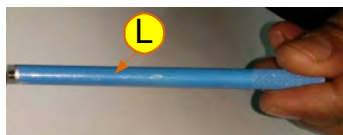


Figure 10-6: Wave Guide Tool

Step 3. While holding the Wave Guide tool (**L**) now attached to the wave guide (**G**), using an appropriate Phillips screwdriver, (**M**) unfasten each of the screws (**K**).

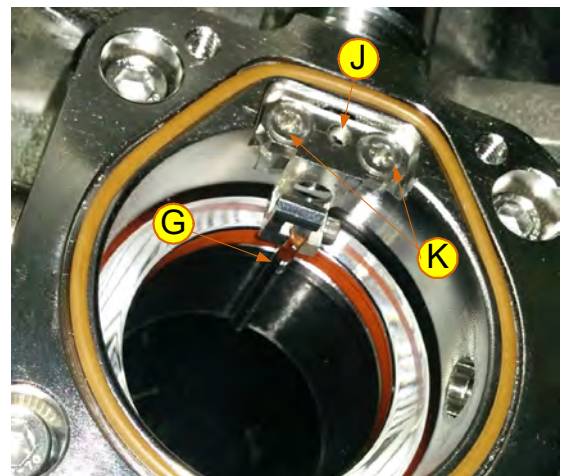


Figure 10-7: Wave Guide Elements-Generic

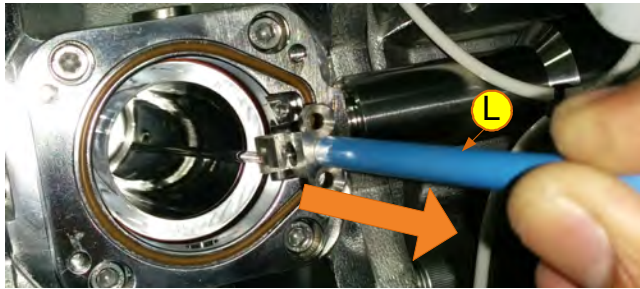


Figure 10-8: Removing the Wave Guide-Generic

Step 4. With the Wave Guide tool (**L**), carefully pull out the wave guide.

Step 5. Unscrew the Wave Guide tool (**L**) and detach it from the wave guide (**G**).

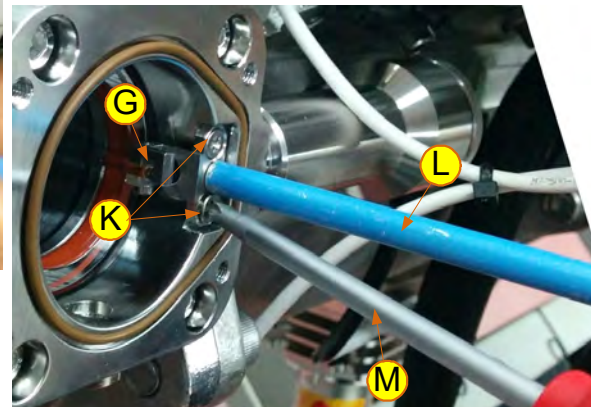


Figure 10-9: Unfastening the Wave Guide -Generic

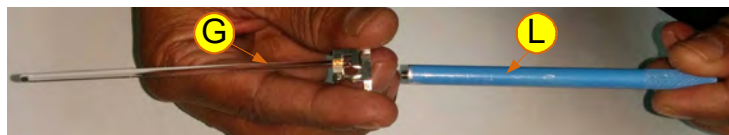


Figure 10-10: Wave Guide Tool and Wave Guide

Cleaning the Wave Guide

Step 6. While wearing white cotton gloves, carefully clean the wave guide with isopropanol and lint-free lens cleaning paper.

Step 7. Ensure that the wave guide is completely dry before returning it to its place.

Step 8. Examine the wave guide for marks and damage. Fingerprints or other stains may cause erroneous UV intensity sensor readings. If the wave guide's surface is damaged, it must be replaced.

Preparing the Replacement Wave Guide

Step 9. Screw the Wave Guide tool (**L**) onto the wave guide socket (**J**).

Step 10. With the Wave Guide tool (**L**), carefully insert the wave guide in place.

Step 11. While holding the Wave Guide tool (**L**) now attached to the wave guide (**G**), using an appropriate Phillips screwdriver, (**M**) fasten each of the screws (**K**).

Step 12. Check that the wave guide is parallel to the side of the quartz sleeve and not pointing at an angle. If it is at an angle, the quartz sleeve is not installed straight. Remove the wave guide and remove and reinstall the quartz sleeve following the directions in *Cleaning/Replacing a Lamp's Quartz Sleeve & O-Rings* on page 111. When complete, install the wave guide again.

ATTENTION!

The correct position of the wave guide is parallel to the side of the quartz sleeve and a distance of 0.75mm from the quartz sleeve.

Step 13. Follow the directions to reinsert the lamp and refasten the lamp's housing and connector See *Replacing a UV Lamp* on page 103.

10.7 Cleaning/Replacing the UV Intensity Sensor's Mirror

Performed
as needed

If your system includes UVIS mirrors:

For each lamp in the Atlantium Unit, there is a small internal mirror **(A)** for directing UV light into the sensor. The UV Intensity Sensor's Mirror must be cleaned every time the lamps are replaced.

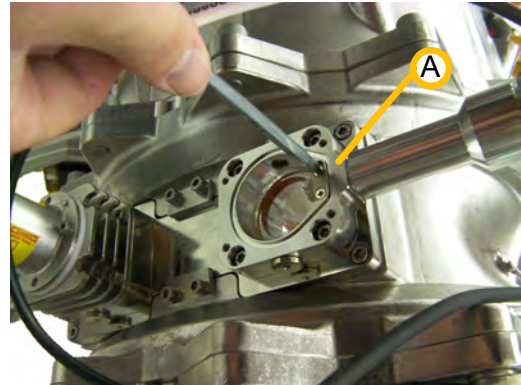


Figure 10-11: Generic Cleaning the UV Intensity Sensor's Mirror

Prepare the following:

- Replacement UV Intensity Sensor's Mirror
- Isopropanol and lint-free lens cleaning paper
- Small Phillips screwdriver



To access the mirror, remove the lamp connector assembly from the side with the UVIS sensor. The Lamp housing on this side has the UVIS sensor connected. See page **103** for instructions on accessing the Lamp housing.

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

➔ To clean the mirror:

- Step 1.** No particular parts need be removed from the Unit in order to access the mirror for cleaning, once the lamp has been removed. See page **103** for instructions on replacing the lamp.
- Step 2.** Clean the reflective side of the mirror – the side facing into the Unit – with isopropanol and lint-free lens cleaning paper.
- Step 3.** Ensure that the mirror is completely dry before re-assembling the UV lamp.
- Step 4.** Examine the mirror's surface after cleaning and before you close the UV lamp' housing. Fingerprints or other stains may cause erroneous UV intensity sensor readings. If the mirror's surface is damaged, it must be replaced.

➤ To replace the mirror:

Step 1. Once the lamp has been removed, you can replace a damaged mirror. See page 103 for instructions on replacing the lamp.

Step 2. Using a small Phillips screwdriver, remove the two screws holding the mirror in place and remove the mirror.

Step 3. Clean the reflective side of the replacement mirror with isopropanol and lint-free lens cleaning paper.

Step 4. Examine the mirror's surface after cleaning and before you install it to make sure it is damage-free.

Step 5. Carefully place the new mirror in position and using the screwdriver, fasten it securely.

Step 6. Before you close the UV lamp's housing, examine the mirror's surface to make sure it is free of fingerprints or other stains, which may cause erroneous UV intensity sensor readings. Clean it again if necessary.

Step 7. Ensure that the mirror is completely dry before re-assembling the UV lamp.



Figure 10-12: Removing the Mirror



Figure 10-13: Cleaning the Replacement Mirror

10.8 Cleaning/Replacing a Lamp's Quartz Sleeve & O-Rings



The quartz sleeve is cleaned as needed according to the UVT Sensor readings and when deposit collects on the quartz sleeve. **The quartz sleeve's rubber O-rings must be replaced when dismantling the quartz sleeve.**

Depending on the quality of the water being treated by the Atlantium Unit, the exterior surface of the quartz sleeve that houses a lamp may require periodic cleaning or replacement if after the optional Cleaning system and the CIP process the quartz sleeve is still not sufficiently clean. This is to be determined by the quality control officer at the facility.

Under normal environmental conditions, the quartz sleeve's rubber O-rings (as shown in **Figure 10-20** below) must be replaced once a year even if they pass visual inspection. For environments where the water temperature is above 70°C (160°F), the quartz sleeve's O-rings must be replaced yearly.

Prepare the following:

- Replacement quartz sleeve (if replacing)
- Two O-rings per sleeve (See view B in **Figure 10-20** below)
- #3 Allen wrench
- #4 Allen wrench
- Protective gloves
- Appropriate cleaning cloth. A mild, household cleaning solution
- Alcohol

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

➞ To clean or replace the quartz sleeve:

Powering Down the System



A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp's switch before you proceed to replace the lamp.

Step 8. For the lamp whose quartz sleeve is to be cleaned or replaced, on the connection box, turn the lamp switch to the **Off** position.

Step 9. Using the viewport on the Atlantium Unit, check that the lamps are indeed turned off.

Step 10. Check the main screen of the **Control Module** and verify that the system parameter for **Power** is **0%**.

Isolating the Unit

Step 11. To isolate the Unit, close the Inlet and Outlet valves.

Step 12. Using the Drain valve, drain the water from the Unit.

Accessing the Quartz Sleeve

Quartz sleeve removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) and the lamp be removed, as well as the quartz sleeve holder.

Removing the Lamp Connector Assembly and Lamp Housing

Step 13. To remove the lamp connector assembly and lamp housing from the upper side of the Unit, as well as the lamp, see **Replacing the Lamp** on page 105.

Removing the Quartz Sleeve's Holder

Step 14. To remove the quartz sleeve's holder on the upper side of the quartz sleeve, using a #4 Allen wrench, remove the four Allen screws (**G**), opening them in a diagonal pattern (i.e. upper left, lower right, upper right, lower left) that secure the sleeve's holding ring to the Unit. Loosen one screw gradually (2 turns) on one side and do the same on the screw at the diagonal opposite side of the sleeve (2 turns).

Step 15. Continue to alternate until both sleeve holders are loose, and then remove the screws completely. Each screw has a washer.

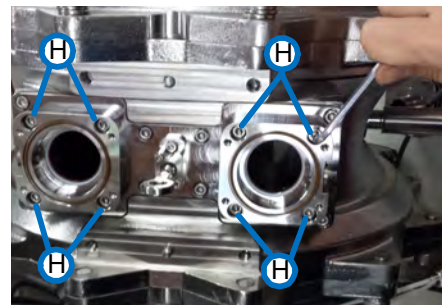


Figure 10-14: Unfastening the Screws Holding the Plate (Generic)

ATTENTION!

While pulling the quartz sleeve's holder away, **DO NOT** tilt it and **DO NOT** pull it away from the lamp/sleeve axis at an angle. The wave guide is fragile and may be damaged if the quartz sleeve is tilted.

Step 16. On the lower side of the Lamp chamber, remove the cable from the UVIS sensor **(B)**.

Step 17. To remove the quartz sleeve's holder on the lower side of the quartz sleeve **(C)**, perform from **Step 14.** on it.

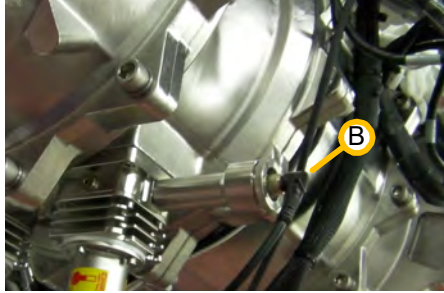


Figure 10-15: Removing the Cable from the UVIS Sensor

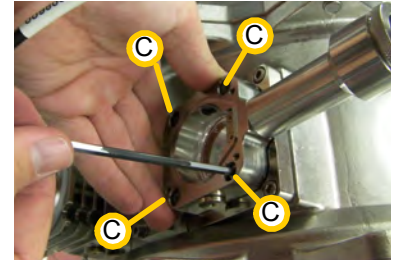


Figure 10-16: Removing the Quartz Sleeve's Holder on the Lower Side

Removing the Quartz Sleeve

ATTENTION!

Wear protective gloves. To protect the quartz sleeve, do not touch the internal surface with your bare hands.

Step 18. On the upper side of the quartz sleeve, place a gloved palm of your hand over the quartz sleeve and gently push in **(D)**. The quartz sleeve shifts inward. The o-ring **(F)** on the side where you pushed slips out (if not, take it off).

Step 19. On the lower side of the quartz sleeve, the sleeve sticks out slightly. **To prevent any damage to the quartz sleeve, make sure you pull the quartz sleeve straight out.** Carefully pull out the quartz sleeve **(G)** and remove the O-ring **(F)**.

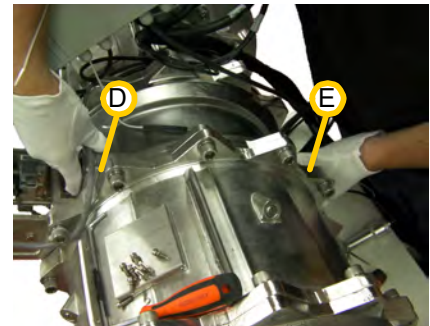


Figure 10-17: Push Quartz Sleeve (Generic View)

Cleaning the Quartz Sleeve

Step 20. If there is a deposit buildup on the sleeve's exterior surface, clean it with a mild, household cleaning solution.

Step 21. Clean the exterior and interior surfaces with alcohol. If the sleeve cannot be cleaned to a pristine condition, replace it with a new one.

Step 22. Ensure that there is no residual moisture or high humidity on the inner surface of the quartz sleeve. If any visible moisture exists, dry it before inserting it in place.

Installing the Quartz Sleeve

Step 23. Mount a new O-ring onto the quartz sleeve (as shown in **Figure 10-20** above)

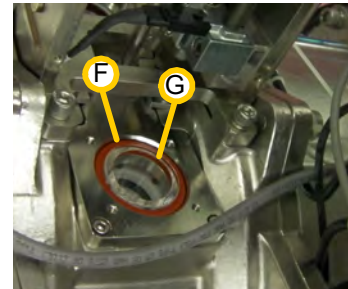


Figure 10-18: Quartz Sleeve and O-ring (Generic View)

Step 24. Holding the quartz sleeve with the serial number pointing down (**H**), insert the other end of the sleeve into the Unit from the lower side until it reaches the other side. Make sure it is centered as you insert it.

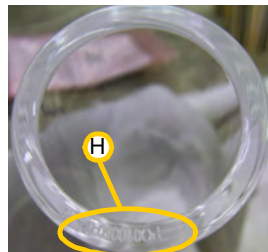


Figure 10-19: Quartz Sleeve Serial Number



Figure 10-20: Inserting the Quartz Sleeve

Step 25. Add the second new O-ring on the other side of the quartz sleeve.

Step 26. Repeat for the other sleeve of the pair.

Assembly

Step 27. Install the quartz sleeve holders on the lower side and close it loosely with 4 screws in a diagonal pattern (upper left, lower right, upper right, lower left). **Do not tighten the screws!!**

ATTENTION!

For systems with wave guide, while reinstalling the sleeve holder, DO NOT tilt it and DO NOT insert it at an angle. the wave guide is fragile. Take care not to damage it when inserting the lamp housing.

Step 28. Install the quartz sleeve holders on the upper side.

Step 29. Using a #4 Allen wrench, close the screws in diagonal pattern and tighten them.

Step 30. To reinstall the lamps, for both lamps of the pair, perform **Step 5.** to on page **107** in the section, **Replacing the Lamp** on page **105**.

Step 31. Reinstall the lamps' connector assemblies pushing each straight in and using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure them to the Unit.

Step 32. Connect the UVIS sensors' cables.

Powering Up the System

Verifying System Performance

Step 33. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 34. Open the viewport and check that there are no air bubbles in the water flow. If necessary, release all air bubbles from within the system.

Step 35. Run the system with the caps of the lamp off as described in the section, 10.5.4. Replacing the Lamp. See **Step 11.** to the end of the procedure, starting on page **107.**

WARNING!



While lamp caps are off, be sure to follow safety requirement:

- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

Step 36. Check the main screen of the **Control Module** and verify that the Atlantium system parameters for **Power, Flow, and UVT** are within wanted acceptable range and record is using the Table in **Appendix A. Checking the System Parameters** on page **142.**

If the values are out of range, notify the executive maintenance engineer.

10.9 Checking/Replacing Sensors

From time to time, sensors may need to be checked (most frequently for condensation). During the Calibration process, the UVIS sensor also requires to be removed and temporarily replaced with a Reference Master sensor during. At times a sensor needs to be replaced. There are two types of UVIS sensors. The sensor itself is the same in both types. The difference is the sensor casing and how it is fastened to the Atlantium Unit. The two types are:

- **Adjustable UVIS sensor** - The Sensor's position is adjustable and its casing is fastened with nuts. See below.
- **Fixed UVIS sensor** - The sensor's casing screws into a fixed position. This type of sensor may contain a pinhole. See **Replacing/Checking a Fixed UVIS Sensor or UVT Analyzer** on page **119.**

10.9.1 Checking an Adjustable UVIS Sensor



Before removing the UVIS sensor to check it for moisture, do one of the following:

If you do not have a caliper measurement tool: You must mark the placement of the sensor so that you can replace it in the correct position.

If you have a caliper: You must measure the position of the sensor. (See on page **103.**) or by using a marker to mark the position before removal.

Prepare the following:

- Wrenches # 24 and # 30
- Caliper measuring tool or thin marker

➔ **To check an adjustable sensor (using a marker):**

Step 1. In the **Control Module**, note the value for UVIS for the lamp on which the sensor to be replaced. For details on finding the **Control Module** screen with this information, see the section, **System Tuning** on page 120.

Step 2. Mark the placement of the sensor so that you can replace it in the correct position. Use the thin marker to mark the position before removal. Mark the sensor in two places, a line along the nut and across the nut and perpendicular to the first line. as shown in (1).

Step 3. Check the sensor's lens for condensation. If present, using a dry lens cloth, dry the lens thoroughly.

Be sure that the lens is completely dry before it is returned to its place.

Step 4. When you return the sensor to its place, make sure the thin lines line up to the nut and align together.

Step 5. In the **Control Module**, check that the value for UVIS for the lamp on which the sensor was replaced is identical to the value before the sensor was removed. For details on finding the **Control Module** screen with this information, see the section, **System Tuning** on page 120.

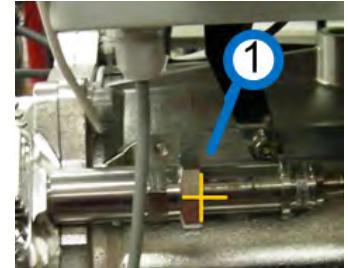


Figure 10-21: Marking the Sensor

➔ **To check an adjustable sensor (using a caliper):**

Step 1. Place the measurement tool on the UVIS sensor as show in **Figure 10-22** below. Use the screw to adjust the placement of the caliper measurement tool.



Figure 10-22: Measuring the UVIS sensor

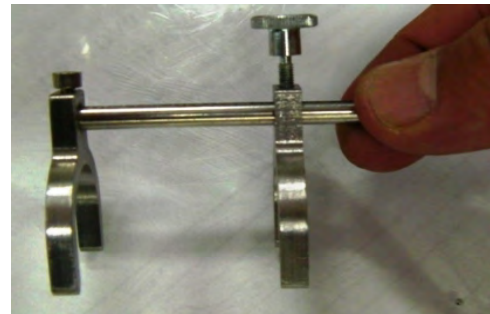


Figure 10-23: Caliper Measurement Tool

Step 2. Carefully remove the caliper, taking care not to accidentally change its position, and set it aside.

Step 3. Remove the sensor according to the relevant steps in *Replacing an Adjustable UVIS Sensor* on page 117.

Step 4. If you are in the midst of a Calibration process for the UVIS sensor, replace the original UVIS sensor with the Reference sensor.

Step 5. Check the sensor's lens for condensation. If present, using a dry lens cloth, dry the lens thoroughly.

Be sure that the lens is completely dry before it is returned to its place.

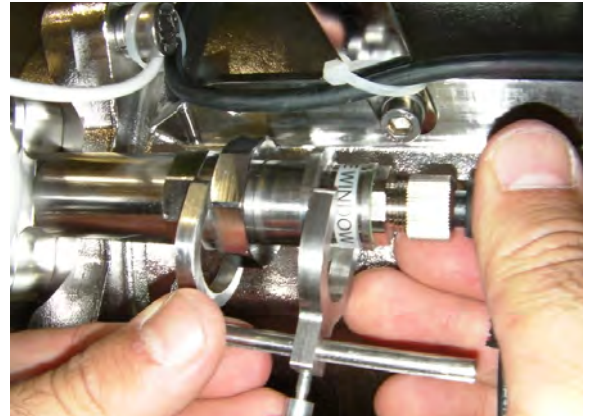


Figure 10-24: Adjusting the Reference sensor

Step 6. When you insert a Reference Master sensor or return a sensor to its place, using the set caliper (calibrated to the original UVIS sensor's position) adjust the sensor so that it is set to the position of the original. Refer to **Figure 10-24** to the left.

10.9.2 Replacing an Adjustable UVIS Sensor



Before removing the UVIS sensor when calibrating it against the Reference Master sensor or checking for moisture, you must mark the placement of the sensor so that you can replace it in the correct position. This is accomplished using the caliper measurement tool (See on page 103.) or by using a marker to mark the position before removal. Replacing the UVIS sensor involve the loosening of the nut holding it in place.

Prepare the following:

- Replacement UVIS sensor or appropriate type
- Wrenches # 24 and # 30
- Caliper measuring tool or thin marker

➔ To replace an adjustable UVIS sensor:

Step 1. To disconnect the harness connector (shown in view **A** in **Figure 10-26**), unscrew the sensor's threaded locking nut.

Step 2. Grip the Sensor casing with wrenches **# 24** and **# 30**, as shown in view **B** below. The inset shows the wrench **#24** in the left hand gripping the casing's left band and wrench **#30** in the right hand gripping the casing's large nut on the right.

Step 3. Holding the left wrench **#24** steady, twist the right wrench **#30** to loosen the nut.

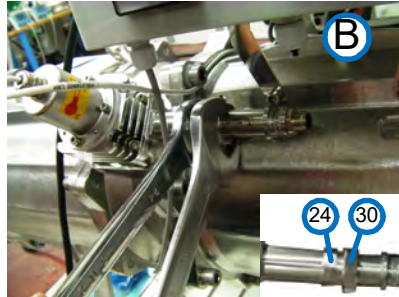


Figure 10-25: Replacing the Sensors (Generic View 2)

Step 4. Carefully pull the sensor out of the housing **(C)**. The Large nut comes off together with the sensor.

Step 5. To enable the system to track relevant data about the sensor, make a note of the serial number of the replacement sensor (for later use).

Step 6. Verify that the sensor lens **(1)** is clean. If needed, using a lens cloth, clean it. (See below.)

Be sure that the lens is completely dry before it is returned to its place.

Step 7. When replacing a sensor, remove the nut **(2)** and the washer **(3)** shown in **Figure 10-28** below.

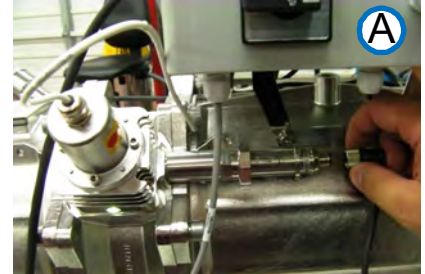


Figure 10-26: Replacing the Sensors (Generic View 1)

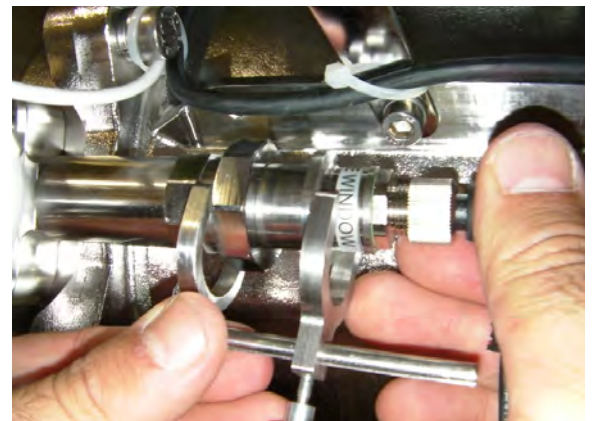


Figure 10-27: Replacing the Sensors (Generic View 3)

- Step 8.** Position the nut (2) on the sensor as shown above. Then position the washer (3) on the sensor as shown above.
- Step 9.** Insert the sensor into the housing.
- If you are replacing the sensor, positioning it about 2/3 the way in. Fasten the nut onto the housing.
 - If you are returning a sensor that was removed for checking or calibration, see **To check a sensor or remove it for calibration** on page 116.
- Step 10.** Grip the Sensor casing with wrenches # 24 and # 30, as shown in view B above. The inset shows the wrench #24 in the left hand gripping the casing's left band and wrench #30 in the right hand gripping the casing's large nut on the right.
- Step 11.** Holding the left wrench #24 steady, twist the right wrench #30 to tighten the nut.
- Step 12.** Reconnect the harness's threaded cap to the new sensor.

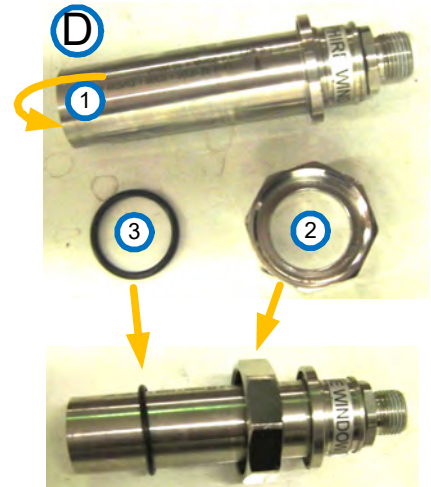


Figure 10-28: Sensor, washer and nut

- Step 13.** For a replacement sensor, in the **Control Module**, In the **Control Module**, perform **System Tuning**, see the section, **System Tuning** on page 120.

10.10 Replacing/Checking a Fixed UVIS Sensor or UVT Analyzer

Performed
as needed

For some of the sensors, there is an pinhole located on the sensor's window. This pinhole is calibrated by Atlantium for the specific measurement point. Do not switch disks between ports. Before inserting the sensor, be sure to include the pinhole if present.

Prepare the following:

- Replacement sensor

➡ To replace a sensor:

- Step 1.** To disconnect the harness connector (shown by the arrow in view A below), unscrew the sensor's threaded locking nut.
- Step 2.** Unscrew the sensor's threaded locking nut from the bottom of the sensor housing, as shown in view B.



Figure 10-29: Replacing the Sensors 2 (Generic View)



Figure 10-30: Replacing the Sensors 2 (Generic View)

- Step 3.** Carefully pull the sensor out of the housing (C).
- Step 4.** If an pinhole (E) is present it slides out.
- Step 5.** Unpack the new sensor (D) and (if it is supplied) the sensor pinhole.



Figure 10-31: Replacing the Sensors 4 (Generic View)



Figure 10-32: Replacing the Sensors 3 (Generic View)

- Step 6.** To enables the system to track relevant data about the sensor, make a note of its serial number.
- Step 7.** Verify that the sensor lens (F) is clean. If needed, using a lens cloth, clean it.
- Step 8.** If an pinhole (E) is present, insert it.
- Step 9.** Insert the sensor into the housing, and screw the threaded cap back into the housing to secure the sensor.
- Step 10.** Reconnect the harness's threaded cap to the new sensor.
- Step 11.** In the **Control Module**, perform **System Tuning**, see the section, **System Tuning** on page 120.

10.11 System Tuning



System tuning is performed on these occasions:

- As the last step in system installation
- As the last step when lamps are changed and after 100 operating hours (as needed)
- Upon advice of an Atlantium Service Representative when the system is reset

➔ To tune the Atlantium system:

- Step 1.** Just prior to tuning, obtain a water sample from the sampling valve on the inlet side of the Atlantium Unit. Test the water sample in an external spectrophotometer. Write down the results. If a spectrophotometer is not available, record the current UVT value shown on the system's display.
- Step 2.** In the **Control Module**, on the top of the screen, tap . The button turns pink. Wait 10 minutes for lamps to turn on and the system to stabilize.
- Step 3.** On the Atlantium Unit, locate each UVIS and UVT Intensity Sensor.
- Step 4.** On each sensor (refer to the figures in the section, **Replacing/Checking a Fixed UVIS Sensor or UVT Analyzer** on page 119):
- a Unscrew the locking nut and gently pull out the sensor.
 - b With a dry and clean lint free cloth, wipe the UVIS sensor's quartz window.
 - c Reinsert the sensor and replace the locking nut.
- Step 5.** To turn on the Atlantium system, on the **Control Module** main screen, tap . The button turns green. Wait approximately three minutes for the **Ready Lamps On** signal to transmit.
- Step 6.** At the top of the screen, tap . The **Monitor Tab** is displayed.

- Step 7.** Tap the **Tuning Tab**. The **Tuning** screen appears.
- Step 8.** Verify that the lamps are on and enabled (the number of lamps listed in the display is according to the number of lamps in your system configuration).
- Step 9.** For each lamp to be tuned, according to **Lamp ID** in column **1**, make a mark in the appropriate checkbox in the last column on the right.
- Step 10.** Under **Status T0 %**, using the **-/+** buttons, set the **Status T0 percentage** up to **100%**.
- Step 11.** Wait a few minutes to allow the system to stabilize.
- Step 12.** To verify that the Sensor readings are stable, observe the reading for a few seconds to see that they remain the same numbers. For the UVIS sensor, see the **UVIS mA** column above. For the UVT Intensity sensor, see the **UVTS mA** column above. If the numbers are not stable, unscrew and remove the two caps on each side of the lamp. (See Figure 10-5, on page 107.)
- Step 13.** Leave the caps open for 30 minutes to make sure no humidity is present in the lamp housing. Then replace the caps and check the sensor reading again.



When the Lamp's caps are removed, do not look directly into the opening without appropriate eye protection. The UV light emitted via the opening is a danger to the eyes.

- Step 14.** At the bottom of the screen, for **UVT To**, enter the current UVT value obtained from the spectrophotometer that you wrote down in **Step 1.** above (or the recorded current UVT value).
- Step 15.** To record the reason for tuning the system, At the bottom of the screen, for **Tuning Reason**, from the dropdown list, select the reason. The options are:
- **System Integration**
 - **Lamp changed**
 - **Maintenance**
- Step 16.** Tap **Start Tuning**. A progress bar appears at the bottom of the screen. The process takes about 6 to 15 minutes.



If the system is not running, or if a lamp is not on, a popup message appears informing you that in order to start a tuning process, the system must be running.

- Step 17.** Verify that the UVT is the number you set previously (see the **UVTS mA** column) and that the Status on the Monitor tab is the same as the number you set in the **Status T0 %** column.
- Step 18.** If the numbers are not stable, unscrew and remove the two caps on each side of the lamp. (See Figure 10-5, on page 107.)
- Step 19.** Leave the caps open for 30 minutes to make sure no humidity is present in the lamp housing. Then return the caps to their position and check the sensor reading again.



When the Lamp's caps are removed, do not look directly into the opening without appropriate eye protection. The UV light emitted via the opening is a danger to the eyes.

10.12 Advanced Maintenance

The advanced maintenance procedures require opening the Connection Box. **Local regulations may require a certified electrician to perform any of these procedures.**

This section contains:

- **Replacing the Temperature Sensor** on page [122](#)
- **Replacing the Temperature Converter Card** on page [123](#)

10.12.1 Replacing the Temperature Sensor



In the event that the Temperature sensor requires replacement, this section details the procedure.



WARNING!

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.



Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature sensor
- #5 Allen wrench
- #3 Flat screwdriver
- #2 Phillips screwdriver

➔ To replace the Temperature sensor:

- Step 1.** Turn the Mains circuit breaker to the **Off** position.
- Step 2.** Using the appropriate Allen wrench, open the Connection Box cover.
- Step 3.** Verify with volt meter that there is no power in first Connection Box card and that no LED is lit on the card.
- Step 4.** Locate and disconnect the Temperature Sensor connector **(A)**.
- Step 5.** Detach the wire from the terminal connector and remove the wire from the Connection Box via the gland.
- Step 6.** Using a #2 Phillips screwdriver, detach the temperature sensor **(C)** from the Atlantium Unit.

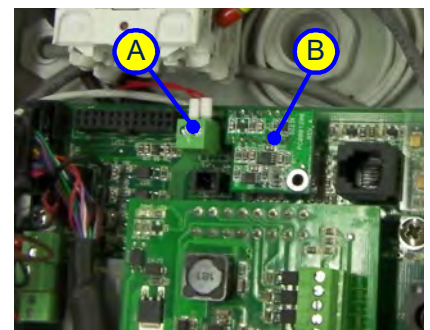


Figure 10-33: Location of the Temperature Sensor Connector

- Step 7.** Attach the replacement Temperature sensor **(C)** to the Atlantium Unit.
- Step 8.** Thread the wire through the gland and attach it to the terminal connector.
- Step 9.** Insert the connector **(A)** to its socket on the terminal block.
- Step 10.** Return the Connection Box cover and using the Allen wrench, fasten the screws. Tighten them in a diagonal pattern.
- Step 11.** Turn the Mains circuit breaker to the **On** position.

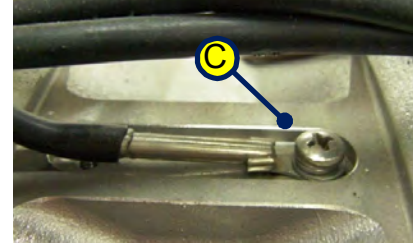


Figure 10-34: Location of the Temperature Sensor on the Atlantium Unit

10.12.2 Replacing the Temperature Converter Card



In the event that the Temperature sensor card requires replacement, this section details the procedure.



WARNING!

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.



Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature Converter card
- #5 Allen wrench
- #2 Phillips screwdriver

➔ To replace the Temperature sensor card:

- Step 1.** Turn the Mains circuit breaker to the **Off** position.
- Step 2.** Using the appropriate Allen wrench, on the first Connection Box, open the Connection Box cover.
- Step 3.** Verify with volt meter that there is no power in first Connection Box card and that no LED is lit on the card.
- Step 4.** Locate and carefully pull off the Temperature Sensor card **(B)**.
- Step 5.** Aligning the pins of the replacement Temperature Sensor card **(B)** with the connector (on the board) on the Connection Box card, carefully insert the small card into the connector of the Connection Box card.
- Step 6.** Return the Connection Box cover and using the Allen wrench, fasten the screws. Tighten them in a diagonal pattern.
- Step 7.** Turn the Mains circuit breaker to the **On** position.

10.12.3 Replacing LM Card in Connection Box

Following replacement of LM card 1300, perform the following settings via the AB Control Module:

Step 1. Go to Configuration Tab



Figure 10-35: Configuration Tab, AB CM

Step 2. Set the following parameters (**only** if changes to factory settings are required):

- ◆ Hot Ignition Delay
- ◆ Ignition Timeout
- ◆ Low Status Warning

Table 10-1: Configuration Screen settings

Feature	Description
Hot Ignition Delay	Minimum time interval [1-10 min] between switch off to re-ignition of lamp. Enables lamp to cool down before re-ignition.
Hot Ignition Delay test	1 Re-ignite lamps immediately after switch off. Lamps should reignite only after set interval. 2 Re-ignite lamps after set interval from switch off. Lamps should reignite immediately.
Ignition Timeout	Time interval [1-30 min] during which system attempts to ignite lamps following switch on. This interval starts at end of Hot Ignition Delay. 1 If one or more lamp(s) fail(s) to ignite during Ignition Timeout interval, ignition attempt stops and lamp ignition fault appears: 21 - ignition failed - lamp not ignited/gone out lamp number 3 2 The faulty lamp is disabled and appears in red on Monitor screen (see Figure 7-10 , on page 76).  3 The Lamp/Ballast fault goes active - green outline (see Figure 7-5) Lamp / Ballast  On successful ignition of all lamps the lamp marker appears in blue and the Lamp/ Ballast fault de-activates.
Low Status Warning	When the system is on and the lamp status is lower than this Low Status Warning value for more than 30 sec the "Low Status Lamp #" warning is raised, the lamp marker goes yellow and the Low Status output is triggered. 



11 Troubleshooting

11.1 Viewing system events/alarms

see section 9.3 page 87.

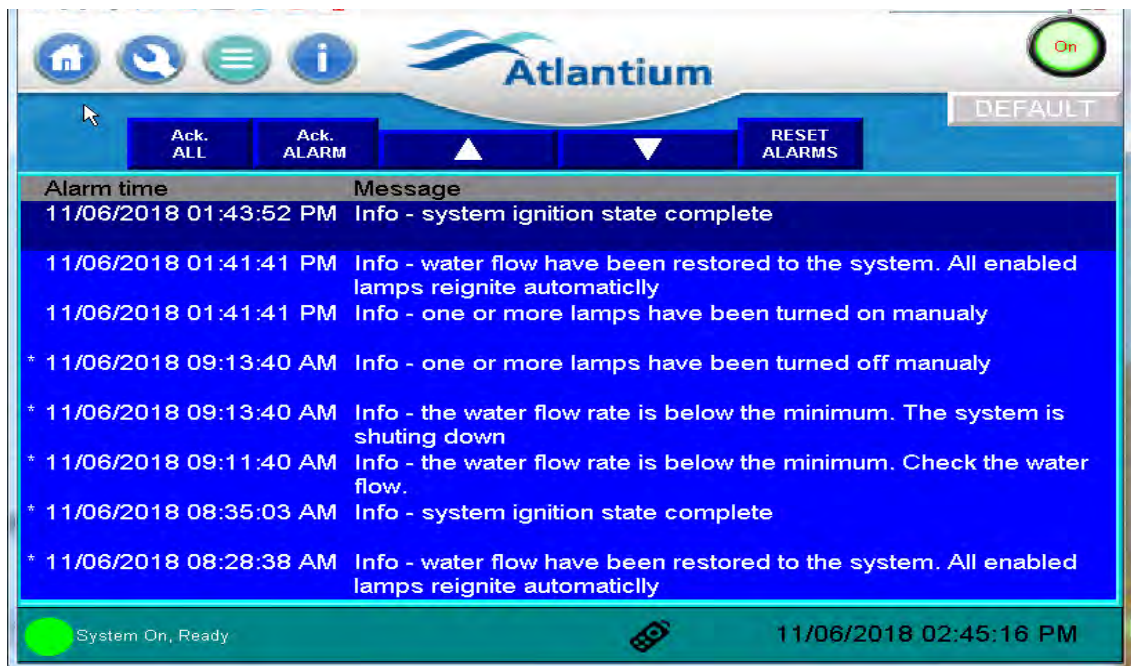


Figure 11-1: AB Control Module, Alarm Logs Screen

11.2 Troubleshooting issues

The UV dose is too low on page 135

The temperature is too high on page 136

Temperature is critically high! on page 136

The UV Lamp unexpectedly shut down on page 140

The rate of the water flow is too low / too high on page 137

UVT is too low! on page 138

UV Lamp ignition failed! on page 138

Control Module power failure on page 140

Ballast Module communication error on page 141

The UV Lamp unexpectedly shut down on page 140

CRC error on page 141

Each event/alarm provides the following additional information:

- System power

- System dose
- System flow
- System status
- System UVT
- System Temperature

Events related to the Ballast Module version 3 contain this additional information:

- Lamp Card (Connection Box Card version)
- State
- Status
- UVT

11.3 Voltage Error Codes

For Ballast Module version C, the Voltage parameter provides a value that functions as an additional Error code and indicates additional information about the fault.

Table 11-2: Voltage Error Codes and Troubleshooting procedures

Lamp voltage value when lamp is off = Error Code	Lamp turn off reason	Probable Cause	Troubleshooting
0 (+24)	LAMP X NOT IGNITE UNKNOWN REASON *	Ballast cable fault	1 Inspect the ballast cable & connectors between the electrical junction box and ballast.
		Faulty ballast/ Faulty lamp	2 Switch the cables between the ballast of the faulty lamp and the nearby ballast of a working lamp. If faulty, replace the Ballast. If not faulty, replace the lamp.
		Lamp connector	3 Check lamp connector and if damaged, replace it and the lamp.

Table 11-2: Voltage Error Codes and Troubleshooting procedures

Lamp voltage value when lamp is off = Error Code	Lamp turn off reason	Probable Cause	Troubleshooting
300 (± 24)	Lamp not ignited / gone out	Ballast cable fault	1 Inspect the ballast cable & connectors between the electrical junction box and ballast.
		Faulty ballast/ Faulty lamp	2 Switch the cables between the ballast of the faulty lamp and the nearby ballast of a working lamp. If faulty, replace the Ballast. In not faulty, replace the lamp.
		Lamp connector	3 Check the lamp connector and if damaged, replace it and the lamp.
350 (± 24)	Lamp short circuit	Cable fault	1 Disconnect the ballast and check that there is no short circuit between lamps wires
		Faulty ballast	2 Replace the ballast and check again. If it is internal ballast short circuit reduce the amount of lamp ignitions at the site and fixed it if it is above six ignitions per day.
		Water inside the sleeve	3 Replace the sleeve's O-rings
		Faulty lamp	4 If problem is not fixed, replace the lamp.

Table 11-2: Voltage Error Codes and Troubleshooting procedures

Lamp voltage value when lamp is off = Error Code	Lamp turn off reason	Probable Cause	Troubleshooting
450 (± 24)	Mains over-voltage	Input voltage is too high (above 565V) or unbalanced voltage supply	Problem in one ballast
		Voltage out of range	1 At the output side of the ballast's Fuses (inside the Junction Box) check that the input voltage of the ballast is in the proper range at all three phases.
			2 If the voltage is valid and the wires are fixed, try to reignite the lamp. If the lamp reignites, wait 10 minutes and see if the ballast is working properly.
		Faulty ballast	3 If the problem appears again at just one ballast, or the lamp does not start properly, replace the ballast.
			Problem in all ballasts at the same time:
		Unbalanced voltage supply	1 Verify that the the voltage supply is balanced (the three phases to PE voltages are equal $\pm 10\%$). If not create balanced supply by isolation transformer
		Voltage out of range	2 Check for the source of the input high voltage and stabilize the voltage.
400 (± 24)	LAMP X MAINS UNDERVOLTAGE	Input voltage of the ballast is low due to a ballast CB fault or wire disconnection at the input of the ballast	Problem in one ballast

Table 11-2: Voltage Error Codes and Troubleshooting procedures

Lamp voltage value when lamp is off = Error Code	Lamp turn off reason	Probable Cause	Troubleshooting
		Low voltage or loss connection	1 At the output side of the ballast's Fuses (inside the Junction Box) , check that the input voltage is according to the allowed range at all three phases. Make sure that the ballast harness and input wires are properly secured tightly.
	Phase failure mains voltage		2 If the voltage is valid and the wires are secured tightly, try to reignite lamp. If the lamp reignites, wait 10 minutes and see if the ballast is working properly.
		Faulty ballast	3 If problem appears again, or the lamp does not ignite, replace the ballast.
			Problem in all ballasts at the same time:
		Low voltage	1 Try to locate the point where the voltage is low in the input phases feeding the ballasts.
			2 If all of the voltage lines are in range, check for the source of the input voltage sags and stabilize the voltage.
200 (±24)	Lamp Internal failure / Phase lost	One phase of the supply is down due to a ballast CB / Fuse fault or wire disconnection at the input of the ballast	Problem in one ballast

Table 11-2: Voltage Error Codes and Troubleshooting procedures

Lamp voltage value when lamp is off = Error Code	Lamp turn off reason	Probable Cause	Troubleshooting
		Input voltage at one of the three phases is low	1 At the output side of the ballast's Circuit Breaker/fuses , check that the input voltage of the ballast is in the proper range at all three phases. Make sure that the ballast harness and input wires are properly secured tightly.
			2 If the voltage is within range, and the wires are secured tightly, try to reignite the lamp. If the lamp reignites, wait 10 minutes and see if the ballast is working properly.
		Internal fault at the ballast	3 If the problem reappears, or the lamp does not ignite, replace the ballast.
			Problem in all ballasts at the same time:
			1 Try to locate the point where the voltage is low in the input phases feeding the ballasts.
			2 If all of the voltage lines are in range, check for the source of the input voltage sags and stabilize the voltage.
50 (±24)	Lamp nominal voltage too low	input voltage is too high	1 Check that the input voltage of the ballast is in the proper range and if yes try to reignite lamp. If the lamp reignites wait 10 minutes and see if the ballast is working properly.

Table 11-2: Voltage Error Codes and Troubleshooting procedures

Lamp voltage value when lamp is off = Error Code	Lamp turn off reason	Probable Cause	Troubleshooting
-		faulty lamp	2 If problem appears again or lamp does not start properly replace the ballast.
		faulty ballast	3 If the problem does not fixed replace the lamp.
150 (±24)	Device cooling *	Room's ambient temperature is above 45°C The ballast fan is faulty or faulty ballast	1 Check the room temperature. If it is too high, reduce the temperature via external means. 2 Reignite the lamp and check that the ballast external fan is working. If not, try to clean it or replace it. If it is still not working replace the ballast

Table 11-2: Voltage Error Codes and Troubleshooting procedures

Lamp voltage value when lamp is off = Error Code	Lamp turn off reason	Probable Cause	Troubleshooting
250 (± 24)	Lamp earth short circuit	One of the lamp's wires is touching the ground	1 Disconnect the faulty ballast from the supply by switching off its switch. Disconnect the Ballast wire connector and check using a DVM if there is short circuit between one of the lamp phases to the ground. If a short circuit is detected, locate the short circuited segment (junction box to ballast or junction box to lamp) and replace it.
500 (± 24)	Manual switch off		NA
600 (± 50)	Loss of ballast's internal parameters***		NA

- * Voltage of lamp is low while the current to the lamp is at the maximum allowed value – in this state, the required power cannot be delivered to the lamp.
- ** The voltage required by the lamp cannot be created by the ballast. The current going to the lamp is zero.
- *** In case of loss of internal ballast parameters, consult with Atlantium service personnel for instructions on how to solve this issue.

You may view the lamp voltage values in the AB Monitor Screen, Figure 9-1 on page 87.

Pay attention to the following safety notes when reviewing the troubleshooting issues:



The solution to the problem might involve exposure to UV light.

The system generates ultraviolet (UV) light within the Atlantium Unit, which can cause serious eye damage or blindness if you stare at it directly when it is working.

Do not look directly into the lamp enclosure during system operation, or while examining or servicing the system's internal components when the system is operating, or when a lamp's breather caps are open.



The solution to the problem might involve working in a high-voltage environment.

Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, **turn the BMmain circuit breaker to the Off position.**

Place a sign on the relevant switch and **Control Module** alerting others **NOT** to touch the switch and touch screen during maintenance so that no one can inadvertently turn it on while maintenance is in progress.



The solution to the problem might involve working with hot surface.

Do not touch the UV lamp with your bare hands. Lamps can reach a temperature of 1,000°C under operating conditions.

Wait at least 10 minutes until the lamp is cooled down before replacing it.

Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.

11.3.1 Lamp status indicators

Check the lamp status indicators on Monitor Screen. See Section 7.10.2 “Operator Monitor Screen” on page 79.

11.4 The UV dose is too low

Control Module displays this warning message

UVT is below the minimum

	Possible Cause	Solution	
1	The Water flow rate is too high.	Check the water flow on the Control Module's main screen. If it is higher than normal, decrease the rate of the water flow through the system. See Configuring the Flow Settings on page 81.	
2	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. See Cleaning In Place (CIP) on page 98. Try to improve the water condition. Check on the status of upstream filtering equipment and upstream chemical processes.	
3	UV lamp efficiency has dropped.		Check the UVIS sensors and clean them if necessary. Check the lamp's efficiency reading. If it is low, replace the lamp. See Lamp & UVIS Assembly on page 106.
4	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. See Cleaning In Place (CIP) on page 98. Try to improve the water condition. Check on the status of upstream filtering equipment and upstream chemical processes.	
5	The UVT sensor has malfunctioned.	Switch between the UVT and UVIS sensors and see if the reading increased. If the reading is increased, switch the sensors back and replace the UVT sensor.	

11.5 The temperature is too high

Control Module displays this warning message
Temperature is too high!

	Possible Cause	Solution
1	The Max temperature value is too low.	Check the system's Max Temperature settings. See <i>Configuring the General & Connection Settings</i> on page 95.
2	The water flow has significantly decreased.	Check the flow reading in the Control Module to see if it has deviated from normal flow. Check the flow via the flow meter. Check the water source for a problem.
3	There are air bubbles or the system is not fully filled with water.	Verify this by looking through the Atlantium Unit's view port.
4	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing a designated air release valve at the highest point on the inlet side of the system.

11.6 Temperature is critically high!

Control Module displays this warning message
Temperature is critically high!

	Possible Cause	Solution
1	If this message appeared right after a BM Communication Error message (and both messages refer to the same lamp) and the temperature display on the monitor is 165°C/329°F, then most likely the communication between the Control Module and the Connection box is lost.	Follow the solutions offered in CRC error on page 141.
2	The temperature reading dropped to 0°C / 32°F due to a short-circuit in the Temperature sensor or Connection Box Card	 Replace the temperature sensor. See <i>Replacing the Temperature Sensor</i> on page 129. If the problem still persists replace the Connection Box card. Contact your Atlantium Service representative.
3	The Max temperature value is too low.	Check the system's Max Temperature settings. See <i>Configuring the Technician Settings</i> on page 84.
4	There is no water flow.	Check the flow reading in the Control Module's main screen to see if it has deviated from normal flow. Check the flow via the flow meter. Check the water source for a problem.
5	There are air bubbles or the system is not filled with water to capacity.	Verify this by looking through the Atlantium Unit's view port.

	Possible Cause	Solution	
12	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing designated air release valve at the highest point on the inlet side of the system.	
13	The Temperature sensor is faulty.	 Replace the temperature sensor. See <i>Replacing the Temperature Sensor</i> on page 129.	

11.1 Temperature sensor fault

Control Module displays this warning message
Temperature sensor fault

	Possible Cause	Solution
1	If this message appeared right after a BM Communication Error message (and both messages refer to the same lamp) and the temperature display on the monitor is 0°C/32°F, then most likely the communication between the Control Module and the Connection box is lost.	Follow the solutions offered in <i>CRC error</i> on page 141.
2	The temperature reading dropped to 0°C / 32°F due to a short-circuit in the Temperature sensor, or Connection Box Card	 Replace the temperature sensor. See <i>Replacing the Temperature Sensor</i> on page 129. If the problem still persists replace the Connection Box card. Contact your Atlantium Service representative.

11.2 The rate of the water flow is too low / too high

Control Module displays this warning message
System turned off automatically - water flow too low/high

	Possible Cause	Solution
1	The water flow rate into the system is too high.	Decrease the rate of water flow from the facility to the Atlantium Unit.
2	Maximum allowable flow rate parameter is too low.	Increase the setting of the Maximum allowable flow rate parameter to the maximum allowable flow rate.

11.3 UVT is too low!

Control Module displays this warning message

The UVT is lower than the minimum UVT defined in the system settings.

	Possible Cause	Solution	
1	Minimum allowable UVT parameter is too high.	Decrease the setting of the Minimum allowable UVT parameter to the minimum allowable UVT.	
2	The UVT sensor's lens is fouled.		Pull out the UVT sensor and see if the sensor's lens is foul or damaged. Clean the lens with a dry lens cloth. If it is damaged, replace the sensor with a new one and perform System Tuning according.
3	Deposit has collected in the UV chamber or lamp's quartz sleeve.	Look through the Atlantium Unit's view port. If the color of the light is in a shade of green or yellow, it probably indicate there is deposit present. Perform CIP. Cleaning In Place (CIP) on page 98	
4	There are air bubbles or the system is not filled with water to capacity.	Verify this by looking through the Atlantium Unit's view port.	
5	You did not install a designated air release valve at the highest point on the inlet side of the system.	Install air release valve at the highest point on the inlet side of the system.	
6	There is a leak inside the UV chamber		Look through the Atlantium Unit's view port and check if there is condensation on the inner side of the view port and/or if there is water collected between the quartz tube sleeve and the stainless steel underneath. if you see this, contact Atlantium support for further instructions.
7	The UVT sensor has malfunctioned.		Switch between the UVT and UVIS sensors and see if the reading increased. If the reading is increased, switch the sensors back and replace the UVT sensor. See page 117

11.4 UV Lamp ignition failed!

Control Module displays this warning message

Lamp ignition failed

Check if it needs to be replaced, or if there is a faulty connection or ballast malfunction.



For Ballast Module version C, check the Voltage value error code. See **Lamp status indicators** on page 134.

	Possible Cause	Solution	
1	The system is set to the wrong model.	In the Control Module , access Technician screen #2 . Verify that the system is set to the correct model, number of lamps and branches. See Configuring the Technician Settings on page 84.	
2	One or more connectors are burnt out.		Turn off the system's main power source On each Connection Box, turn the switch to the Off position. Disconnect and thoroughly inspect the Ballast connectors. IF replacement is needed, contact Atlantium Technical Support.
3	The power supply inside the Ballast Module is faulty.	 	In a system with more than one lamp, if possible, switch between the Ballast Module connectors in order to verify the Ballast is faulty, if so replace the Ballast Module. In a system with only one lamp, replace the Ballast Module with a new one. See Connecting the Cables to the Electrical Cabinet on page 61.
4	A lamp is faulty/burned out.	 	In a system with more than one lamp, switch between the lamps in order to verify the lamp is faulty, if so replace the lamp. In a system with only one lamp, replace the lamp with a new one. See Inserting the new lamp on page 112.
5	The ID number of the Connection Box card does not match ID number of the Ballast Module.		On each Connection Box, open the top cover and locate the 3-digit display (as shown on the right). Check that the set number is identical to the number marked on corresponding Ballast Module. If not identical, use the switch to change it to the correct number. 
6	One or more of the fuses in the Connection Box is burned out.		Replace the burned out fuse in the Connection Box. Contact Atlantium support for further instructions.
7	The Connection Box card is faulty.		On the Connection Box, open the top cover and check that the 3-digit display and LEDs are lit. If they are dark, replace the Connection Box card. Contact Atlantium support for further instructions.

11.5 Control Module power failure

The system turned off due to Control Module power failure.

	Possible Cause		Solution
1	No power is feeding into the Connection Box.		Open the Connection boxes and verify that all Connection Boxes have power. See Connection Box Connections Diagram provided.
2	A wire is loose.		With a small flat screwdriver go over all the communication screws in the Control Module and Connections Box s and make sure all wires are located properly in their place and are well tightened. Control Module Connections on page 56
3	A jumper is missing.		A communication jumper might need to be installed, contact Atlantium Support for further instruction.
4	Ground is not connected to the Control Module's power inlet.		A certified electrician is required to verify that the Control Module is properly grounded. Control Module Connections on page 56
5	There is a Hardware problem in the Control Module's I/O (ISO) or in the Connection Box.		Contact Atlantium support for further instructions.

11.6 The UV Lamp unexpectedly shut down

Control Module displays this warning message

A lamp shutdown unexpectedly.



For Ballast Module version C, check the Voltage value error code. See **Lamp status indicators** on page 134.

	Possible Cause	Solution
1	A short power failure (up to 15 seconds) to the Atlantium Unit only, while the Control Module still had power.	After 3 minutes, turn the lamps back on using the On/Off button on the Control Module's main screen.
2	Ballast, lamp, connection failure	Check for faulty connection, ballast or lamp.



This situation can happen only in installations were the **Control Module** and the Atlantium Unit have separate power sources.

11.7 CRC error

Event
Cyclic Redundancy Check error. Check if the parameters and counters have been reset to factory defaults. If so, contact Atlantium Technical Support.

	Possible Cause	Solution
1	While writing parameters into the Control Module's flash storage, the system encountered a problem due to an unexpected power failure or software upgrade.	Reenter all system parameters from scratch. Contact Atlantium support for further instructions. See Configuring the General & Connection Settings on page 95.



For Ballast Module version C, check the Voltage value error code. See **Lamp status indicators** on page 134.

11.8 Ballast Module communication error

Control Module displays this warning message
There is lost communication between one of the Connection Boxes and the Control Module . The system is shutting down.

	Possible Cause		Solution
1	No power is feeding into the		that all Connection Boxes have power.
2	A wire is loose.		With a small flat screwdriver go over all the communication screws in the Control Module and the BM and make sure all wires are located properly in their place and are well tightened. Control Module Connections on page 56 and See AB Control Module Alarm List on page 88..
3	A jumper is missing.		A communication jumper might need to be installed, contact Atlantium Support for further instruction.
34	Ground is not connected to the Control Module's power inlet.		A certified electrician is required to verify that the Control Module is properly grounded. Control Module Connections on page 56
45	There is a Hardware problem in the Control Module's I/O (ISO) or in the Connection Box.		Contact Atlantium support for further instructions.



Table A-1: Atlantium System Parameters Checklist

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Appendix B Glossary of Terms

Term	Definition
Auxiliaries	Valves, flow devices, etc. essential to the workings of the Atlantium system but not directly a part of the Atlantium system.
Ballast	The power supply that translates the electrical power to UV power.
Connection Box	The box that sits on top of each lamp unit and holds the circuit board for the Electrical Cabinet or Ballast Module electrical configurations. Not relevant for Ballast Module/Cable-style.
Control Module	Control Module that controls operations and dose measurement. Comes in two versions: Standard and Premium. The Standard has a monochromatic screen and several key features. The Premium has a color screen can accommodate a full complement of reporting and trending features.
External On/Off	This enables an external source (SCADA, PLC, Process) to turn the Atlantium system on and off.
IP56	Protected from limited dust ingress. Protected from high pressure water jets from any direction
<i>Electrical configuration styles: Atlantium provides a choice of electrical configurations.</i>	
Electrical Cabinet-style (EC)	EC - Electrical Cabinet.
Ballast Module	Waterproof case for the ballast which connects via a consolidated plug from the circuit board which sits on top of each lamp unit.
General Alarm	This enables the Atlantium unit to send an alarm to a buzzer, cell phone or other destination.
Inlet/Outlet Valves	Isolating Valves on the Inlet/outlet sides.
Lamp Chamber	The chamber which holds the lamp.
Lifting Support	The lifting element which can also be used to help support the Unit.
Pinhole	This is a small disc that is fitted for each sensor port to ensure that the right amount of light penetrates.
Portable CIP System	The portable unit Atlantium sells that connects to the Unit to clean and disinfect it without disassembling. It includes several kits – a valve kit, an accessories kit and the CIP kit itself.
Quartz Chamber	The quartz pipe in which the disinfection occurs.
System Ready Signal	The signal the Atlantium unit provides to tell the production system that the Atlantium System is ready to supply properly disinfected water.

Term	Definition
System Tuning	This is process is performed when key components are replaced or changed to synchronize them.
Triangle Support Brackets	RZ-Series only- brackets that can be used to support and balance a unit.
UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds the real time dose calculation.
UVT Analyzer	The sensor that measures the lamp UV intensity through the water. Together with the UV Intensity sensor and the Control Module, it analyzes, tracks and reports the water quality.



Appendix C Consumables & Spare Parts



The Part Numbers and Descriptions and unit prices of the Consumables & Spare Parts shown in these tables are subject to change.

Table C-1: Spare Parts for RZB300

Part Number	Description
Consumables - Required	
MP0023000	O-RING 2-327 V0680-70 FOR SLEEVE
OPH000120	UV LAMP 250mm 4.2kW LONG NECK
MM0050410	TEFLON GRAFIT WIPER FOR DPM
MP0022600	DYNAMIC SEAL, 8X15X6, H-PU-DMH 8X1 FOR DPM
MPH000100	O-ring FOR RZB PLATE FOR DPM
Recommended Spare parts	
HSL010950	PT100 TEMPERATURE SENSOR, 1.1m
OPH000200	QUARTZ SLEEVE RZB
OPB000100	UV SENSOR 10W/m2
SAE013300	LAMP CONNECTOR ASSY, RZ
PS0000340	BALLAST MODULE 4.2KW, 380-480V
PS0000500	POWER SUPPLY, AC/DC, 48V, 5A, 240W, DIN RAIL
AP0001200	TEMPERATURE CONVERTER CARD
AP0001350	LM CARD, RZB SYSTEMS
AP0001400	DC/DC MODULE, 48V TO 24V
HA0000321	TIMING BELT, 280x10, RTT5, PU
PS0004000	POWER SUPPLY, AC/DC, 12V, 2.1A, 25.2W, HOLES